

COLO-RECTAL SURGERY

Functional Colorectal Disorders

CONSTIPATION

Physiology of Defecation

- The stimulus to defecation is distention of the rectum.
- Relaxation of the internal anal sphincter (IAS)
- Contraction of the external sphincter (EAS)
- Sampling response
- Squatting position
- The angulation between the rectum and the anal canal is straightened out (Figure 1).
- Valsalva maneuver coordinated with pelvic floor relaxation.
- Inhibition of the EAS, which permits passage of the fecal bolus.

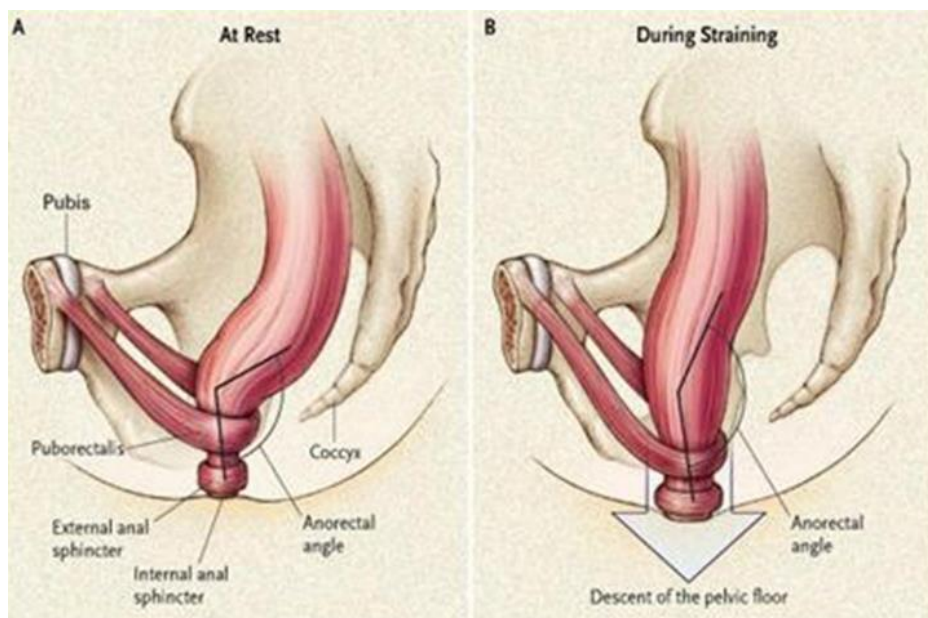


Figure 1. Normal bowel evacuation at rest (A) and during straining (B): pelvic floor and anal function

What is the Normal Defecation Frequency?

- It ranges from 2-3 time/week

Definition of Constipation

- Constipation is the infrequent passage of stool.
- It is generally defined as ≤ 3 bowel movements per week, which may be associated with straining to defecate, the passage of hard stools, tenesmus, or the need for self-digitation to evacuate stool.

Epidemiology

- *Prevalence*: 14% of the general population experiences chronic constipation.
- *Gender*: ♀ > ♂ (3:1)

Classification

According to Course

- Acute or chronic
- Chronic Functional Constipation
Subtypes
 1. Slow-transit constipation
 2. Pelvic floor dys-synergia = paradoxical contraction or inadequate relaxation of the pelvic floor muscles during attempted defecation. (is any disturbance of muscular coordination, resulting in uncoordinated movements) obstructed defecation (dys-synergia or anismus)
 3. Inflammatory bowel disease (IBS) with constipation

According to Etiology

- Causes and differential diagnosis of constipation are summarized in Table 1.
 1. Primary constipation (functional constipation)
 - It means constipation in the absence of an identifiable medical disorder.
 - It is most commonly due to poor diet and insufficient exercise.
 2. Secondary constipation
 - It means constipation due to a medical disorder or medication

Table 1: Causes and differential diagnosis of constipation

Type of Constipation	Causes	Examples
<i>Primary Constipation / Functional constipation</i>	Normal transit constipation Slow transit constipation Pelvic floor dysfunction	- Pelvic floor dys-synergia
<i>Secondary Constipation</i>	Gastrointestinal Causes	- Colorectal carcinoma - Diverticulosis / diverticulitis - Anal cancer - Stricture (colonic – rectal) - Thrombosed hemorrhoids - Incomplete bowel obstruction - Complete bowel obstruction - Volvulus - Ileus - Obstructed hernia - Irritable bowel syndrome (IBS)
	Neurological Causes	- Autonomic neuropathy (DM) - Stroke - Hirschsprung disease - Parkinson disease - Chaga's disease (megacolon) - Botulism - Multiple sclerosis - Peripheral neuropathy (DM, vitamin B₁₂ deficiency) - Spinal cord injury - Pelvic dys-synergia
	Endocrine Causes	- Hypothyroidism - DM - Electrolyte imbalance (hypokalemia, hypercalcemia) - Hyperparathyroidism (HPT)
	Connective Tissue Disorders	- Scleroderma - SLE - Dermatomyositis - Amyloidosis
	Drug-Induced	- Opioid analgesics - Iron supplements - Antacids - Anti-cholinergics - Anti-depressants (tricyclic) - Calcium channel blockers - Bile acid resins - NSAIDs - Beta-blockers - Calcium supplements - Anti-psychotics - Anti-parkinsonism - Anti-convulsants - 5-HT₃-receptor antagonists

Pathophysiology

- Both primary and secondary constipation can cause changes in stool consistency and defecation habits. External factors such as lack of exercise or inadequate fluid and fiber intake (primary constipation)/internal factors such as changes within the colon or rectum (secondary constipation) → slow passage of stool → prolonged absorption of water by the bowel → dry, hard stool → painful defecation → sensation of incomplete and irregular bowel emptying → constipation.
- Mechanism of altered bowel motility
 1. Effective peristalsis of the bowel is controlled by intrinsic (e.g. myenteric plexus) and extrinsic (e.g., sympathetic, and parasympathetic) innervation.
 2. Any alteration in bowel innervation may lead to ineffective peristalsis.
- Drugs (e.g. calcium channel blockers, opiates, antispasmodics, antidepressants) → altered autonomic outflow and bowel muscle contraction.
- Endocrine pathology (e.g. hypothyroidism) → down-regulated bowel motility
- Neurological pathology (e.g. spinal injury, enteric neuropathy) → disease or trauma of bowel innervation
- Ineffective peristalsis → difficult passage of stool regardless of stool consistency → sensation of incomplete and irregular bowel emptying.

Causes of Obstructed Defecation (Mechanical Causes)

Recto-anal Intussusception (Internal or Occult Rectal Prolapse / Procidentia)

- *Definition*

Circumferential, full thickness, enfolding of the mid-rectum during straining on defecography (Figure 2). This enfolding extends into the anal canal, without reaching the anus.
- *Classification*
 1. First-degree when the protrusion is invisible.
 2. Second-degree when it is visible on straining,
 3. Third-degree when visible externally
- *Etiology*
 1. Diastasis of the levator
 2. Deep Douglas pouch
 3. Redundant recto-sigmoid
 4. Loss of rectal horizontal position (due to lose attachment to the sacrum)

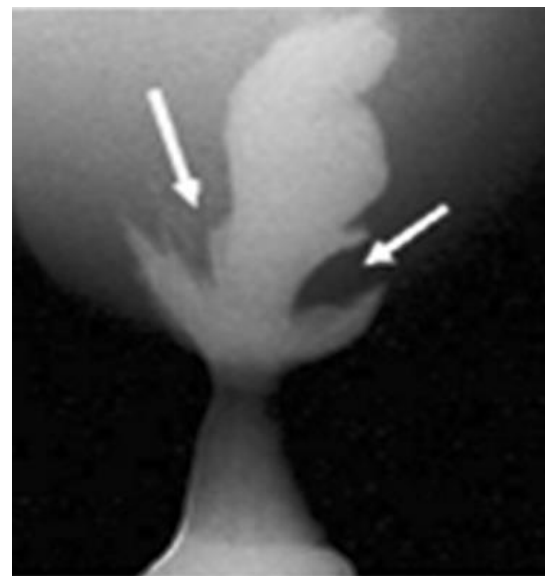


Figure 2. Defecography shows Recto-anal intussusception (arrows)

Rectocele

- **Definition:** Herniation of the rectal wall through a defect in the recto-vaginal septum in the direction of the vagina (Figure 3).
- **Classification**
 1. First-degree: 2-4 cm
 2. Second-degree: 4-6 cm
 3. Third-degree: > 6 cm
- **Etiology**
 1. Vaginal childbirth,
 2. Chronic ↑ in abdominal pressure
 3. Congenital or inherited weakness in the pelvic support system
- **Treatment:** Most surgeons advocate operative repair when a symptomatic rectocele is large (> 3 cm) or if it fails to empty sufficiently on defecography.



Figure 3. Rectocele (herniation of the rectal wall)

Enterocoele

- **Definition:** Prolapse of the small bowel into the recto-genital space (Figure 4).
- **Classification (Pathological)**
 1. Congenital (unusual deep Douglas pouch)
 2. Pulsion-mediated (caused by chronic ↑ of abdominal pressure)
 3. Traction (associated with a loss of support of the pelvic floor)
- **Etiology**
 1. Primary: Multiparity, advanced age, general lack of elasticity, obesity & ↑ intra-abdominal pressure.
 2. Secondary: After gynecological surgical procedures, especially hysterectomy.

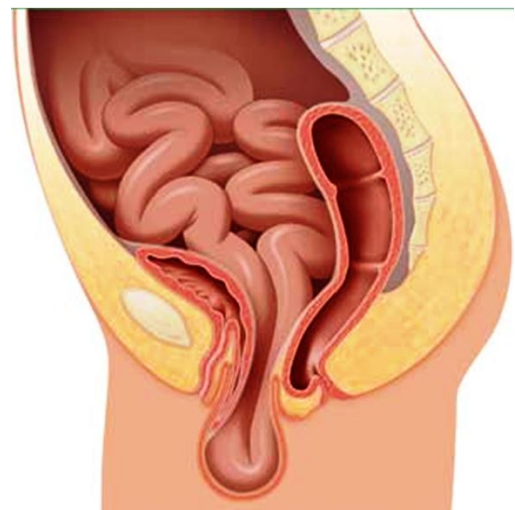


Figure 4. Enterocoele (prolapsed small bowel)

Diagnosis

Patient History

- Diagnostic criteria for *functional constipation* (Rome diagnostic criteria for functional constipation) (Table 2) must include *two or more* of the following:
 1. Straining during > ¼ (25%) of defecations
 2. Lumpy or hard stools (Bristol Stool Form Scale 1-2) > ¼ (25%) of defecations
 3. Sensation of incomplete evacuation > ¼ (25%) of defecations
 4. Sensation of anorectal obstruction/blockage > ¼ (25%) of defecations

5. Manual maneuvers to facilitate > ¼ (25%) of defecations (e.g. digital evacuation, support of the pelvic floor)
 6. Fewer than three SBM per week
 7. Loose stools are rarely present without the use of laxatives.
 8. Insufficient criteria for irritable bowel syndrome
- Criteria fulfilled for the last 3 months with symptom onset at least 6 months prior to diagnosis

Table 2. Rome diagnostic criteria for functional constipation in adults

Symptoms / Diagnosis	Rome I	Rome II	Rome III	Rome IV
Straining to evacuate	>25% of defecations	>25% of defecations	>25% of defecations	>25% of defecations
Lumpy or hard stools	>25% of defecations	>25% of defecations	>25% of defecations	>25% of defecations
Sensation of incomplete evacuation	>25% of defecations	>25% of defecations	>25% of defecations	>25% of defecations
Sensation of anorectal obstruction/blockage	-	>25% of defecations	>25% of defecations	>25% of defecations
Manual maneuvers to facilitate defecations	-	>25% of defecations	>25% of defecations	>25% of defecations
Less than three evacuations/week	Yes	Yes	Yes	Yes
Number of criteria for diagnosis	≥2	≥2	≥2	≥2
Chronological factor	3 months	12 weeks/12 months	3-6 months	3-6 months

Physical Examination

- Inspect the anorectal area for possible fissures or hemorrhoids
- Check the anal wink reflex: An absent anal wink reflex suggests a pathology (e.g., sacral nerve injury).
- Digital examination of the rectum
- Check for rectal carcinoma.
- Test the sphincter tone to evaluate for pelvic floor dysfunction.

Investigations

- *Laboratory investigations:* exclude hypokalemia, hypothyroidism, diabetes mellitus
- *Imaging*
 1. Abdominal X-ray
 2. Colon transit time
 - Procedure: The patient ingests a number of radiopaque markers (plastic rings containing radiopaque material) (Figure 5) in a meal and abdominal radiographs are obtained to monitor the clearance of the rings from the colon.
 - Normal colonic transit time is 20-56 hours, and most adults will clear all the markers in 4-5 days.
 - Pre-procedural evaluation: Laxatives (and possibly GI-active medications) should be temporarily dis-continued, otherwise, no special preparation is needed.
 - Technique: 20 or 24 markers are ingested in a single dose, and then serial daily abdominal radiographs are obtained (Figure 6)



Figure 5: Colon transit time markers



Figure 6. Day one after oral intake of the capsule markers.

- *Colonoscopy*: to exclude mechanical obstruction (e.g. tumor, stenosis).
- *Anorectal manometry and electromyography [EMG]* (lack of anorectal inhibitory reflex for Hirschsprung disease, high rectal volume tolerability for rectal inertia and internal sphincter achalasia due to lack of nitrinergic fibers)
- *Cineodefecography* (occult prolapse, sigmoidocele, perineal descent & paradoxical puborectalis contraction)
- *Suction capsule rectal biopsy* (for absence or presence of ganglia, and plexus of nerves and possible histochemical staining procedures)

Treatment

Laxatives (Table 3)

Table 3: Laxative agents, mechanism of action and adverse effects

Class	Agents	Mechanism of Action	Adverse Effects
<i>Osmotic Laxatives</i>	- Polyethylene glycol (PEG) (very effective and well-tolerated, best initial treatment) - Glycerin - Magnesium hydroxide - Magnesium citrate - Lactulose - Sorbitol	- Increase of osmotic pressure draws water into the intestinal lumen → stimulation of intestinal motility. - Lactulose is degraded by intestinal microbiota into lactic acid and acetic acid: - Induces nitrogen (NH_4^+) excretion. - Used in the treatment of hepatic encephalopathy	- Diarrhea - Dehydration - Hypernatremia - Hypermagnesemia - Severe flatulence (lactulose and sorbitol) - Osmotic laxative misuse is frequently seen in patients with bulimia nervosa - Overuse causes metabolic acidosis.
<i>Stimulant / Secretory Laxatives</i>	- Senna - Bisacodyl	- Stimulation of nitric oxide-mediated epithelial cell secretion of electrolytes into the colonic lumen - Myenteric neuronal depolarization → colon contractions	- For short-term use only - Diarrhea - Hypokalemia - Bisacodyl overuse can cause metabolic acidosis - Senna may cause melanosis coli.
<i>Emollient stool softener</i>	- Docusate	- Emulsification (i.e. integration of water and fat) of stool → softening of stool → easier passage through the intestinal tract	- Diarrhea - Metabolic alkalosis - Bloating - Cramping
<i>Bulk-forming Laxatives</i>	- Methylcellulose - Psyllium husks - Polycarbophil	- Bulk-forming laxatives are indigestible, not systemically absorbed. - Soluble fibers increase water absorption in the intestinal lumen → stretching of the bowel wall → stimulation of peristalsis	- Bloating - Worsening constipation or ileus if the patient does not take enough water with doses

- *Biofeedback Techniques for Treatment of Anismus*
Most studies report good short-term efficacy with improved psychological quality of life, whereas long-term follow-up studies showed a fading effect over time. They include:
 1. Sensory training
 2. Electro-myographic feedback
 3. Manometric feedback.
- *Surgery* (Not for psychological patients)
 1. Surgical treatment of rectocele: Trans-vaginal, trans-anal, trans-perineal, and abdominal repair
 2. Surgical treatment of rectal intussusception: perineal procedures (Delorme's procedure)
 3. Surgical treatment of rectal prolapse (intra-rectal intussusception): Abdominal procedures (open technique) – laparoscopic procedures.
 4. Refractory mechanical or functional outlet problems (fecal diversion)
 5. Refractory slow transit (subtotal colectomy + ileo-rectal anastomosis)

INCONTINENCE

Definition

- Incontinence is defined as “recurrent uncontrolled passage of fecal material for at least one month in an individual with a developmental age of at least 4 years”.
- It occurs in 1.4-18% of population.

Factors Necessary for Maintaining Fecal Continence (Figure 7)

- Highly compliant rectal wall.
- Normal anorectal angle.
- Formed fecal materials.
- Normal anal canal and sphincter function.
- Anal cushions.

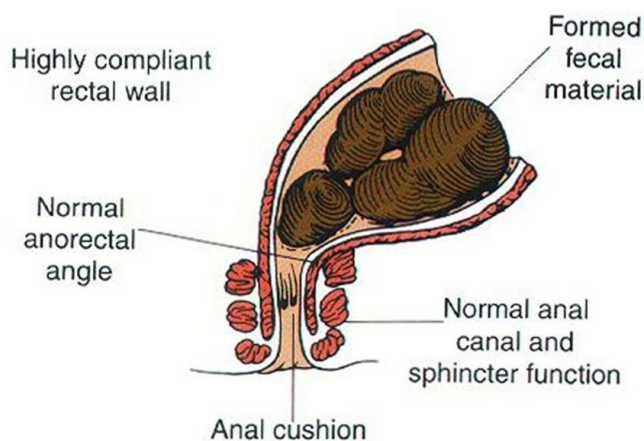


Figure 7: Factors necessary for maintenance of fecal continence (+ anal and rectal sensation)

Causes of Incontinence

Any change or inadequacy of the previous factors will lead to a degree of incontinence.

Inadequate Rectal Compliance

- Absent rectal reservoir (ileo-anal anastomosis, low ant. resection)
- Rectal ischemia
- Rectal neoplasms
- Altered stool consistency
- Inflammatory bowel disease (IBD)
- Infectious diarrhea
- Laxative abuse
- Radiation enteritis
- Short bowel syndrome (SBS)
- Malabsorption syndrome

Inadequate Rectal Sensation

- Dementia,
- Cerebrovascular accident (CVA)
- Multiple sclerosis (MS)
- Brain or spinal cord injury/neoplasm
- Sensory neuropathy
- Tabes dorsalis
- Overflow incontinence
- Fecal impaction – leading cause of incontinence in institutionalized elderly patients
- Diabetes Mellitus (DM)

Anatomical Sphincter Defect (Internal or External Sphincter)

- Traumatic
- Obstetric injury – prolonged difficult labor with forceps application, episiotomy complications, third or fourth-degree lacerations
- Anorectal surgery – anal fistula surgery (the most common operative procedure that results in fecal incontinence), hemorrhoidectomy.

Pelvic Floor Denervation

- Degenerative neurogenic factors are a common cause of non-surgically related incontinence.
- Primary (idiopathic neurogenic incontinence)
- Pudendal neuropathy: Denervation of puborectalis muscle and external sphincter muscles
- Secondary
- Injuries to spinal cord, cauda equina
- Congenital abnormalities: spina bifida – myelo-meningocele

Rectal Prolapse

- Approximately, 60-70% of patients with rectal prolapse suffer from incontinence.

Types of Incontinence

- Gas / soft or liquid stool /solid stool
- Passive or with awareness
- Mild (soiling); moderate; severe

Scoring

Clinical scoring was adopted by Wexner (Cleveland clinic) for detection of the condition severity (Table 4).

Table 4 Cleveland Clinic Scoring System (Wexner`s scoring)

	Never	Rarely	Sometimes	Usually,	Always
<i>Solids</i>	0	1	2	3	4
<i>Liquids</i>	0	1	2	3	4
<i>Flatus</i>	0	1	2	3	4
<i>Use of Pad</i>	0	1	2	3	4
<i>Lifestyle Alteration</i>	0	1	2	3	4

Diagnostic Evaluation

- *Clinical examination* especially perineal and neurological examinations.
- *Laboratory investigations*
- *Procto-sigmoidoscopy*: Rigid or flexible procto-sigmoidoscopy can be performed in the office setting and identify characteristic mucosal inflammatory changes e.g. IBD.
- *Endoanal ultrasonography (US)* (Figures 8 and 9): It is a non-invasive tool that is used with increasing frequency to examine anal sphincter integrity. It is a relatively comfortable procedure for the patient and is capable of accurately identifying sphincter defects.

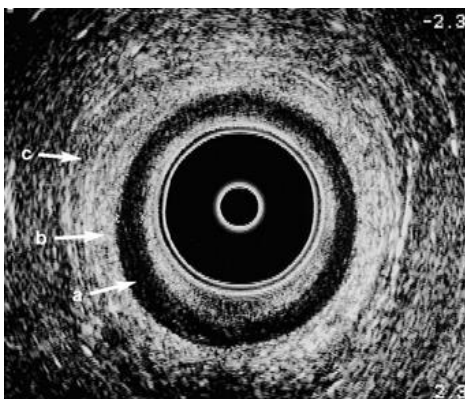


Figure 8: Normal endoanal US



Figure 9: Anterior defect IAS and EAS (arrow)

- *Magnetic resonance imaging (MRI) and dynamic MRI* (Figures 10 and 11):
- It is valuable in detection of anal sphincter s defects. MRI should be considered in potential candidates for anal sphincters repair.

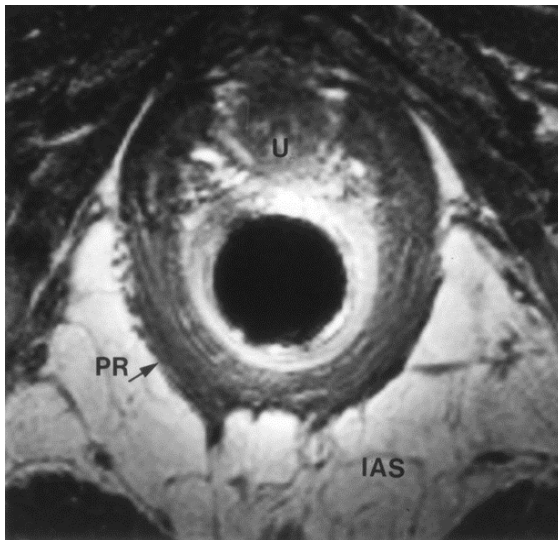


Figure 10: Normal MRI and sphincters

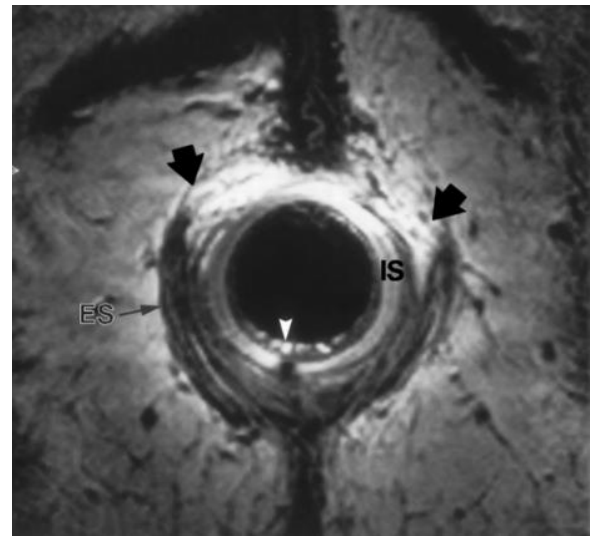


Figure 11: Defect of IAS and EAS

- *Ano-rectal Manometry:*
It is an objective method of studying the physiology of the ano-rectal sphincter mechanism. The principle is based on measurement of the resistance in terms of pressure that the sphincters offer to a constant flow of water.
- *Electromyography (EMG):*
Anal EMG is the recording of myo-electrical activity from the striated sphincter at rest, during voluntary & reflex contraction, and simulated defecation.
- *Pudendal Nerve Terminal Motor Latency (PNTML):*
It is a simple method of assessing pudendal nerve function. This test indicates the integrity of the distal motor innervation of the pelvic floor musculature. It is important in the evaluation of neurogenic fecal incontinence.
- *Defecography and video defecography* (Figure 12):
Barium, thickened to the consistency of stool, is introduced into the rectum.
 - Evacuation is monitored with fluoroscopy.
 - Assessment of the anorectal angle at rest and during defecation.
 - The following may be detected:
 1. Excessive perineal descent
 2. Failure of the puborectalis muscle to relax
 3. Rectocele
 4. Internal intussusception.

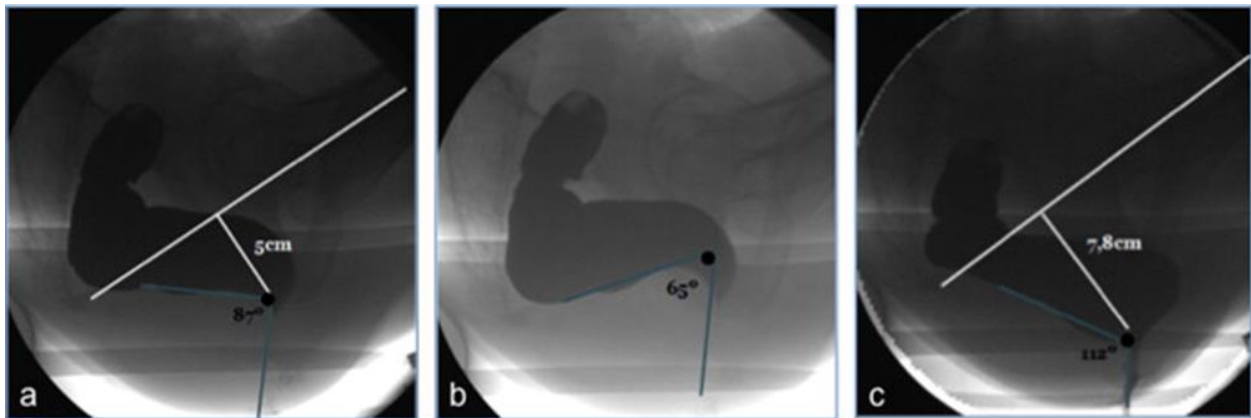


Figure 12: Video defecography

Treatment Options

The choice of treatment needs to be individualized according to the patient, etiology and severity of incontinence.

Non-invasive Treatment

1. *Sphincter exercises:* These can help improve the bowel control, especially if the main problem is urgency and there is no major defect in the muscles.
2. *Bowel habit regulation:* Some people find that the bowel responds well to a regular habit.
3. *Skin (peri-anal) care* and coping with soreness or irritation with careful personal hygiene.
4. *Medications:* Constipating agents, such as loperamide, codeine phosphate and co-phenotrope (diphenoxylate and atropine), are valuable agents in the treatment of fecal incontinence
5. *Biofeedback:* It is a behavioral technique that captures the “bio” logical information about an individual. This information is "feed" back to the individual using auditory or visual feedback. The individual uses this information to gain sensitivity, and with practice, control over bodily functions that were dysfunctional or over-functioning and previously thought to be uncontrollable.

Minimally-invasive Procedures

1. *Evacuation enema:* Another alternative for conservative treatment of anal incontinence is the daily use of colonic irrigation. Usually, patients are instructed to fill their rectum with 500-1000 ml of water while they are seated on a toilet. However, colonic irrigation can potentially cause severe adverse effects and must be carefully administered.
2. *Antegrade continent enema (Malone procedure):* For patients with rectal evacuation disorders, regular antegrade irrigation of the colon is employed to achieve evacuation.
3. *Anal encircling procedure (Thiersch Procedure):* It was originally advocated for treatment of rectal prolapse, then for the treatment of anal incontinence. Complications of the procedure include infection, stricture, fecal impaction, persistent incontinence and pain.
4. *Anal plugs:* Anal plug is another non-surgical option for patients who want to avoid surgery.

5. *Radiofrequency Energy (SECCA procedure)*: The addition of temperature-control technology in the last decade has allowed radiofrequency to be utilized for more specific therapeutic applications.
6. *Tissue augmentation*: Injections directly fill the asymmetric anal canal defect using a bulking agent to augment the internal anal sphincter (IAS) muscle. Bulking agents include Polytetrafluoroethylene paste (PTP implants), Teflon, Polytex, Autologous fat, Micro-carbon coated beads, Collagen & silicone injection (Figure 13). These are injected trans-sphincteric ally to the IAS. This has been recently shown to ↑ anal canal resting pressure in healthy volunteers. This novel agent may have a future role in the treatment of passive fecal incontinence associated with a weak IAS.

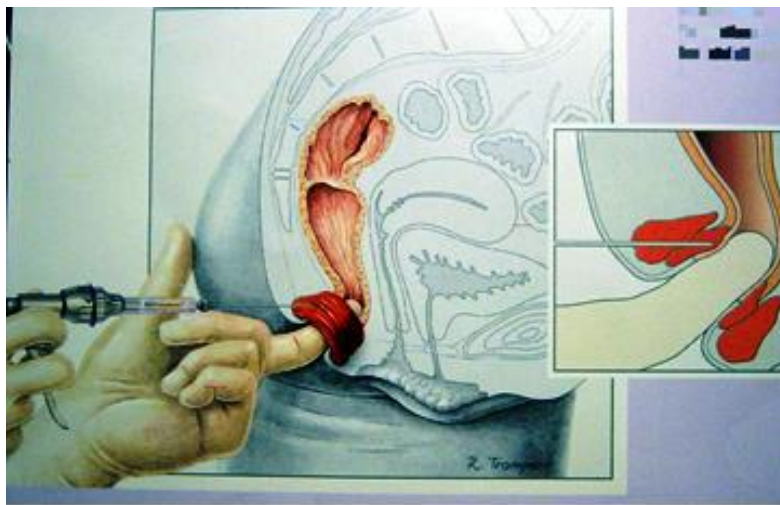


Figure 13: Internal anal sphincter (IAS) defect treated by injectable agents (silicone phenylephrine gel)

Invasive (Surgical) Treatment

A. Sphincter Repair

Sphincter repair is considered in patients with an isolated sphincter defect because the remaining muscle usually maintains the ability to contract. There are 3 approaches to sphincter repair: namely, apposition, placcation, and overlapping.

1. *Direct apposition*
 - It involves mobilization of the external anal sphincter (EAS) and suture of the muscle ends in an end-to-end fashion.
2. *Post-anal repair*
 - It was devised by Parks for the treatment of idiopathic and neurogenic incontinence.
 - The puborectalis limbs are approximated and the EAS is plicated.
3. *Overlapping repair* (Figure 14)
 - It is preferable to a simple apposition of the sphincter muscles.
 - Patients best suited are those in whom incontinence is 2^{ry} to an anterior sphincter defect.
 - Obstetric and iatrogenic causes are among the commonest surgically correctable faecal incontinence.

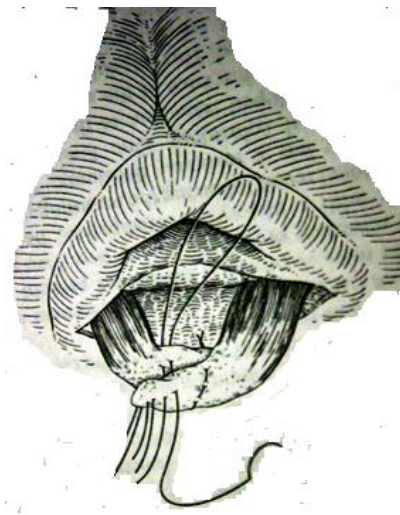


Figure 14: Overlapping repair

4. *Total pelvic floor repair*

- It is the combination of anterior levator plasty and IAS plication with post-anal repair.
- This operation was proposed by Keighley for patients who had a suboptimal outcome after post-anal repair.

5. *Anterior levator plasty and EAS plication*

- It is preferred by some surgeons > post-anal repair, which has a poor functional outcome, especially on long-term results.
- It is used mainly in post-obstetric fecal incontinence).

6. *Puborectalis plication*

- Physical finding of lax puborectalis musculature, obtuseness of the ano-rectal angle and bulging of the anterior rectal wall into the proctoscope suggest incontinence.

B. Muscle Wrap

The most widely used for the creation of a new anal sphincter are transposed skeletal muscles. Muscles that may be suited for muscle wrapping are the following:

1. *Sartorius muscle*

- It has segmental vascularisation, which restricts the arch of rotation if the muscle is transposed, the sartorius muscle seems to be less-well suited.

2. *Gluteus maximus*

- The choice of this muscle to recreate an anal sphincter is logical because it is a powerful muscle that often is used as an accessory muscle to continence.

3. *Gracilis muscle*

- First used for sphincter-wrapping in 1952 by Pickerel in the treatment of children with faecal incontinence due to a neurogenic cause.
- It has been applied on a wide scale with mixed results. Although active contraction of the muscle is forceful enough to hold up stool, still all-day continence is seldom achieved. This is also because the gracilis, like all skeletal muscles, is unable to have a sustained contraction.

C. *Artificial Sphincter Replacement* (Figure15)

- Artificial sphincters have been used to treat urinary incontinence since 1973.
- In 1987, Christiansen and Lorenzen applied a device (called the American Medical System) to a patient with fecal incontinence. The implantation procedure involves the surgical insertion of a cuff in a tunnel around the anal canal, tubing between the cuff and an abdominal incision, a balloon behind the rectus abdominis, and a pump in the labia majora or scrotum.

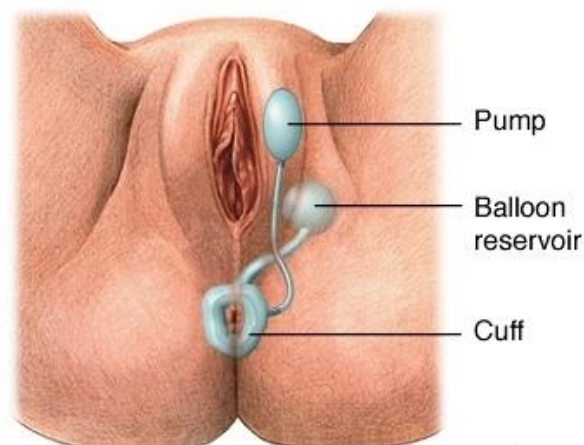


Figure 15: Artificial anal sphincter device in place

D. *Neuromodulation*

1. *Dynamic graciloplasty* (Figure16)

- Gracilis muscle is wrapped around the anal verge and its nerve supply is connected to a neuro-stimulator via electrodes (Figure 17) and the neurostimulator placed subcutaneously. This neuro-stimulator maintains muscle contraction and continence. This overcomes the problem of skeletal muscle fatigue.
- The average patient accepted for dynamic graciloplasty has a severe incontinence for many years, has had maximal conservative treatment, biofeedback training and one or more previous operations for restoration of continence.

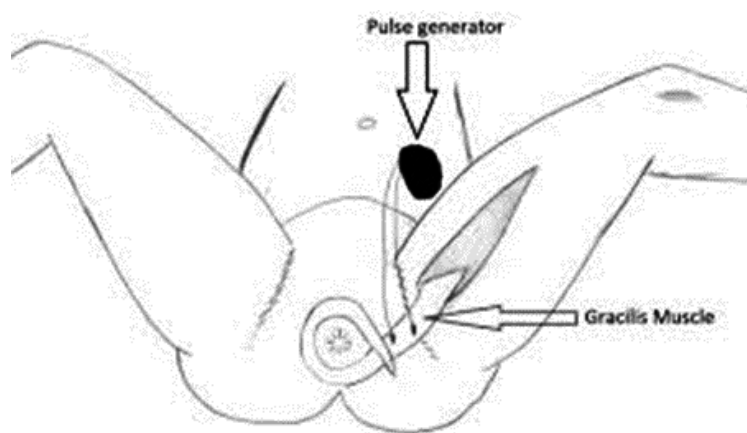


Figure 16: Dynamic graciloplasty



Figure 17: Neuro-stimulator

2. *Sacral Nerve Stimulation (SNS)* (Figure 17)

- Nowadays, stimulation of the sacral nerve roots is successfully used to control urinary incontinence, but only recently has experience gained with this technique been transferred to the field of ano-rectal disturbances, above all fecal incontinence.
- The concept is to recruit residual function of the ano-rectal continence organ by electrical stimulation of its peripheral nerve supply.

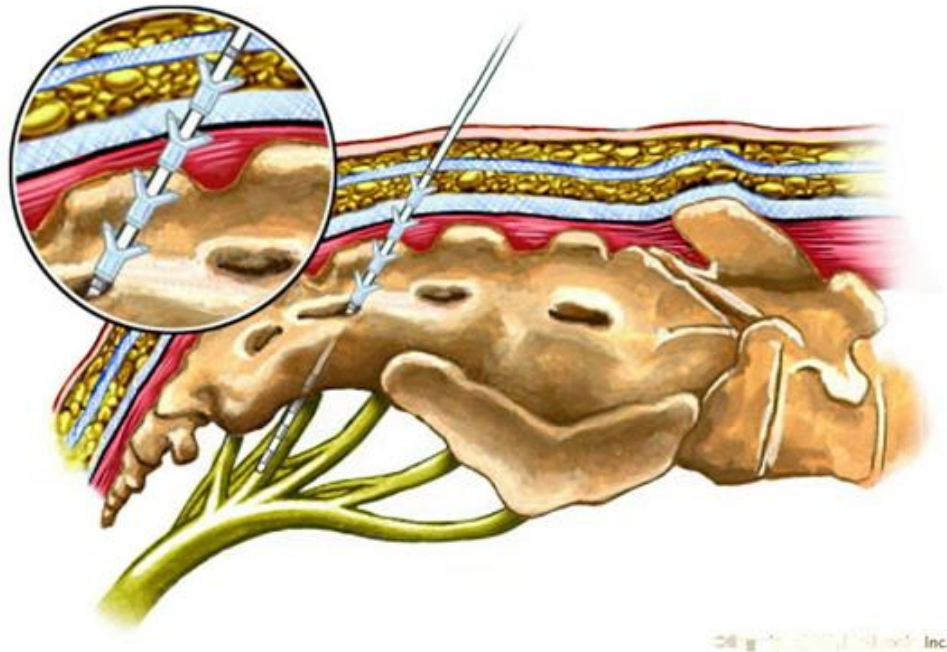


Figure 17: sacral nerve stimulation.

E. Diverting Stoma

- Although fecal diversion may be considered a failure of fecal incontinence treatment, it is an effective, safe, and appropriate operation for certain patients with severe incontinence.

Diverticular Disease and Polyps

DIVERTICULAR DISEASE

Types

- Diverticula can occur in a wide number of positions in the gut, from the esophagus to the recto-sigmoid.
- There are two varieties:
 1. *Congenital*: All 3 coats of the bowel are present in the wall of the diverticulum, e.g. Meckel's.
 2. *Acquired*: The wall of the diverticulum lacks a proper muscular coat. Most alimentary diverticula are thought to be acquired.

A. DIVERTICULA OF THE SMALL INTESTINE

Most of these diverticula arise from the mesenteric side of the bowel, probably as the result of mucosal herniation through the point of entry of blood vessels.

Duodenal Diverticula

There are two types:

1. *Primary*.

Mostly occurring in older patients on the inner wall of the 2nd and 3rd parts (Figure 1). These diverticula are found incidentally on barium meal and are usually *asymptomatic*. They can cause problems locating the ampulla during ERCP.

2. *Secondary*.

Diverticula of the duodenal cap result from long-standing duodenal ulceration.



Figure 1. Duodenal diverticula on the inner wall of the 2nd part of duodenum.

Jejunal Diverticula

- These are usually multiple and of variable size (Figure 2). Clinically, they may (1) be symptomless, (2) cause abdominal pain, (3) produce a malabsorption syndrome or (4) present as an acute abdomen with acute inflammation and occasionally perforation.
- They are more common in patients with connective tissue disorders. In patients with major malabsorption problems, they give rise to anemia, steatorrhea, hypoproteinemia or vitamin B₁₂ deficiency.
- Resection of the affected segment with end-to-end anastomosis (R-EEA) can be an effective treatment.



Figure 2. Multiple jejunal diverticula

Meckel's Diverticulum

- Meckel's diverticulum represents the patent intestinal end of the vitello-intestinal tract (VIT) (Figure 3)

Incidence and Location

- It is present in 2% of the population.
- It is situated on the *anti-mesenteric* border of the small intestine, commonly 60 cm (2 feet) from the ileo-cecal valve, and is usually 3–5 cm (2 inches) long. Many variations occur
- "2% – 2 feet – 2 inches" is a useful *aide-mémoire* (Figure 4).

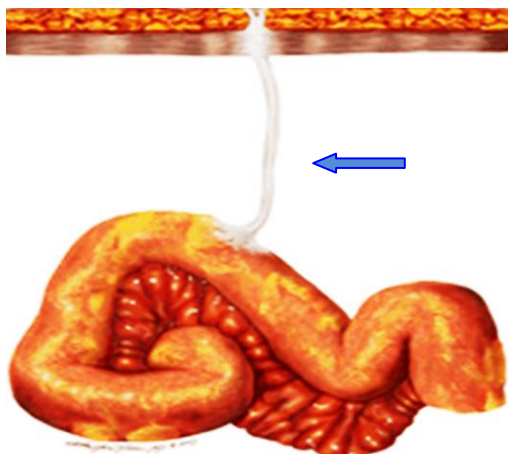


Figure 3. Vitello-intestinal tract (blue arrow)

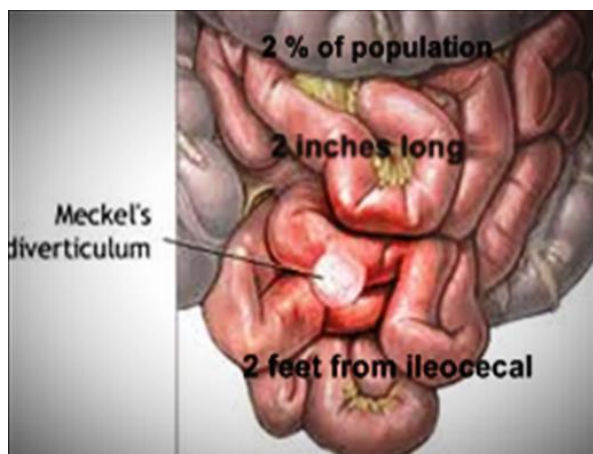


Figure 4. Meckel's diverticulum: The rule of "two"

Pathology

- A Meckel's diverticulum possesses *all 3 coats* of the intestinal wall & has its own blood supply (Figure 5). It is $\therefore$ vulnerable to infection & obstruction in the same way as the appendix. Indeed, when a normal appendix is found at surgery for suspected appendicitis, a Meckel's diverticulum should be sought by inspection of an appropriate length of terminal ileum.
- In 20% of cases, the mucosa contains *heterotopic epithelium*, namely;
 1. Gastric
 2. Colonic
 3. Sometimes pancreatic tissue.

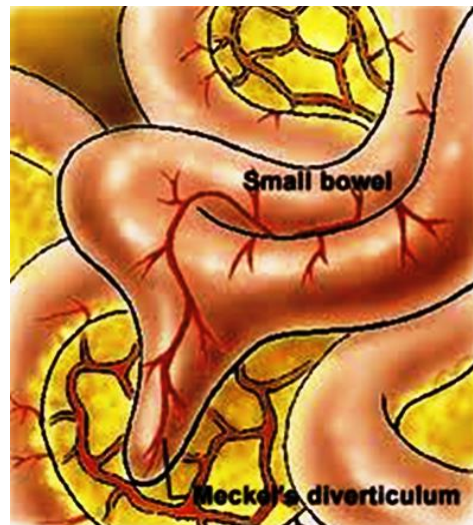


Figure 5. Meckel's diverticulum has all 3 coats of the intestinal wall

Clinical Presentation

- In order of frequency, these symptoms are as follows:
 1. *Severe hemorrhage* (Figure 6) caused by peptic ulceration. Painless bleeding occurs per rectum and is maroon in color. An operation is sometimes required for serious progressive gastrointestinal (GI) bleeding. When no lesion in the stomach or duodenum can be found, the terminal 150 cm of ileum should be carefully inspected.
 2. *Meckel's diverticulitis* (Figure 7) may be difficult to distinguish from the symptoms of acute appendicitis. When a diverticulum perforates, the symptoms may simulate those of a perforated duodenal ulcer (DU). At operation, an inflamed diverticulum should be sought as soon as it has been demonstrated that the appendix and Fallopian tubes are not at fault.



Figure 6. Hemorrhage from Meckel's diverticulum



Figure 7. Inflammation of Meckel's diverticulum

3. *Chronic peptic ulceration (PU)*

- As the diverticulum is part of the mid-gut, the pain, although related to meals, is felt around the umbilicus.
- It is caused by the presence of ectopic gastric epithelium in the wall of the Meckel's diverticulum (Figure 8).
- The base of the diverticulum should be palpated intra-operatively to rule out any thickening that may represent ectopic tissue.

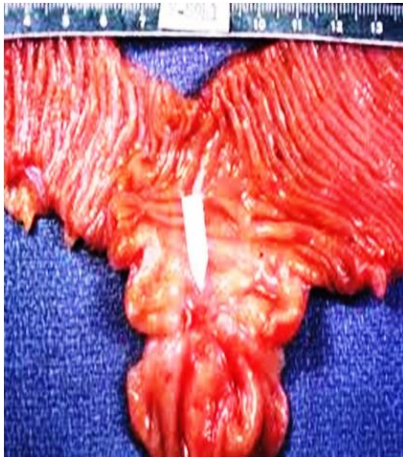


Figure 8. Ectopic gastric tissue in a Meckel's diverticulum (white arrow)

4. *Intussusception* (Figure 9): In most cases, the apex of the intussusception is the swollen, inflamed (Figure 10), with heterotopic epithelium at the mouth of the diverticulum.



Figure 9. Intussusception of Meckel's diverticulum



Figure 10. Swollen & inflamed apex of the intussusception

5. *Intestinal obstruction.* The presence of a band between the apex of the diverticulum & the umbilicus (Figure 11) may cause obstruction either by the band itself or by a volvulus around it. If neglected, it may cause gangrene of the bowel. (Figure 12).



Figure 11. a band between the apex of the diverticulum & the umbilicus



Figure 12. If neglected, it may cause gangrene of the bowel.

- *'Silent' Meckel's diverticulum*

An aphorism attributed to **Dr. Charles Mayo** is: '*a Meckel's diverticulum is frequently suspected, often sought & seldom found*'. A Meckel's diverticulum usually remains symptomless throughout life and is found only at necropsy. When a *silent Meckel's* diverticulum is encountered in the course of an abdominal operation, *it can be left* provided it is wide mouthed and the wall of the diverticulum does *not* feel thickened (Figure 13). Where there is doubt and it can be removed without appreciable additional risk, it should be resected.

- *Exceptionally, a Meckel's diverticulum is found in an inguinal or a femoral hernia sac – Littre's hernia* (Figure 14).



Figure 13. "Silent Meckel's diverticulum" could be left alone



Figure 14. Meckel's diverticulum in an inguinal hernia (*Littre's hernia* - rare)

Investigations (Imaging)

- Meckel's diverticulum can be very difficult to demonstrate by contrast radiology; *small bowel enema* would be the most accurate investigation (Figure 15).
- *Technetium-99m (^{99m}Tc) scanning* may be useful in identifying Meckel's diverticulum as a source of GI bleeding (Figure 16).

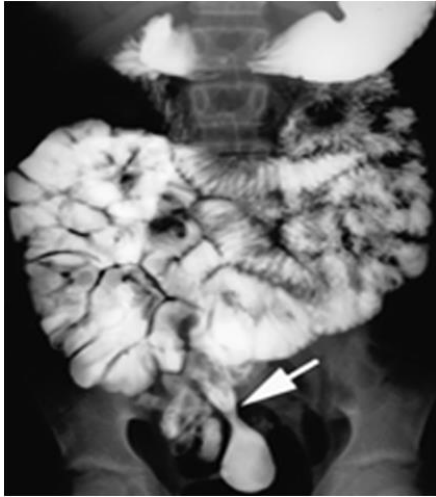


Figure 15. Small bowel enema showing Meckel's diverticulum (white arrow)



Figure 16. *Technetium-99m (^{99m}Tc) scanning* showing bleeding from Meckel's diverticulum (black arrow)

Treatment

- *Meckel's diverticulectomy*
 - A Meckel's diverticulum that is *broad-based* should not be amputated at its base & invaginated in the same way as a vermiform appendix, because of the risk of stricture. Furthermore, this does not remove heterotopic epithelium when it is present.
 - Alternatively, a linear stapler device may be used (Figure 17).
- Where there is induration of the base of the diverticulum extending into the adjacent ileum, it is advisable to resect a short segment of ileum containing the diverticulum, restoring continuity with an end-to-end anastomosis (Figure 18).

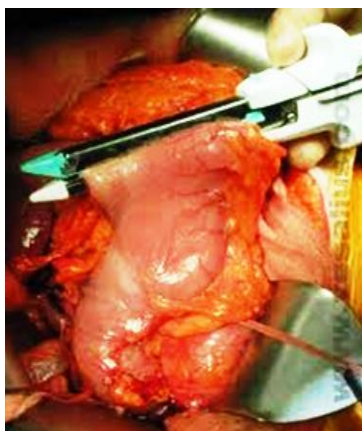


Figure 17. Meckel's diverticulectomy using stapler

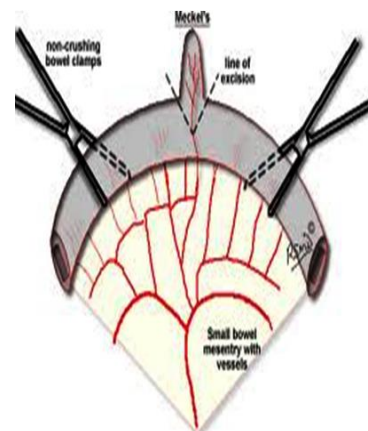


Figure 18. Resection of adjacent segment of small bowel

B. DIVERTICULAR DISEASE OF THE COLON

Prevalence (by Age)

< 50 years	5-10 %
> 50 years.	30 %
> 70 years	50 %
> 85 years	65 %

Pathogenesis

- The start is disordered colonic motility, which leads to segmentations of the colon into "bladders" (Figure 19) separated by contraction rings with pressures reaching 90 mmHg inside these bladders. Segmentation causes high pressure and the pulsion force responsible for diverticulosis (Figure 20) i.e. herniation of colonic mucosa through the circular muscle at the points where the blood vessels penetrate the colonic wall (Figure 21).



Figure 19. disordered colonic motility → segmentations of the colon

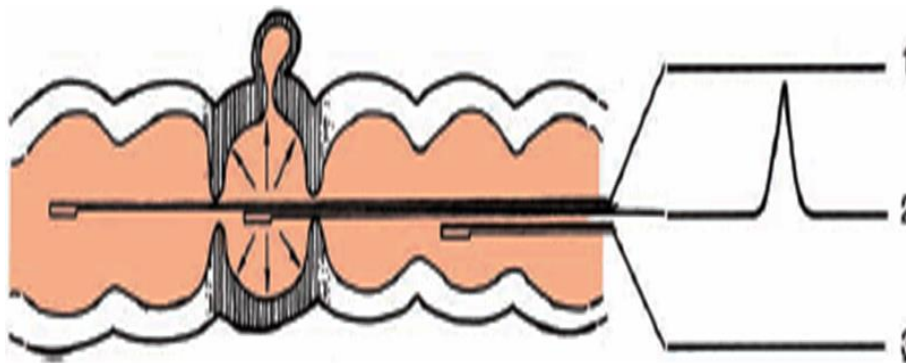


Figure 20. Segmentation of the colon causes high pressure and the pulsion force responsible for diverticulosis

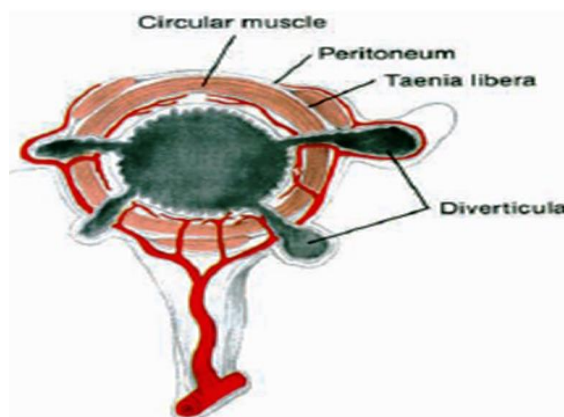


Figure 21. Herniation of colonic mucosa through the circular muscle at the points where blood vessels penetrate the colonic wall

Clinical Presentation (Table 1)

- Approximately, 90% of cases are **asymptomatic**
- It is important to distinguish between **diverticulosis**, which may be asymptomatic, and clinical diverticular disease in which the diverticula cause symptoms.
- **Diverticulitis** is the result of inflammation of one or more diverticula, usually with some peri-colitis. *It is not a precancerous condition, but cancer may coexist*
- **Pain** is due to diverticulosis or due to inflammation (diverticulitis).

Pain due to diverticulosis

- Colicky, of variable severity and duration (may be for days)
- Felt in left iliac fossa (LIF) and hypogastrium
- Precipitated by food & stress, and relieved by flatus
- Stools are small hard pieces (*sheep droppings*)

Pain due to diverticulitis

- Simulates left-sided appendicitis. Well localized + rigidity, tenderness and rebound tenderness, with a possible palpable inflammatory phlegmon

Table 1: Clinical Picture of Sigmoid Diverticular Disease

Elective Cases (Mild)	Emergency Cases
- Distension, - Flatulence - Sensation of heaviness in the lower abdomen that may be indistinguishable from those of irritable bowel syndrome (IBS).	- Persistent <i>lower abdominal pain</i>, usually in the LIF ± peritonitis, could be caused by diverticulitis. Fever, malaise & leucocytosis can differentiate diverticulitis from painful diverticulosis. - The patient may pass <i>loose stools</i> or may be <i>constipated</i>; - <i>Tender left lower abdomen</i>, but occasionally also in the RIF if the sigmoid loop lies across the midline. - The sigmoid colon is often <i>palpable</i>, tender and thickened. - PR exam may, but <i>not</i> usually, reveal a <i>tender mass</i>. - Any urinary symptoms may denote a vesico-colic fistula, which → <i>pneumaturia</i> (flatus in urine) & even <i>fecaluria</i> (stools in urine).

Clinical Picture of Complications

1. Perforation

- Localized → left-sided peritonitis, and if walled off, a fluctuant tender mass may be palpable (*peri-colic abscess*)
- Free → *generalized peritonitis*, with rapid general condition deterioration

2. Diverticulitis

3. Obstruction

- Large bowel obstruction (LBO) is due to repeated inflammation with dense fibrous tissue constriction
- Small bowel obstruction (SBO) may be due to adhesions at the site of large bowel inflammation

4. Fistula (5%)

- Abscess will burst into a nearby organ causing fistulae (small bowel, uterus, vagina, abdominal wall or perineum, and urinary bladder)
- Colo-vesical fistula is heralded by urgency and dysuria, and its manifestations are pneumaturia & fecaluria

5. Hemorrhage (15%)

- The two common causes of massive lower GIT bleeding (that may require blood transfusion) in elderly patients are diverticulosis and vascular ectatic lesions.

Natural History of Diverticulosis (Figure 22)

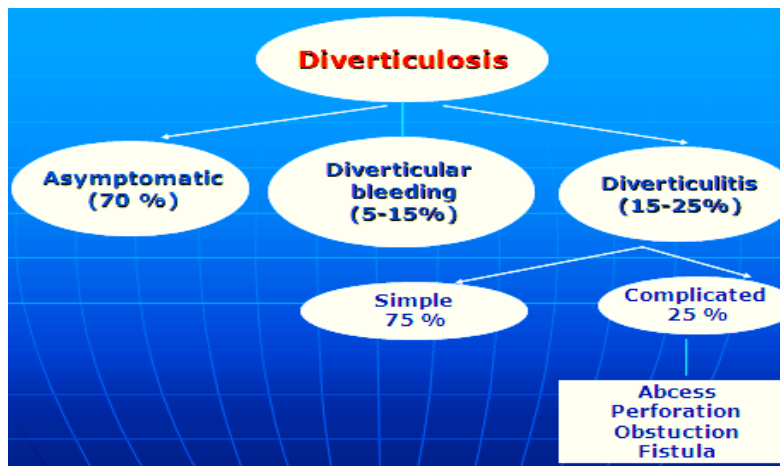


Figure 22. Natural history of diverticulosis. It may be asymptomatic (70%) or present with profuse bleeding (5-15%) or inflammation (diverticulitis often with peri-colitis) that may be complicated with abscess formation, perforation, obstruction or fistula formation.

Hinchey Classification of Diverticulitis (Table 2) (Figure 23)

Table 2: Hinchey classification (staging) of diverticulitis

Stage	Severity	Pain	Systemic	Investigation	Management
I	Peri-colic abscess or phlegmon	LIF	Possibly no change	Delayed barium enema, endoscopy	Bowel rest, IV antibiotic, DVT prophylaxis & fluids
II	Pelvic or intra-abdominal abscess	Severe, fullness - LIF	Mild toxic	CT scan	Percutaneous drainage
III	Non-feculent peritonitis	Peritonitis	Toxic	CT scan	Resuscitation + operation
IV	Feculent peritonitis	Peritonitis	Severe toxicity, shock	Proceed to operation	Resuscitation + immediate operation

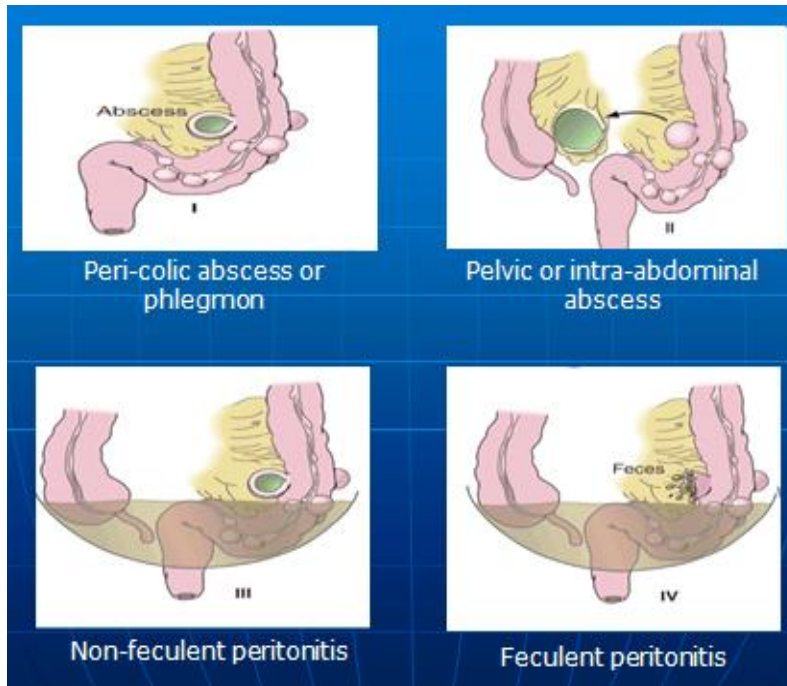


Figure 23. Classification of diverticulitis (Hinchey Classification System).

Diagnosis / Investigations

1. *In the acute phase*, computerized tomography (CT) scan (Figures 24 & 25) is particularly useful (high specificity) by identifying the following:
 - Bowel wall thickening,
 - Abscess formation (Figure 26)
 - Extra-luminal disease or other disease.
 - In addition, CT-guided percutaneous drainage may be done and postpone further operative procedures.



Figure 24. Multi-slice CT scan showing diverticula

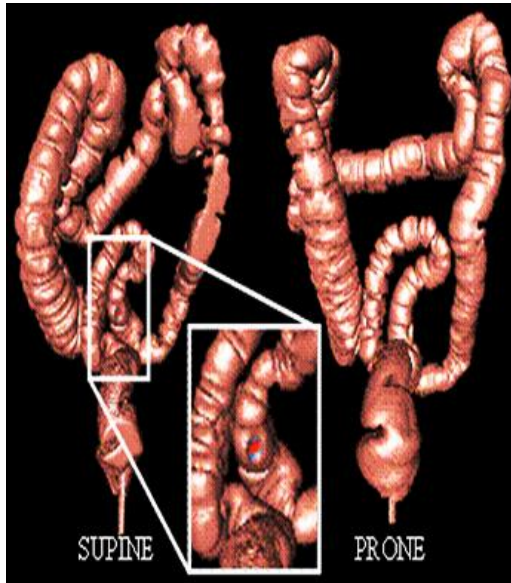


Figure 25. Multi-slice CT scan with virtual colonography showing diverticula

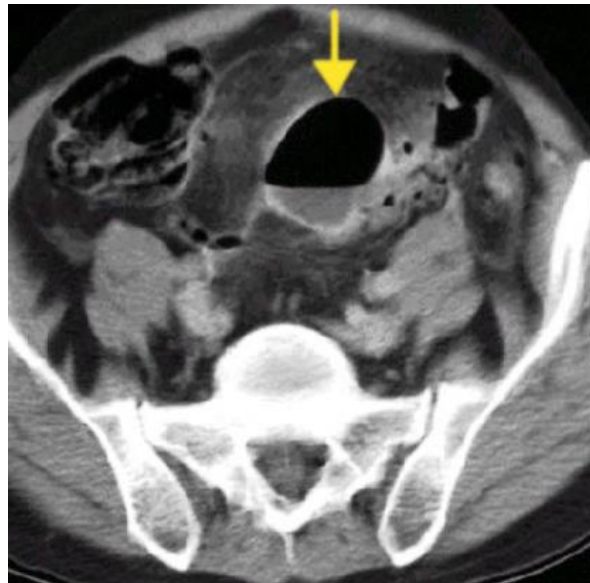


Figure 26. CT scan with showing diverticular abscess with fluid level (yellow arrow).

2. *In elective cases:*

- *Barium enema* (Figures 27-34) and *endoscopy* (sigmoidoscopy and colonoscopy) are the main methods of diagnosis. Barium enemas are usually reserved for patients who have recovered from an attack of acute diverticulitis for fear of causing perforation or peritonitis. Barium enema is done to exclude a carcinoma and to assess the extent of the disease.
- *Water-soluble contrast enemas* (Figure 35) are helpful in sorting out patients with large bowel obstruction. In the acute situation, it is good at detecting intra-luminal changes & leakage (Figure 36).

3. *In presence of inflammation or localized perforation:*

- *CT scan* is the *investigation of choice*.

4. *In presence of obstruction and free perforation:*

- *Plain X-ray abdomen* is diagnostic; also *US* is informative as regards fluid collections.

5. *In presence of fistula:*

- *Barium enema* is the best method + *colonic endoscopy* (Figures 37 and 38).
- In case of colo-vesical fistula, cystoscopy and biopsy should be added.

6. *In the presence of hemorrhage:*

- *Angiography* and *colonoscopy* are of value.

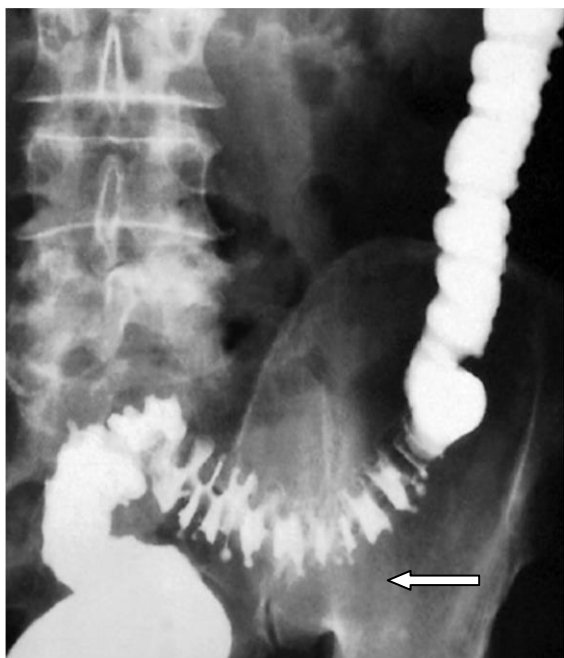


Figure 27. Barium enema showing sigmoid diverticular disease (*saw-teeth appearance*) white arrow).



Figure 28. Barium enema showing a large filling defect in the sigmoid colon caused by a peri-colic abscess (white arrow).



Figure 29. Barium enema showing diverticular disease involving the whole colon.

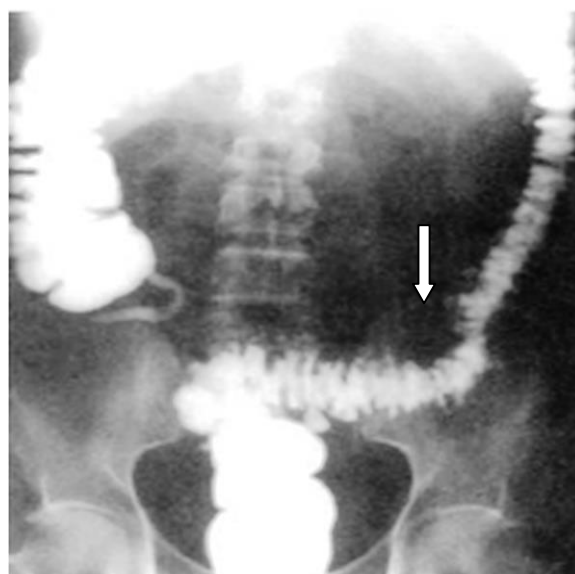


Figure 30. Barium enema showing area of spasm with a *zigzag appearance* in the sigmoid colon (white arrow)

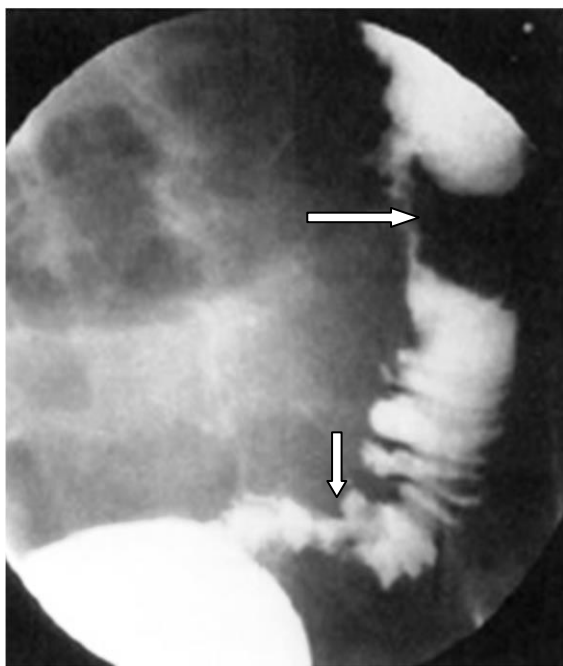


Figure 31. Barium enema showing 2 areas of narrowing (white arrows), one in the sigmoid colon & the other in the mid-descending colon, each representing a *stricture 2^{ty}* to diverticulitis.

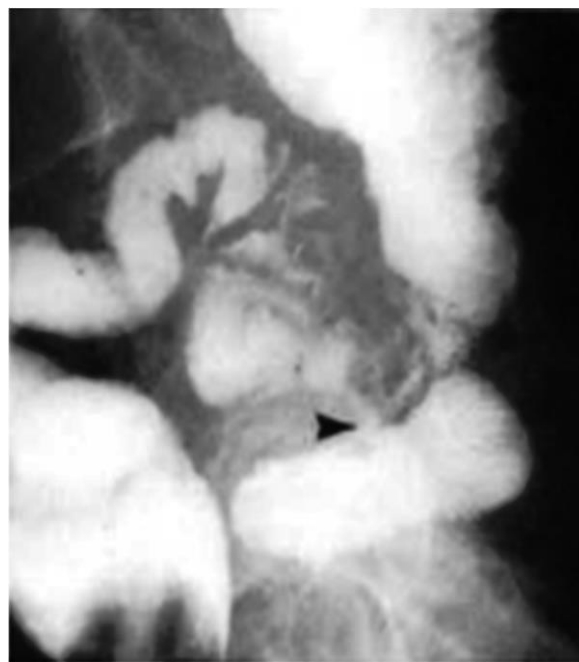


Figure 32. Barium enema showing a colovesical fistula denoted by the black arrow.



Figure 33. Barium enema showing colo-enteric fistula denoted by the black arrows.



Figure 34. Barium enema showing a walled-off peri-colic abscess with a colo-vaginal fistula.



Figure 35. Water-soluble contrast enema showing multiple diverticula of the sigmoid colon (white arrows).

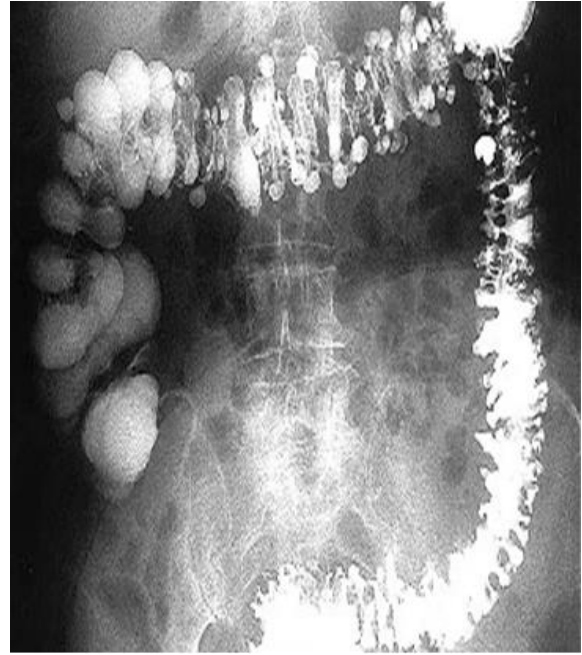


Figure 35. Water-soluble contrast enema showing leakage of contrast due to diverticular disease.



Figure 35. Colonoscopy showing the "mouth" of multiple diverticula.



Figure 36. Colonoscopy showing the "mouth" of multiple diverticula with inflammation.

Differentiation of Diverticulitis from Carcinoma of the Colon (Table 3)

Table 3: Differences between diverticulitis and carcinoma of the colon

Criteria	Diverticulitis	Carcinoma
- History	Long	Short
- Pain	More common	25% painless
- Mass	25% have tenderness over the inflammatory mass	Sausage shaped, felt in left iliac fossa, sometimes the above loaded colon is the only thing palpable
- Bleeding	17% often perfuse, periodic	65% usually small amounts persistently present
- Radiograph	Diffuse change, with the characteristic saw teeth and multiple diverticula appearance	Localized, shouldering, characteristic apple core appearance
- Sigmoidoscopy and/or colonoscopy	Inflammatory change over an area, with mouths of diverticulae seen scattered	No inflammatory signs, however, diverticula can be also present. Carcinoma is <i>biopsied</i>

Treatment of Diverticular Disease

1. Patients with Diverticular Pain

- Bulk laxative + antispasmodics
- Sigmoid colectomy if conservative treatment fails

2. Patients with Uncomplicated Diverticulitis

- Conservative treatment (bowel rest, IV fluids, bed rest, antibiotics for 4-5 days, pain control with Meperidine [better than Morphine]. After 3 weeks, endoscopy is performed to rule out malignancy, then 4-6 weeks later, elective resection with primary anastomosis is performed (if appropriate).
- Indications of elective surgery
 - a. ≥ 2 acute attacks, successfully treated medically
 - b. One attack requiring hospitalization in patient < 40 years old.
 - c. One complicated attack
 - d. One attack in an immuno-compromised patient
 - e. Inability to rule out colonic carcinoma
- Operative procedure
 - o The ideal operation carried out as an interval procedure after careful preparation of the gut is a one-stage resection.
 - o This involves removal of the affected segment and restoration of continuity by end-to-end anastomosis (Figure 37).

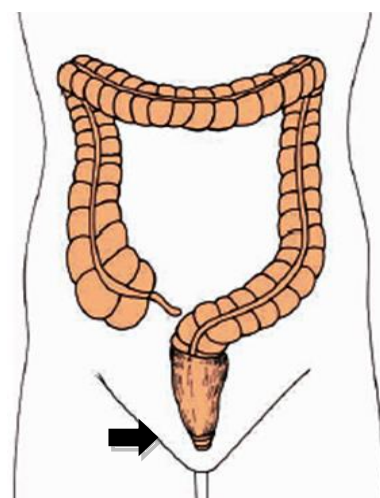


Figure 37. Primary resection and end-to-end anastomosis

3. Treatment of Complicated Diverticular Disease

Diverticulitis with perforation

- Stage 1 (contained peri-colic abscess) → resection + end-to-end anastomosis (EEA) ± protecting colostomy
- Stage 2 (walled-off pelvic abscess due to leaking peri-colic abscess) → *Hartmann* procedure (Figure 38)
- Stage 3 (generalized purulent peritonitis due to perforated pelvic abscess) → (1) *Hartmann* procedure, or (2) 1^{ry} resection and anastomosis after on-table lavage in selected cases (Figure 39), or (3) 1^{ry} resection and anastomosis after on-table lavage with covering ileostomy
- Stage 4 (generalized fecal peritonitis due to perforated diverticulum) → exploration and *emergency Hartmann* procedure

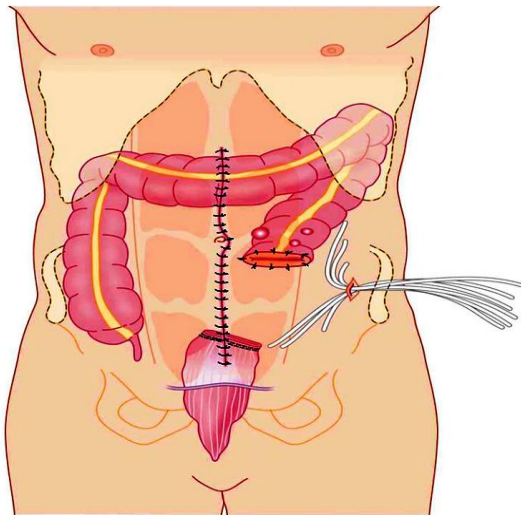


Figure 38. Hartmann procedure (end-colostomy in the left iliac fossa & closure of the distal rectal stump).

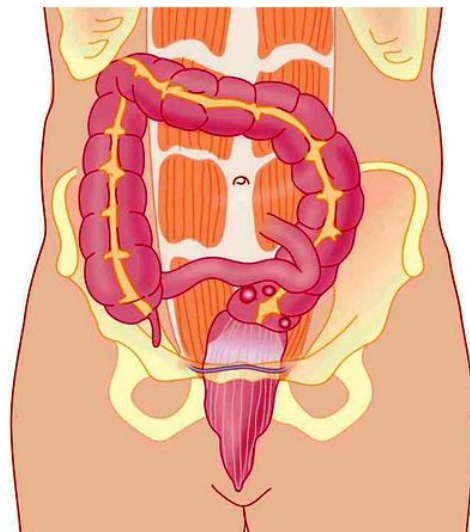
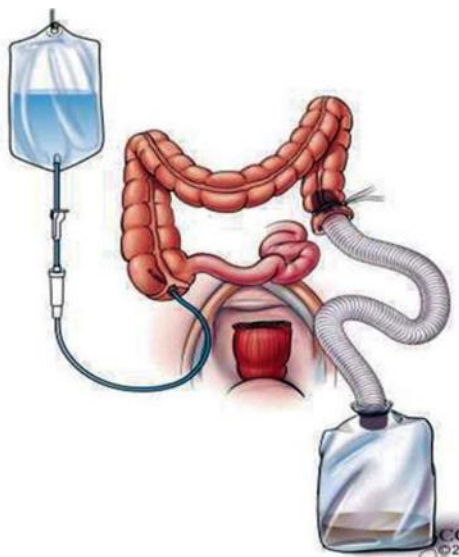


Figure 39. Resection and end-to-end anastomosis after on-table colonic lavage.

Obstruction

- If the obstruction is due to small bowel adhesions → *adhesolysis* and *resection* of the affected segment with EEA.
- In colonic obstruction in the presence of inflammatory edema and adhesions, or the bowel is loaded with feces → it is better to perform *Hartmann procedure* and *avoid* EEA. A *primary anastomosis with a covering proximal stoma* can also be used.

Fistulation

- Most patients are best treated by elective sigmoid resection with 1^{ry} anastomosis and repair of the fistulous opening in the affected organ.
- In the case of a *colo-vesical fistula*, it is usually possible to ‘pinch off’ the affected bowel from the bladder, close it & then resect the sigmoid (elective resection, 1^{ry} re-anastomosis + insertion of Foley catheter for 7-10 days).
- In very difficult cases, a staged procedure with a preliminary defunctioning stoma may be necessary.

Hemorrhage

- Fortunately the hemorrhage, although severe, is rare to cause shock; only profound anemia requiring transfusion is common. It usually responds to conservative treatment, occasionally it requires resection. *Emergent / elective resection, 1^{ry} re-anastomosis*
- Endoscopy can locate the area of hemorrhage (rectum, sigmoid or above the sigmoid) in a good % of cases. If the source cannot be located, then *subtotal colectomy and ileostomy* is the safest option

Diverticular disease with carcinoma

- Diverticular disease and carcinoma coexist in 12% of cases.
- *Exploration* may be necessary but, even then, differentiation may be difficult until histological investigations are available.

Laparoscopic Surgery

- In selected cases, laparoscopic surgery has been used for *sigmoid resection*.
- This has the benefit of decreased hospital stay and costs.
- However, there is little high-quality research in the field to advocate its true merits.



Figure 40. Diverticula seen during laparoscopy

POLYPS OF THE COLON AND RECTUM

Definition

A polyp is a swelling arising from colonic mucosa on a single pedicle producing a visible protrusion above the surface of the surrounding normal large bowel mucosa (Figure 41)

Classification (Tables 4 and 5)

Polyps can be classified according to different variables as follows:

- | | |
|---------------|------------------|
| - Congenital | - Acquired |
| - Neoplastic | - Non-Neoplastic |
| - Benign | - Malignant |
| - Symptomatic | - Asymptomatic |
| - Sessile | - Pedunculated |
| - Single | - Multiple |
| - A Disease | - Syndrome |

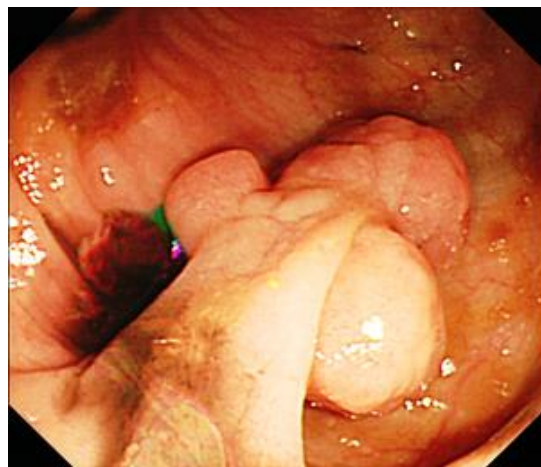


Figure 41. A polyp is merely a descriptive term of any lesion elevated above the mucosal surface of the large bowel.

A. BENIGN COLO-RECTAL POLYPS

Table 4: Histological classification of benign colo-rectal polyps

Polyps	Single	Multiple
<i>Inflammatory</i>	Benign lymphoid polyp	Pseudopolypi may be multiple in IBD, diverticular disease, Bilharziasis
<i>Hamartomatous</i>	- Juvenile polyp (70%) - Peutz-Jeher polyp	- Juvenile polyposis (30%) - Peutz-Jeher syndrome - Cronkhite-Canada syndrome - Cowden disease
<i>Hyperplastic or metaplastic</i>	Metaplastic polyp	Often multiple
<i>Neoplastic (epithelial)</i>	Adenoma (tubular, tubulo-villous or villous)	Familial adenomatous polyposis (FAP)
<i>Neoplastic (submucosal)</i>	Leiomyoma – Lipoma – Neuro-fibroma – Hemangioma	Often single

Table 5: Classification according to their malignant potential

<i>Polyps with NO Malignant Potential Non-neoplastic</i>	<i>Polyps with NO Malignant Potential Neoplastic (Adenoma)</i>
- Inflammatory - Hamartomatous - Hyperplastic - Lymphoid	- Tubular (0-25% villous tissue) - Tubulo-villous (25%-75% villous tissue) - Villous (75%-100% villous tissue).

Inflammatory Polyps (Pseudo-polyps)

- These lesions develop as by-products of the ulcers that penetrate into the submucosa, leaving islands of adjacent regenerative mucosa (Figure 42).
- They are associated with ulcerative colitis (UC), Crohn's disease, chronic dysentery, ischemic colitis and schistosomiasis. They can be associated with bilharzial colonic mass.
- They are multiple, concentrated in the sigmoid area, different in sizes and stalk length, bleed on touch, can be easily avulsed leaving a bleeding ulcer, and are associated with excessive mucus and blood-tinged mucus.
- Diagnosis rests on histopathology.

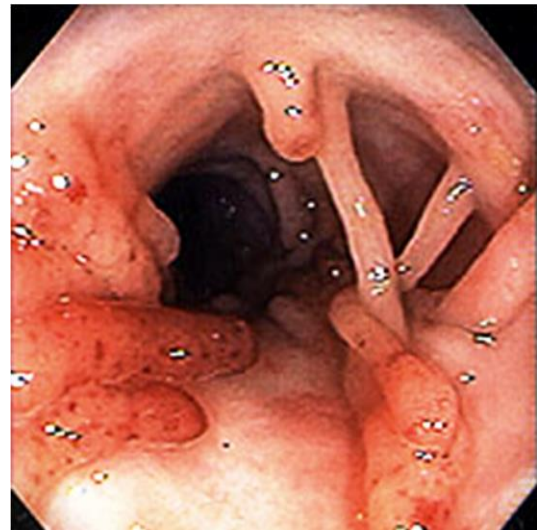


Figure 42. Inflammatory colonic polypi

Hamartomatous Polyps

Hamartomatous polyposis syndromes include (1) juvenile polyposis syndrome, (2) Peutz-Jeher syndrome and (3) Cronkhite-Canada syndrome

Juvenile Polyposis Syndrome

- *Juvenile polyps* occur in infants and children <10 y, usually single (70%) and often multiple, pedunculated (Figure 43), usually within the reach of the finger. They affect males > females, with familial tendency, present by bleeding/rectum and treated by excision through the endoscope.
- *Juvenile polyposis* means the presence of ≥ 10 juvenile polyps in the GIT. It is inherited as autosomal dominant (AD) disease due to mutation of SMAD4 gene on chromosome 18. Thus, they are "developmental malformations" affecting the glands and lamina propria, occurring commonly in the rectum of children <5 years. In adults, they are called "*retention polyps*". The risk of colon cancer is 50%. There is also $\uparrow$ risk of gastric, duodenal and pancreatic cancers

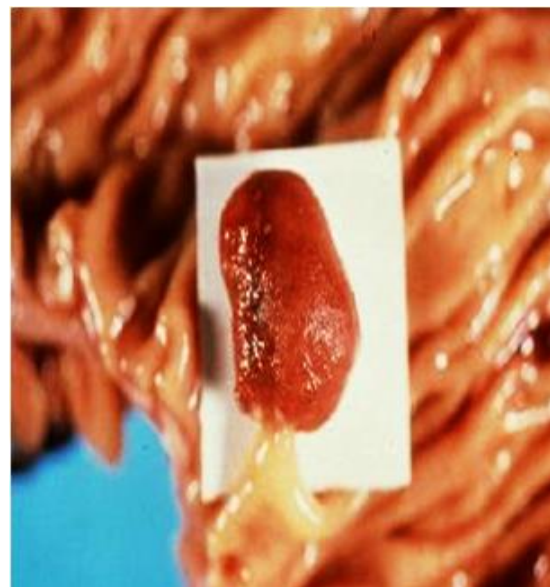


Figure 43. Juvenile colonic polypi (pedunculated)

Peutz-Jeher Syndrome

- It is autosomal dominant, due to mutations of the STK11 gene on chromosome 19. There is a very strong familial tendency
- Characterized by peri-oral and buccal mucosa *pigmentations* (Figure 44) and *hamartomatous polyps* throughout the GIT (Figure 45) mainly in the small bowel (78%), colo-rectal (70%) and stomach (40%).
- Both, gastro-intestinal (GI) and non-GI cancers are common. The GI cancers include colon, pancreas, stomach, small bowel and esophagus. Non-GIT cancers include breasts, ovaries, uterus and lungs.



Figure 44. Peri-oral melanosis



Figure 45. Melanosis & small polyp

Cronkhite-Canada Syndrome

- *Gastrointestinal (GI)* hamartomatous polyposis causes diarrhea, weight loss and abdominal pain
- *Extra-GI* manifestations include alopecia, cutaneous hyper-pigmentation and onychodystrophy
- *Treatment:* Nutritional support, corticosteroids, acid suppression and antibiotics
- The 5-year mortality rate (MR) may reach 55% with most deaths resulting from GI bleeding, sepsis and congestive heart failure (CHF).

Metaplastic (Hyperplastic) Polyps

- They constitute the majority of non-neoplastic polyps, with prevalence rates of 20-34% at autopsy and screening colonoscopic studies.
- Predominantly located in distal colon & rectum.
- Plaque-like protrusions, small (<0.5cm), with smooth glistening surface (Figure 46).
- After biopsy, they only require *observation*.

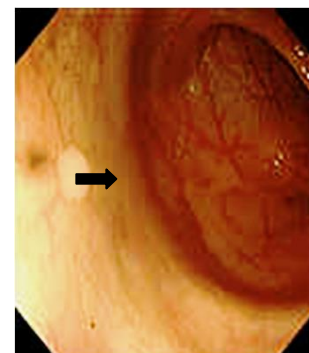


Figure 46. Hyper-plastic (meta-plastic) polyp (black arrow)

Lymphoid Polyps

- They present with mucosal nodularity (Figure 47) representing lymphoid hyperplasia (Figure 48)

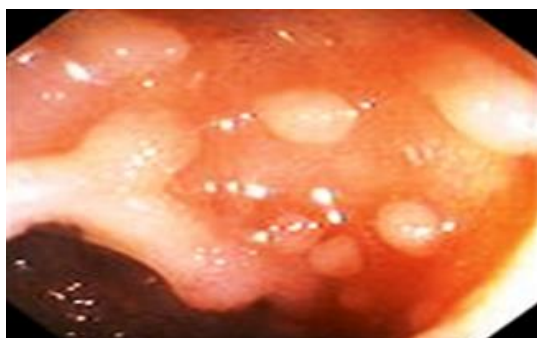


Figure 47. Lymphoid polyps

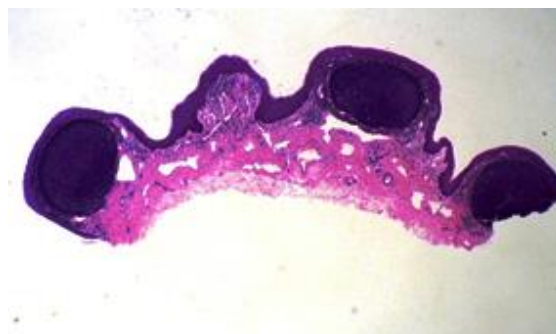


Figure 48. Lymphoid hyperplasia

Neoplastic / Epithelial (Adenomatous) Polyps (three main types)

1. Adenomas

Facts and Figures

- ↑ with age (50% by 70 years as compared to 33% of population by 50 years)
- 70 % of all colo-rectal polyps are adenomatous
- 70 % are located in the left colon
- 70 % are solitary (30% synchronous)
- 70 % are small (<1 cm in size)
- 7 % have severe dysplasia, 3-5% have invasive cancer

Adenoma-Carcinoma Sequence

Regardless of etiology, most colo-rectal cancers (CRC) arise from adenomas as shown in Figure 49.

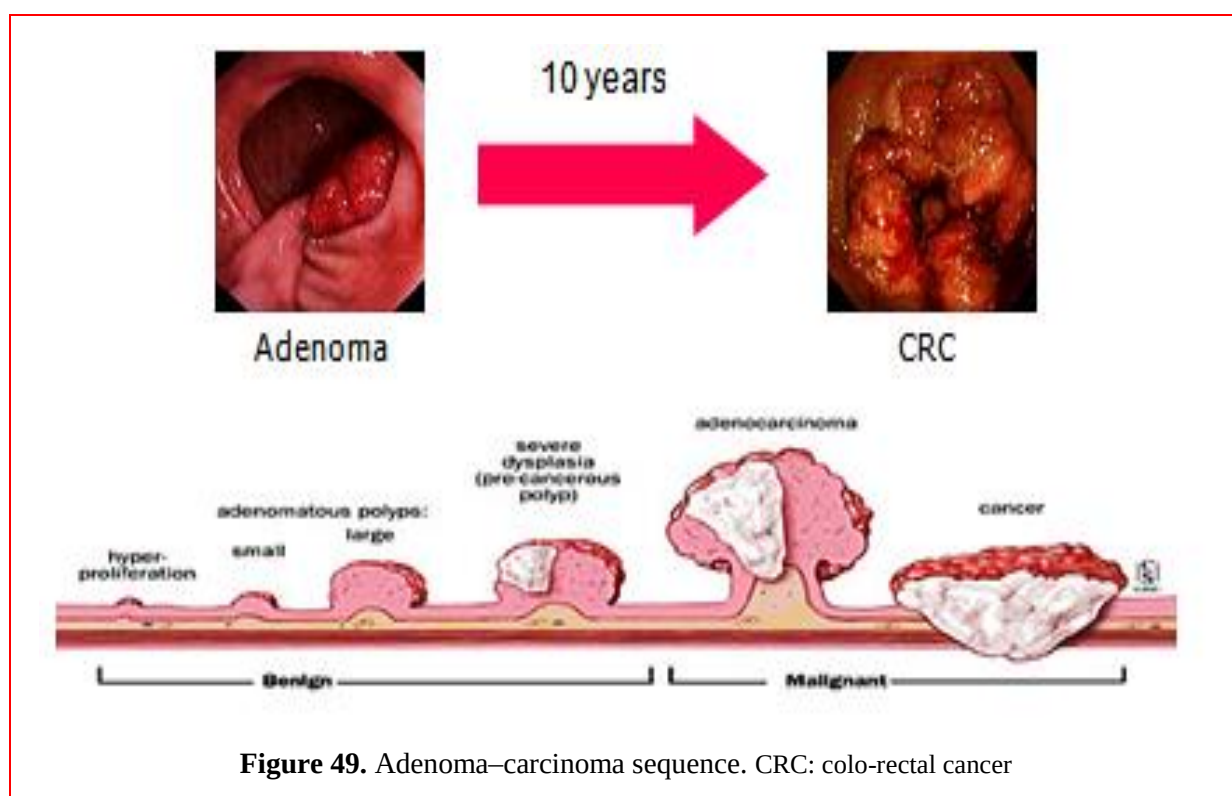


Figure 49. Adenoma–carcinoma sequence. CRC: colo-rectal cancer

Factors determining the risk of malignant transformation (Table 6)

Table 6: High- and low-risk factors of malignant transformation of adenomatous polyps

High Risk	Low Risk
- Large size (> 1.5 cm) - Sessile or flat - Severe dysplasia - Villous architecture (>25% villous component) - Polyposis syndrome (multiple polyps - >3 of any size)	- Small size (<1 cm) - Pedunculated - Mild dysplasia - Tubular architecture - Single polyp

2. *Tubular or Tubulo-villous adenoma*

- Vary in size from few mm to 7 cm.
- As they grow they attain a darker color and look more vascular and become pedunculated

3. *Villous Adenoma*

- Sessile, large 3-7 cm and expand in a carpet like fashion around the lumen and vertically.
- The surface is coarsely granular, there is always very much mucus and it bleeds on touch.
- It usually affects the rectum.
- They carry a high risk of colorectal cancer (CRC)
- It should be managed as carcinoma in-situ.

Differences between Tubular and Villous Polypi (Table 7)

Table 7: Characteristic differences between tubular and villous polypi

Tubular	Villous
>85%	<15%
Males > females	Females > males
Often multiple	Single, spread in a carpet-like fashion around the lumen of the bowel
70% are pedunculated	90% are sessile
Varies in size from 1-7 cm	Same
MP: consists of closely packed gland tubules with varying grades of differentiation	MP: covered with a single layer of colonic mucosa, which may show changes of carcinomatous degeneration
About 5% turn malignant	About 30% turn malignant

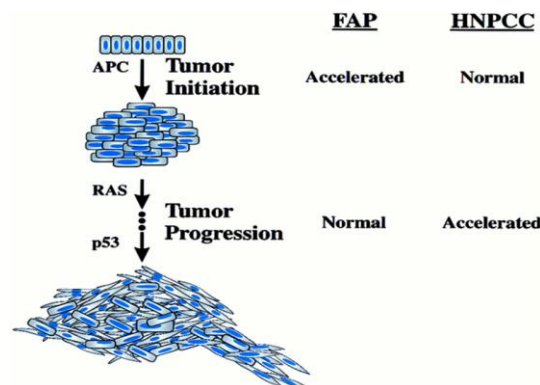
Familial Adenomatous Polyposis (FAP) (and its variants)

- They present 1% of CRC.
- Hundreds of different sized polypi (Figure 50).
- Familial inheritance as an autosomal dominant gene with $\approx 100\%$ penetrance.
- Patients develop adenomas by the mean age of 15 y (puberty), and develop CRC by the age of 40 y.
- It is a disease of *abnormal tumor initiation*, caused by mutations of APC gene (*tumor suppressor gene*) on chromosome 5q21 that encodes for a protein, which functions in cell adhesion and signal transduction. Mutations results in truncated protein & affects cell growth (Figure 51)



Figure 50. Familial adenomatous polyposis (FAP) coli. (>100 polyps)

Figure 51. Mechanism of carcinogenesis in FAP



- *Screening of FAP*
 1. Genetic screening of family members for APC mutations (Figure 52)
 2. Annual flexible sigmoidoscopy starting at 10-12 y until the age of 40 years, then every 3-5 years. *If polyposis is present, colectomy should be considered*
 3. Upper GIT Endoscopy every 1-3 y is also recommended to evaluate for upper GI adenomas
- *Diagnosis of FAP*
 1. Endoscopy / biopsy
 2. Genetic tests: Protein truncation test (will show mutation) - DNA sequencing

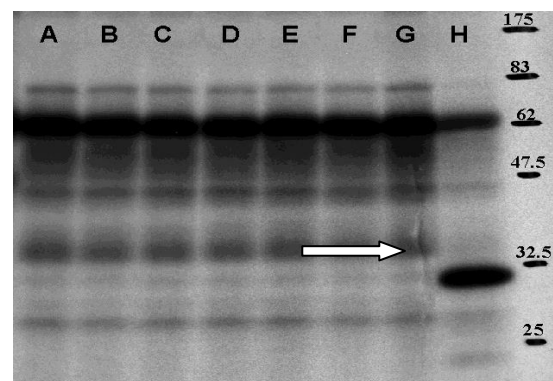


Figure 52. Protein truncation test showing mutation (arrow)

- *Extra-colonic manifestations of FAP*

1. Congenital hypertrophy of retinal pigmented epithelium (CHRPE) (Figure 53)
2. *Gardner's syndrome*: Desmoid tumors (Figure 54), osteomas (Figures 55 and 56) and multiple epidermoid (sebaceous) cysts and connective tissue tumors.
3. *Turcot syndrome*: CNS malignancies including medulloblastoma and glioblastoma
4. Duodenal, hepato-biliary-pancreatic and thyroid tumors

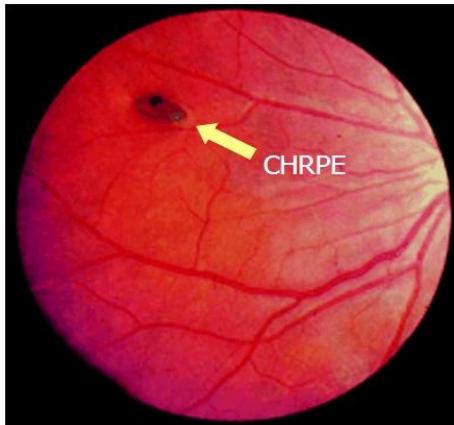


Figure 53. CHRPE in FAP (arrow)

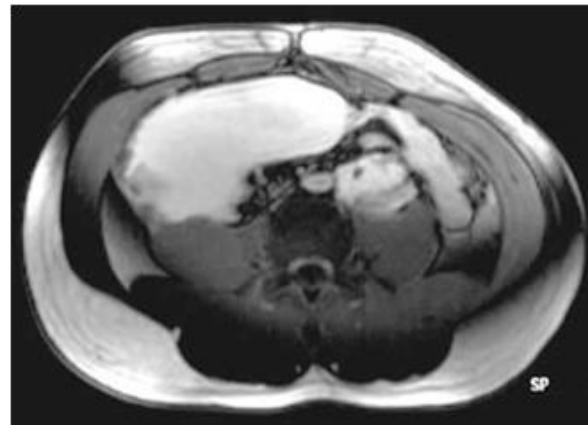


Figure 54. CT showing desmoid tumors in FAP

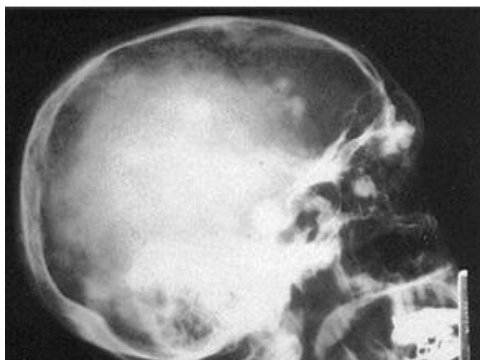


Figure 55. Plain X-Ray showing skull osteoma in FAP



Figure 56. Plain X-Ray showing mandibular osteoma in FAP

Attenuated FAP

- It is a variant of FAP
- *Number*: < 100 adenomas
- *Age*: Late age of onset: Adenomas at 44 years and CRC at 56 years
- *Location*: Proximal distribution of adenomas
- *Surveillance*: Colonoscopy is used for surveillance
- *Treatment*: The infrequent involvement of the rectum supports the role of total colectomy & ileo-rectal anastomosis (IRA).

Hereditary Non-Polyposis Colo-rectal Cancer (HNPCC)

- It represents 2-5% of all CRC
- Autosomal dominant with 70-80% penetrance
- It takes only 3-5 ys to develop CRC from adenomas

- *Molecular genetics of HNPCC (accelerated progression)* (Figure 11): Mutations of MMR genes → replication errors during DNA synthesis (*micro-satellite instability*) → *acceleration of genetic mutations*. HNPCC patients develop adenomas at the same rate as the general population. Once these adenomas develop, however, defective DNA repair ensues & mismatches accumulate. Thus, it takes only 3-5 y to develop CRC from adenomas.
- *HNPCC: Lynch Syndromes*
 1. Lynch syndrome I
 - Early onset of CRC (40-45 y).
 - Occurs predominantly proximal to the splenic flexure (60-70%).
 - Increased frequency of synchronous and metachronous lesions (33%)
 2. Lynch syndrome II
 - Same features of Lynch I syndrome + *extra-colonic malignancies* such as endometrial (most common, gastric, small bowel, hepato-biliary, ovarian, ureteral and renal tumors).
- *Screening for NHPCC*
 1. *Colonoscopy* every 2 years starting at ages 20-25 or years younger than the earliest diagnosis of CRC, whichever is earlier until 40y, and then annually
 2. Flexible sigmoidoscopy is *not* acceptable, due to the proximal location of tumors
 3. Trans-vaginal US and endometrial aspiration annually starting at 25-35 years
- *Diagnosis of NHPCC*
Due to lack of phenotypic markers like polyps, diagnosis is based on *family history* of CRC only
Amsterdam Criteria 1
 1. One member less than 50 years of age
 2. Two involved generations
 3. Three family members affected, one of whom is a first-degree relative of the other two*Amsterdam Criteria 2*
Same as Amsterdam 1, but includes all HNPCC-related tumors

Clinical Features and Diagnosis of Polyps

- Most polyps are *asymptomatic*; many of them can be diagnosed by *occult blood in stools*.
- *Bleeding per rectum* is the most common symptom (ranges from mild to severe)
- Bouts of *colonic colic* due to intussusception.
- *Rectal manifestations* (protrusion, tenesmus and excessive mucus discharge) (if in excessive amounts it may lead to hypokalemia and metabolic acidosis)
- Diagnosis is by either *barium enema* and *double contrast* or *endoscopy* (better for *biopsy*). There is a definite malignant potential to colorectal polyps (adenomatous).
- Predicting severity by endoscopy:
 1. Mild disease: < 5 rectal adenomas
 2. Moderate: 3-19 adenomas
 3. Severe: 20 or more adenomas.

Treatment of Polyps

Factors affecting Treatment of Colorectal Polyps

1. *Location*: Colon or rectum
2. *Number*: Single or multiple
3. *Morphology*: Pedunculated or sessile
4. *Histology*: Benign or malignant.

Excision (Polypectomy)

- Pedunculated polyps

Diathermy snaring or coagulation using the flexible sigmoidoscope or colonoscope with CO₂ insufflations, and insulated snares may safely remove polyps (Figure 57).



Figure 57. Endoscopic snaring of a small polyp

- Sessile polyps

If possible → *colonoscopic polypectomy*

- Larger polyps may require → piecemeal removal
- 5-8 polyps → snaring polypectomy
- > 8 polyps → removal of affected segment (*segmental colectomy*)

If endoscopic removable is not possible → operative removal

- Colon → colectomy
- Rectum → staged with EUS or MRI
 - a. Benign / early malignant (T1No) → Trans-anal local excision or TEMS (may need further radical surgery)
 - b. Other malignant polyps → radical excision (APR /anterior resection)

- *Complications of Polypectomy*

1. Risk of immediate bleeding & also delayed bleeding depending on vasculature of the stalk
2. Colonic perforation (uncommon, right colon > left due to thinner bowel wall).
3. Polyps may be missed due to incomplete examination, blind spots or poor preparation

Radical Surgery

- *Factors determining the need for radical surgery (unfavorable histology):*

1. Cancer within 2 mm of cut edge of stalk.
2. Poorly differentiated cancer.
3. Lympho-vascular invasion.
4. Unfavorable shape

- *Adenomatous polyps containing cancer* present a dilemma for the endoscopist. Should the patient undergo colectomy or not? Polyps with "unfavorable" histology (mentioned above) are an indication for *bowel resection* while those with "favorable histology" can be spared.

B. MALIGNANT COLO-RECTAL POLYPS

Definition

- A malignant polyp is one that contains invasive cancer. Malignant cells that have invaded through the mucularis mucosa *into* the submucosa (Figure 58)

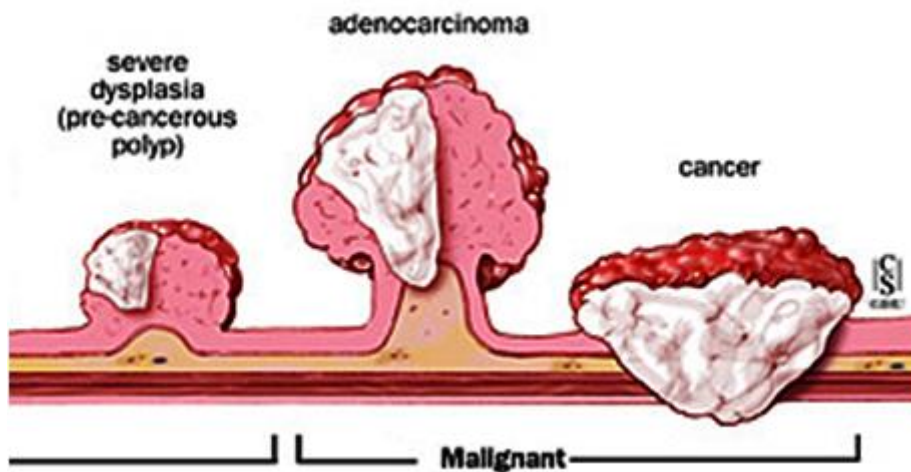


Figure 58. Malignant polyp with malignant cells reaching the submucosa.

Conventional Treatment of Malignant Polyps

- If an experienced colonoscopist is *not* available, *open colotomy* provides a way of surgical *polypectomy* (better to use a frozen section to proceed for *segmental colectomy* in case malignancy is shown)
- *Villous papilloma* (broad base, lower position and tendency to turn malignant or to harbor malignancy, 25% of which is often missed except with total excision biopsy)
 - a. For the usual *low lesion* and that of *moderate size* → total excision; *trans-anally* or *trans-sphincterically* (York Mason).
 - b. For high or extensive lesions → *low anterior resection* (low-stapled colo-anal anastomosis)
- *Familial Adenomatous polyposis (FAP)*
Timing of surgery

In FAP, timing of the operation depends on the clinical presentation as shown in Table 8

Table 8: clinical presentation and timing of surgery for treatment of FAP

Clinical Presentation	Timing of Surgery
- Asymptomatic patient with modest number of small adenomas	May wait for a few years, as long as colonoscopic surveillance is performed yearly
- Symptomatic patient with large number of adenomas	Early surgery
- Suspicious of CRC	Very early or urgent surgery

Surgical options

- Total procto-colectomy + ileal pouch-anal anastomosis (TPC-IPAA):

It is usually suitable for most patients with FAP (Figure 59).

- Total colectomy with ileo-rectal anastomosis (TC-IRA).
- Other surgical options include:
 - o TC-IRA for attenuated FAP (AFAP).
 - o Total procto-colectomy + terminal ileostomy (Figure 60): Indicated for low rectal cancers, poor sphincters and desmoid tumors

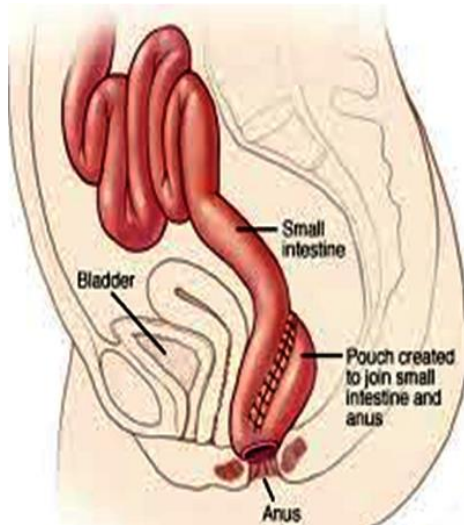


Figure 59. Total procto-colectomy (TPC) + ileal pouch anal anastomosis (IPAA)

Role of Medical treatment in FAP ?

- Sulindac (NSAID) and Celecoxib (COX-2 inhibitor) shown to control and decrease the number of colorectal adenomas in FAP
- It is *not* a definitive treatment, but a temporizing treatment (e.g. when surgery needs to be delayed)
- It may control pouch and rectal polyposis after initial prophylactic surgery

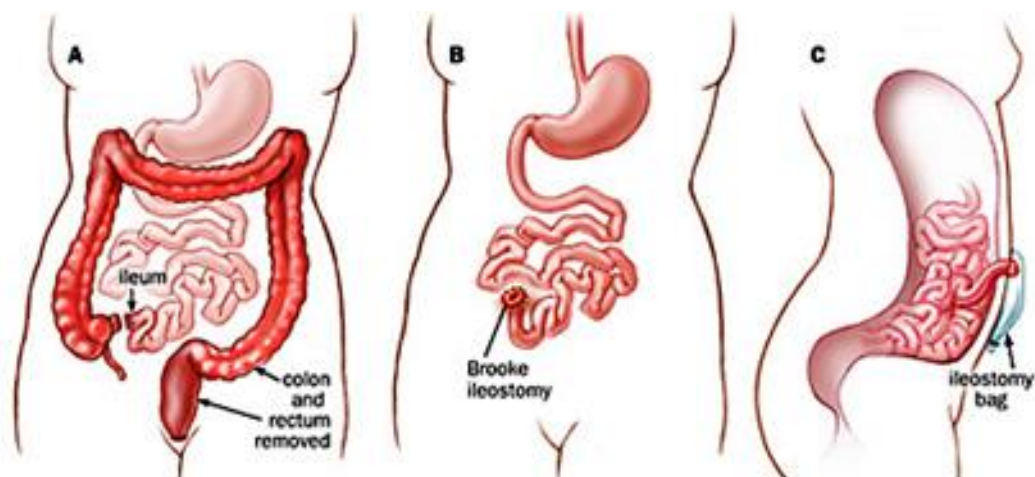


Figure 60. Total procto-colectomy (TPC) + terminal ileostomy

- Surgical Treatment of HNPCC

1. Total colectomy with ileo-rectal anastomosis (TC-IRA)
2. Restorative proctocolectomy with ileal pouch-anal anastomosis (TPC-IPAA)
3. Segmental colectomy *not* recommended because of high rate of metachronous CRC
4. TAHBSO (trans-abdominal hysterectomy-bilateral salpingo-oophorectomy) for endometrial cancer

Inflammatory Bowel Disease (IBD)

ULCERATIVE COLITIS (UC)

Definition

- Ulcerative colitis (UC) is a type of inflammatory bowel disease (IBD) that results in inflammation and ulcers of the colon and rectum.

Pathology

- The disease starts in rectum (90%) and spreads continuously proximally to involve:
 1. The rectum only (25%)
 2. The left side only (25%)
 3. The colon sub totally (15%)
 4. The colon totally (25%) with back-wash ileitis in 10-20% of cases
- It is predominantly a *mucosal* disease with the characteristic crypt abscesses and specific goblet cell mucus depletion. Chronic inflammatory cell infiltration and proliferative granulation tissue (GT) and edematous mucosa causing *Pseudo-polypi* are present in chronic cases.
- *Endoscopic picture (gross appearance)*
 1. Loss of normal vascular pattern
 2. Red, edematous mucosa with friable granular changes
 3. Contact bleeding on sigmoidoscopic or colonoscopic passage
 4. In chronic cases, pseudo-polypi (these are edematous mucosa between denuded undermined ulcerations), confluent ulcerations

Clinical Features

- Patients generally fall into following categories:
 1. Severe acute colitis
 2. Intermittent relapsing colitis
 3. Chronic persistent colitis
 4. Asymptomatic disease
- Approximately, 85% of patients follow a chronic relapsing course, 10% may follow chronic continuous course and 5% may suffer from single attack

Severity of the Attack

- Assessment of disease severity
 1. Mild = < 4 stools/day. Systemically well
 2. Moderate = > 4 stools/day. Systemically well
 3. Severe = > 6 stools/day. Systemically unwell (, fever, anemia, hypo-albuminemia)
 4. Toxic colitis (Cleveland criteria)
- Severe colitis + at least two of the following
 1. Heart rate >100/ m
 2. Temperature > 38.5
 3. Leukocytosis >10.500
 4. Hypo-albuminemia <3g/dl

Cardinal Features are "Intermittent Attacks of"

1. Rectal bleeding
2. Diarrhea
3. Mucous discharge
4. Abdominal pains
5. Weight loss
 - There is complete absence of symptoms between attacks in 60% of cases.
 - Relapses occur with variable frequency (5% of patients suffer single attack)
 - Symptoms correlate well with the severity of the disease (bleeding and diarrhea)
 - a. Mild: < or 4 motions/day
 - b. Moderate: > 4 motions /day without systemic signs of illness
 - c. Severe: > 6 motions with systemic signs (fever >37.5°C, tachycardia >90, low albumin <3g or loss of weight >3Kg).

Differential Diagnosis of Bloody Diarrhea

- Amebic colitis
- UC or other forms of IBD
- Spurious diarrhea of cancer rectum
- Rarely, bilharzial colitis

Investigations

Laboratory

- CBC (complete blood count), ESR (erythrocyte sedimentation rate), CRP (C-reactive protein)
- Liver function test (LFTs), mainly serum albumin
- Electrolytes

Endoscopy

- Colonoscopy with rectal biopsy (or sigmoidoscopy, as rectal involvement is present in > 95%, however, the extent of the disease should be determined by barium enema)

Radiology (barium enema is contraindicated in toxic mega colon)

- Plain X-ray for toxic mega colon (transverse colon >5.5 cm) and for perforations
- Barium enema and double-contrast (double wall sign, loss of haustrations, mucosal irregularities). *CT scan is not as useful in UC as in Crohn's disease (CD)*

Medical Treatment

5-Aminosalicylic Acid

- Used in mild / moderate UC & CD.
- 5-ASA block production of prostaglandins & leukotrienes
- Sulfasalazine was first agent described
- Topical preparation may be used in those with left-sided colonic disease
- Maintenance therapy of proven benefit in those with UC

Corticosteroids

- Often used in those in whom 5-ASA therapy is inadequate
- Also used in those presenting with acute severe disease
- Can be given orally, topically or parenterally
- Use should be limited to acute exacerbations of the disease
- Of *no* proven value as maintenance therapy in either UC or CD
- Use must be balanced against side effects

Immunosuppressive and Immune Modulators Agents

- Often used in those in whom steroids *cannot* be tapered or discontinued
- Agents used include:
 - o *Azathioprine*: Effective in both UC & CD.
 - o *Methotrexate*: Effective in CD
 - o *Cyclosporine*
 - o *Infliximab*: Anti-TNF- α therapy

Antibiotics as metronidazole

Surgical Treatment

The ideal Operation for UC would:

1. Remove the diseased bowel.
2. Return the patient to health.
3. Lower the risk of developing cancer.
4. Obviate the need for permanent ileostomy.
5. Preserve the anus for defecation.
6. Maintain continence.
7. Have few complications.
8. Be done in one stage.

For Toxic Megacolon

Anti-diarrheal medications, narcotics, hypokalemia, colonoscopy or barium enemata are prohibited

Indications of surgery

1. Failure of medical treatment (a trial of 3 or 5 days)
2. The presence or occurrence of complications like mega colon, hemorrhage or perforation

Procedure

- *Subtotal colectomy (STC) with ileostomy and Hartmann pouch*

The colon is divided at the distal sigmoid rather than the rectum and suture the stapled distal stump above the fascia and below the skin. In case it blows-up, a mucous fistula is created. The mobilization of distended friable colon is always associated with occasional perforation which, increase the morbidity and mortality of the operation

To maintain remission

- No role for steroids systemically
- Retention enemata (Budesonide *enemacort*) are used
- 5-ASA is the cornerstone of drugs (Olsalazine *Dipentum* is the best)
- Immunosuppressive drugs (Azathioprine or 6-Mercaptopurine) are steroid-sparing drugs and are occasionally used in UC contrary to CD where it is usually used.
- Cyclosporine may be used in severe cases that fail to respond to conventional treatment.

For Non-acute Cases

Indications of surgery

1. Severe disease not responding to medical treatment (*intractability*).
2. Chronic disease with anemia, frequent stools, urgency and tenesmus
3. Steroid-dependent disease (with failure of steroid sparing drugs)
4. Risk of neoplastic changes during review colonoscopy (severe dysplasia or villous adenomatous changes)

Operation Options

- The four current surgical options for treating ulcerative colitis are:
 1. Total (pan)-procto-colectomy & ileostomy (Brooke ileostomy) (TPC, ileostomy)
 2. Total procto-colectomy & continent ileostomy (Barnett or Kock pouch) (TPC, KP),
 3. Total procto-colectomy + ileal pouch + ileo-anal anastomosis (TPC, IPAA).
 4. Total abdominal colectomy with ileo-rectal anastomosis (TAC, IRA),
- A covering ileostomy *is a must in*:
 1. Patients receiving steroids at the time of surgery.
 2. Patients who are nutritionally compromised.
 3. When the vascularity or tension of the pouch is suspected.
- Anal transition zone (ATZ) is usually spared if *stapled anastomosis* is used. It gives better functional results (sampling is better). *Hand-sewn anastomosis* after mucosectomy is done if there is dysplasia, or in cases of FAP (familial adenomatous polyposis) to avoid cancer.

Total Procto-colectomy with Ileostomy (TPC, ileostomy)

- *Indications*: TPC with ileostomy remains the *benchmark procedure* for treatment of UC.
- *Advantages*
 1. The diseased colon is removed
 2. UC medications can be stopped
 3. Patients are able to return to normal health and activities
 4. The risk of cancer is obviated
 5. The procedure is done in one stage.

- *Disadvantages*
 1. The need for a permanent ileostomy and stoma appliance with its potential complications (skin irritation, intestinal obstruction, stoma retraction, or para-stomal herniation).
 2. Lack of control over stool evacuation
 3. The complications associated with pelvic dissection
 4. The potential for a perineal wound.

Total Procto-colectomy with Continent (Kock) Pouch (TPC, KP)

- *Indications:* Prior to 1983 "TPC, KP" was a popular alternative to TPC with Brooke ileostomy. Though several centers continue to promote its advantages as a treatment for UC, the popularity of KP has waned. Currently KP is limited to those patients who are:
 1. Not candidates for an IPAA
 2. As a modification for patients who currently have a Brooke ileostomy.
 3. As an alternative to Brooke ileostomy for those patients requiring ileal pouch excision
- *Advantages*
 1. All of the diseased bowel is removed
 2. Patients are off of medications
 3. Patients return to health and normal activities
 4. The risk of cancer is obviated
 5. Patients are continent and do not have to wear a stoma appliance.
- *Disadvantages*
 1. It is a 2- or 3-stage operation
 2. The risk of pouchitis (inflammation of the pouch)
 3. Complications of pelvic dissection
 4. Patients must catheterize the pouch to evacuate the contents
 5. A 50% reoperation rate is reported because of nipple valve slippage (continence is maintained by the creation of a nipple valve fashioned by intussuscepting a portion of the ileum into the pouch. Several techniques have been employed to prevent nipple valve slippage including the use of fascia, mesh and intestines in the Barnett modification. Nipple valve slippage results in either incontinence of the pouch or the more urgent problem an inability to catheterize and empty the pouch. Other valve specific complications include bleeding, ischemia, and valve necrosis).

Total Procto-colectomy with Ileal Pouch Anal Anastomosis (TPC, IPAA)

- *Advantages*
 1. All diseased bowel is removed
 2. Patients may stop UC medications
 3. Patients return to health and normal activity
 4. The cancer risk is obviated
 5. Stool is passed through the anus
 6. Continence is preserved.

- *Disadvantages*
 1. It requires 2 or 3 stages
 2. Complications associated with pelvic dissection
 3. Patients have a mean stool frequency of 6/day one of which is nocturnal
 4. Incontinence rates vary from 5-20%
 5. Many patients must take anti-diarrheal preparation
 6. Pouchitis.
- *Results:* Patients with an IPAA report the highest levels of satisfaction compared to patients with either a Brooke or continent ileostomy. However, complications such as stool frequency, urgency & incontinence can detract from the quality of life.
- *Pouch Design*
 Stool frequency is affected by stool volume, pouch compliance, pouch capacity, & the patient's response to distention. As shown in Figure 1, Pouches may be constructed from either 2 limbs ("J" pouch), 3 limbs ("S" pouch), or 4 limbs ("W" pouch). Though J-pouches are the easiest to construct, they have the lowest capacitance & compliance. S-pouches have a longer reach into the pelvis than J-pouches, but the efferent limb must be < 2 cm in length to prevent outlet obstruction. Anal transition zone (ATZ) is usually spared if *stapled anastomosis* is used (Figure 2). W-pouches have a greater capacitance and compliance than either J- or S-pouches, but this does *not* translate into a functional advantage.

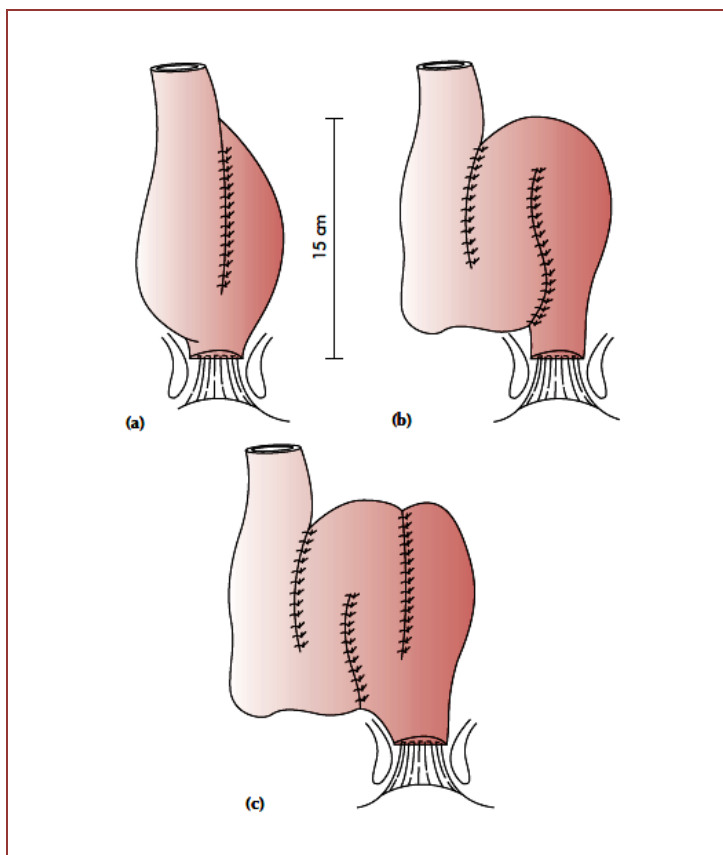


Figure 1. Ileo-anal anastomosis with pouch. A substitute rectum is made from joined folds of ileum to form an expanded pouch of small intestine. The pouch is then joined directly to the anus at the level of the dentate line, all other rectal mucosa having been removed. Three ways of forming a pouch are illustrated: (a) a simple reversed 'J'; (b) an 'S' pouch; (c) a 'W' pouch.

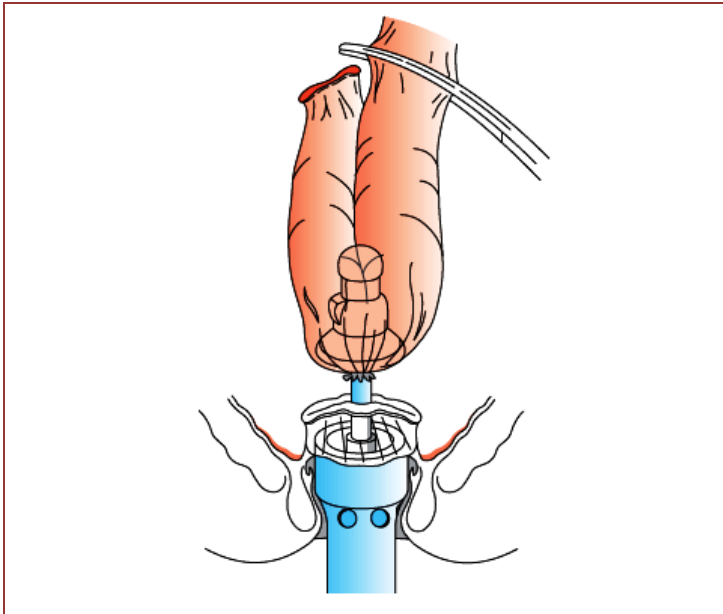


Figure 2. Stapled 'J' pouch with stapler creating a pouch–anus anastomosis.

- *Diverting Ileostomy*

Diverting loop ileostomy has been an integral part of TPC, IPAA. It protects the anastomosis and reduces pelvic sepsis. However, closure of an ileostomy is *not* without complications + added cost of a 2nd surgery and hospitalization. TPC, IPAA can be performed without an ileostomy, particularly in patients in good nutritional condition and not on steroids or 6-MP.

- *Pouchitis and Other complications*

Pouchitis is the most common complication and patients present with increased stool frequency, urgency, incontinence, cramping abdominal pain and a flu-like generalized malaise. On *pouchoscopy* the mucosa is inflamed and may be ulcerated. The fast response to Metronidazole supports the role of anaerobic bacteria overgrowth as a cause or significant factor in the etiology. Fortunately few patients' symptoms are severe enough to require pouch excision. Other conditions to be considered include:

1. Idiopathic pouchitis
2. Anastomotic stricture with bacterial overgrowth
3. Pouch anal abscess or fistula
4. Ischemia
5. Cytomegalovirus (CMV) infection
6. Recurrent UC in retained rectal mucosa
7. Crohn's disease (CD)
8. Irritable bowel syndrome (IBS).

Total Abdominal Colectomy with Ileal Rectal Anastomosis (TAC, IRA),

This procedure removes the abdominal colon, but leaves the rectum intact.

- *Indications:* It was an alternative to TPC and ileostomy for younger patients in an effort to delay an ileostomy until a more acceptable time. Now, it is used in those patients who:

1. Have rectal sparing, or are not suitable candidates of IPAA
2. Have a picture suggestive of indeterminate colitis.

- *Advantages*
 1. Improvement in health
 2. No pelvic dissection
 3. It is done in one stage
 4. Patients have about 3 stools/day
 5. Continence is preserved
 6. Patients do not have a stoma.
- *Disadvantages*
 1. All diseased bowel is *not* removed
 2. Risk of cancer is reduced, but *not* eliminated (after 25 years, 13% of patients with IRA for UC developed cancer in the rectal remnant).

Complications of UC

1. Toxic Mega-Colon or Acute Dilatation (10%)

- Usually occurs in the first attack of total colitis
- Predisposed to buy drugs inhibiting motility and hypokalemia, barium enema and colonoscopy.
- The condition is heralded by general condition deterioration and diagnosed by plain X ray
- Because of high risk of perforation, treatment is by surgery after 5 days trial of medical treatment.

2. Colonic Perforation (2%)

- Can occur $\pm$ dilatation due to use of barium enema or colonoscopy with aggression

3. Massive Hemorrhage (3%)

- Responds to conservative treatment and transfusions

4. Colonic Stricture (10%)

- Benign strictures occur commonly in the rectum and transverse colon (usually with total colitis)
- It is an absolute indication of colonoscopy and biopsy if seen be enema

5. Colonic Dysplasia and Carcinoma (5%)

- The risk depends on (1) extent of disease, (2) duration (>10 years) and (3) onset during childhood
- The majority lies in recto-sigmoid and transverse colon
- Flat plaques of dysplasia (high grade) are treated as carcinomata

6. Peri-anal Suppurative Diseases (15%)

- Apart from hemorrhoids (frequently present) suppuration and fissure are more common in Crohn's disease (CD)

CROHN'S DISEASE (CD)

Definition

Crohn's disease is a type of inflammatory bowel disease (IBD) that may affect any segment of the gastrointestinal tract (GIT) from the mouth to the anus.

Pathology

- Common sites
 1. Ileocolic 40%
 2. Colon 25%
 3. Small intestine 20%
 4. Ano-rectal (alone) 5%
- It is a chronic *trans-mural* inflammatory disorder affecting all GIT with *segmental* distribution and *skip lesions*
- The pathology is characterized by trans-mural infiltration by chronic inflammatory cells, edema, non-caseating epithelioid cell granulomas and angitis. The affected segment is extremely thickened and has a woody feeling (hose-pipe lesions). Mesenteric lymph nodes (LNs) are enlarged and mesenteric fat wraps the gut
- Anal lesions (anal CD) are common and are manifested as dusky edematous skin tags, indolent ulcers and deep fissures, recurrent perianal suppuration, fistulae or sinuses)
- There are 2 forms of CD (present clinically): perforating & non-perforating.
- *Endoscopic picture*
 1. Earliest lesion is pallor of mucosa + discrete aphthoid (aphthous-like) ulcers
 2. Later cobblestoning, mucosal thickening and linear ulcerations
 3. Deep fissures may penetrate to adjacent structures causing abscesses and fistulae
- *Comparison of pathological features between UC and CD* (Table 1)

Table 1: Comparison of pathological features between UC and CD

Ulcerative Colitis (UC)	Crohn's Disease (CD)
Lesions continuous - superficial	Lesions patchy – penetrating
Rectum always involved	Rectum normal in 50%
Terminal ileum involved in 10%	Terminal ileum involved in 30%
Granulated ulcerated mucosa	Discretely ulcerated mucosa
No fissuring	Cobblestone appearance with fissuring
Normal serosa	Serositis: common
Muscular shortening of colon	Fibrous shortening
Fibrous strictures: rare	Strictures: common
Fistulae: rare	Entero-cutaneous or intestinal fistulae in 10%
Anal lesions in <20%	Anal lesions in 75%. Anal fistulae and chronic fissures.
Malignant change well recognized	Possible malignant change

Degrees of CD

Mild-to-Moderate

- Ambulatory patient
- Tolerating oral feeding
- No manifestations of
 1. Toxicity (high fever, rigors, prostration)
 2. Abdominal tenderness and painful mass
 3. Obstruction
 4. >10% weight loss

Moderate-to-Severe

- Failed medical treatment of mild (5-ASA, antibiotics "Quinolones or Metronidazole" and Budesonide)
- Patients with more prominent symptoms of:
 1. Fever (mild fever)
 2. Abdominal pains and tenderness without a mass
 3. Intermittent nausea and vomiting without frank obstruction
 4. Weight loss and/or significant anemia

Severe to Fulminate (should be hospitalized)

- Symptoms despite steroid or Infleximab therapy
- Patient presenting with
 1. High fever, rigors or prostration
 2. Persistent vomiting with evidence of obstruction
 3. Evidence of abdominal mass or rebound tenderness
 4. Significant weight loss or cachexia

CD in Remission (whether induced medically or surgically)

- Patients asymptomatic or without inflammatory sequelae or evidence of residual disease
- Patients who need *steroids* for well-being are *not* in remission

Clinical Picture

- *Systemic* features in CD are > in UC. The presentation of CD can vary widely according to the *age* of patient, *location* (50% ileo-cecal and 25% present with colitis) and *behavior* of the disease (Vienna classification **A L B** system) as shown in Table 2).

Table 2: Vienna classification **A L B** system

Age	Location	Behavior
A1 <40 y A2 >40 y	L1: Terminal ileum L2: Colon L3: Ileo-colic	B1: Non-stenosing, Non-penetrating B2: Stricturing B3: Penetrating - other features of CD.

Intestinal Manifestations

The following are the most common

1. *Chronic regional enteritis*: The lesion is usually in the ileo-cecal area and causes: (a) chronic ill-health, (b) abdominal pains due to obstructive attacks, (c) anemia, (d) change of bowel habits, attacks of obstruction and others of diarrhea, and (e) a mass may be palpable
2. *Acute regional ileitis*: simulates acute appendicitis
3. *Enteric fistula*: It is common to occur if deep fissures penetrate through the gut wall-giving rise to entero-enteric, or entero-colic, or entero-cutaneous or entero-vesical fistulae
4. *Peri-anal CD*: Edematous cyanotic skin tags (elephant ears), eccentric multiple fissures, painful cavitating ulcers, complex anal fistulae, recurrent perianal suppurations and strictures
5. *Acute intestinal obstruction*: There is always a history of recurrent similar attacks, which recover with or without treatment.
6. *Colonic CD*: Bleeding and mucus per rectum, diarrhea, abdominal pains and weight loss.

Extra-Intestinal Manifestations

Extra intestinal manifestations are however, more common than UC; they may be associated with disease activity or not (Table 3)

Table 3: Extra-Intestinal manifestations of Crohn's disease

Associated with Disease Activity	Not Associated with Disease Activity
- <i>Skin</i> - o Erythema nodosum - o Pyoderma gangrenosum - <i>Joints</i> - o Asymmetrical non-deforming arthropathy - <i>Eyes</i> - o Anterior uveitis - o Episcleritis - o Conjunctivitis - <i>Hepatobiliary conditions</i> - o Acute fatty liver - <i>Thrombo-embolic disease</i>	- <i>Joints</i> - o Sacroilitis - o Ankylosing spondylitis - <i>Hepatobiliary conditions</i> - o Primary sclerosing cholangitis (PSC) - o Cholangiocarcinoma - o Chronic active hepatitis - o Gallstones: due to interrupted entero-hepatic circulation at the terminal ileum - <i>Amyloid disease</i> - <i>Nephrolithiasis</i>: Obstructive uropathy due to retro-peritoneal infiltration and ↑ renal calculi specially Ca oxalate due to ↑ oxalate absorption - ↑ incidence of cancer colon and small bowel cancer

Investigations

Laboratory (same as UC)

Endoscopy (same as UC with some differences)

- Rectum is spared in 50% hence colonoscopy is preferred to sigmoidoscopy & barium enemata

- The presence of skip lesions and terminal ileal involvement mandates thorough complete colonoscopy and multiple biopsies
- Colonoscopy is done with caution in case of the presence of deep fissures.

Radiology

- Barium enema, small bowel enema (superior to follow-through), CT and sinography are all used
- The useful diagnostic signs in barium studies are edema with separation of the affected loops, *string sign of Kantor*, spike or thorn signs and fistulae with skip lesions.
- In CT scan, abscesses and fistulae with the presence of inflammatory masses and bowel obstruction are the main features

Treatment

General Rules

- Neither medical nor surgical treatment is curative
- Medical treatment is employed initially and continued until the medications fail to alleviate symptoms or produce unacceptable side effects or a complication arises that requires surgery
- Objectives of surgery are to:
 1. Deal with the immediate adverse effects of the disease,
 2. Restore health and function
 3. Permit withdrawal of steroids or immunosuppressive agents

Induction of Remission

Mild-to-moderate CD (ileal, ileo-colic and colonic)

- Oral 5 ASA: Mesalazine (3-4 g/d) - Sulfasalazine (3-6g/d)
- Alternative treatment (*antibiotics are specially used in peri-anal CD*): Metronidazole (10-20 mg/kg/d) - Ciprofloxacin (1 g/d) - Controlled ileal release: Budesonide (3-9 mg /d)

Moderate-to-severe CD

- Prednisone 40-60 mg /d **or**
- Budesonide (9 mg/d) until symptom resolution or weight gain resumption (*usually 7-28 d*)
- Appropriate antibiotics or drainage for inflammation or abscess
- Infleximab: Effective adjuvant treatment - possible alternative to steroids when not tolerated or ineffective and better to use in presence of fistulae

Severe-to-fulminate CD

- Hospitalization
- CT and/or US ±PAD (percutaneous abscess drainage) and appropriate antibiotics
- Surgical consultation for drainage in abscess formation.
- If abscess is excluded: Parental steroids equivalent to 40-60 mg prednisone and TPN

Peri-anal CD

- Acute suppurating drainage with or without a Seton
- Non-suppurating fistula or fissures: (1) Antibiotics (Ciprofloxacin and/or Metronidazole), (2) Immuno-modulators: AZA/6-MP, (3) Infleximab.

CD in Remission

- Never use steroids
- The use of 5-ASA is following induction with 5-ASA or post-operative, *not* effective after steroid induction (in general it is of unproven benefit as maintenance of remission).
- Immunotherapy with AZA/6-MP after steroid induction in steroid-dependent disease is the usual way of maintaining a remission.
- The use of *antibiotics* to maintain remission is in peri-anal disease and post-operatively.
- *Infliximab* is used also to maintain remission
- 5-ASA or AZA/6-MP are used after ileo-colonic resection to reduce the risk of symptomatic recurrence

Nutritional Support

- Defective absorption and fear of food are among causes of malnutrition and anemia, and sometimes electrolyte imbalance that should be corrected.
- TPN indications
 1. Intestinal failure with gross nutritional deficiencies and severe hypo-albuminemia
 2. Mal-nourished patients prior to surgery
 3. In obtaining remissions in severe active cases
 4. In patients with entero-cutaneous fistulae (30% heal, 6 weeks is the time limit)
 5. As long-term TPN in short bowel syndrome (SBS)

Surgical Treatment of Crohn's Disease

Indications of Surgery (Table 4)

Table 4: Absolute and relative indications of surgery for treatment of Crohn's disease

Absolute Indications	Elective Indications
1. Perforation with generalized peritonitis 2. Massive hemorrhage 3. Carcinoma 4. Fulminant or non-responsive acute severe colitis	1. Chronic obstructive symptoms 2. Chronic ill-health or debilitating diarrhea 3. Intra-abdominal abscess or fistula 4. Complications of peri-anal disease 5. Growth retardation in children or amyloidosis 6. Failure of medical treatment with persistent symptoms 7. Complications of medical treatment (side effects of steroids or AZA), with inability to stop treatment 8. Surgery should be as conservative as possible 9. No evidence that increased resection margins reduces risk of recurrence 10. If possible, improve pre-operative nutritional state

Preparation of the Patient for Surgery

- *Patient preparation is similar no matter the location of the disease.*
- *Pre-operative counseling* regarding the nature and course of disease with discussion about the rationalization of surgery, hospitalization, peri-operative morbidity including the possibility of ileostomy and future implications of recurrence is important.
- *Comprehensive assessment* with endoscopic and radiological evaluation of the small and large bowel is important to define the road map of the diseased areas and the operative strategy.
- *Correction of restorable metabolic defects* (anemia, electrolyte deficiencies, dehydration, and coagulation defects) is important. Pre-operative optimization of the nutritional status with TPN is controversial. In our practice, we use pre-operative hyper-alimentation for 5 to 7 days, if surgery is elective or semi-elective and malnutrition is considered severe.
- *Cessation of immuno-suppressive drugs* (AZA, 6-MP and Cyclosporine) 3-4 weeks prior to elective procedure is advised because of concerns about tissue or anastomotic healing.
- *Stress-dose steroids* are administered peri-operatively if the patient has been treated with steroids within 6 m prior to operation.
- *Additional preparation* includes *IV antibiotics* with post-operative antibiotic selection based on operative findings, likely flora, pending cultures and sensitivity. After anesthesia, a *nasogastric* tube is passed and *Foley catheter* is placed in the bladder. Central venous lines & peripheral arterial lines may be placed on the likely difficulty and duration of the operation. *Ureteric stents* are *not* used routinely but can be helpful with re-operative pelvic surgery. Finally, *prophylaxis against DVT* is employed using pneumatic stockings with or without mini-dose heparin.

Surgical Strategy

- *Conservation in surgical resection:* Small bowel conservation in CD is imperative to minimize the risk of excessive bowel resection, which can cause disability from malabsorption, including short bowel syndrome (SBS). Thus, segments of small bowel affected by non-obstructing, non-hemorrhaging CD do *not* warrant resection.
- *Resected margins:* Important to the concept of conservation in surgery is the issue of resected margins. Extended resection margins (12 cm) are unnecessary and have no advantage over limited (2-cm) margins in reducing cumulative recurrence rates of the disease.
- *Regarding re-operation:*
 - Surgery is deferred for any elective indication until at least 3m, preferably 6m, since the last laparotomy. Otherwise, formidable adhesions may be encountered resulting in increased risks of inadvertent enterotomy.
 - A small window of time (7-12 d) exists following laparotomy during which reoperation for complications such as entero-cutaneous fistula can be attempted with reasonable success.
 - Reoperation is justifiable for complications of intestinal ischemia, peritonitis & major hemorrhage between 12-90 d after the initial laparotomy.

- For patients with post-operative ileus/bowel obstruction, it is safer to provide long-term TPN and a percutaneous gastrostomy than to risk enteric injury, fistulas or substantial loss of intestinal length in a well-intentioned, but dangerous venture. If an enterocutaneous fistula complicated with intra-abdominal sepsis or pouching and skin excoriation problems occur during the high-risk period, a small laparotomy in the left upper quadrant with a proximal high-loop jejunostomy isolates the septic source. Home TPN is continued for 4-6 m to maintain nutrition, pending restoration to correct the fistula & restore intestine continuity.
- *The use of temporary ostomy* may be indicated in situations where anastomotic healing is compromised such as excessive blood loss during a long operation, severe hypoalbuminemia, incompletely drained sepsis, or when a phlegmonous pyogenic membrane is in proximity to an anastomosis. These are not necessarily absolute indications or even the only indications for such diversion, but in the examples mentioned, temporary ostomies appear reasonable.

Surgical Procedures

- Several types of operations are used in the surgery of CD. These may be classified as:
 1. Resection with or without anastomosis.
 2. Bypass procedures (internal and external such as gastro-duodenal bypass or ileostomy).
 3. Strictureplasty.
- The surgical procedure depends on the site of the disease.

Ileo-colic Disease (40% of patients undergoing surgery)

- *Resection of the diseased bowel with ileo-ascending anastomosis* is the preferred procedure. As much as normal right colon as possible is preserved for increased water absorption and avoiding anastomosis sitting over the duodenum with the risk of fistula development. *Wrapping* the anastomosis with omentum may avoid this problem.
- If disease is associated with perforation and peritonitis, creation of diverting ileostomy proximal to the ileo-colic anastomosis decreases the morbidity associated with the next procedure.

Jejuno-ileal Disease (Small Bowel) (10-20% of patients)

- *Surgical resection*
 - The most common surgical procedure is *resection* with end-to-end anastomosis (EEA), keeping as much of the bowel as possible. The leakage rate is > 15%
- *Strictureplasty*
 - Since CD is a pan-intestinal disease, surgical treatment has shifted toward conservative approaches. Strictureplasty has become a valuable option for treatment of small bowel CD.
 - *Types of strictureplasty: Heinecke-Mikulicz* strictureplasty (most commonly used), is suitable for short strictures up to 10 cm. *Finney* strictureplasty is used for longer strictures.

- *Indications of stricturoplasty*
 1. Diffuse involvement of the small bowel with multiple strictures
 2. Stricture(s) in a patient with previous major resection(s) of small bowel (>100 cm)
 3. Rapid recurrence of CD manifested as obstruction.
 4. Stricture in a patient with short bowel syndrome (SBS)
 5. Non-phlegmonous fibrotic stricture
- *Contraindications of stricturoplasty*
 1. Free or contained perforation of the small bowel
 2. Phlegmonous inflammation, internal fistula, or external fistula involving the affected site
 3. Multiple strictures within a short segment
 4. Stricture in close proximity to a site chosen for resection
 5. Colonic strictures
 6. Hypo-albuminemia (< 2.0 g/dl)
 7. Rigid thickened non-supple bowel segments, either short or long
 8. Extensive ulceration of mesenteric margin especially bleeding or friable
 9. Suspicion of cancer in a strictured-segment.

Colonic Disease (20-30% of patients)

- The choice of operation for colonic CD depends on age, disease distribution, extent of involvement, previous resections, rectal compliance and adequacy of fecal continence.
- *Segmental resection with ileo-colic or colo-colic anastomosis and abdominal colectomy with ileo-proctostomy* are the procedures most commonly performed for Crohn's colitis.
- For pan-colitis a *total procto-colectomy with permanent ileostomy* is the operative choice.
- Although isolated centers are offering *procto-colectomy and ileal pouch-anal anastomosis* with reasonable results, patients requiring surgical treatment of proctitis ± accompanying colitis usually undergo procto-colectomy and end-ileostomy.
- *Indications of ileo-rectal anastomosis (IRA)*
 1. The rectum should be spared when at least it has no major features of the disease (ulcerations and strictures)
 2. Normal anal sphincter function
 3. No active perianal disease
 4. No rectal dysplasia or malignancy
 5. Properly distensible rectum (assessment of rectal compliance is essential)
 6. Despite the high rate of reoperation, IRA is important in Crohn's colitis because it postpones the need for proctectomy with the risk of sexual dysfunction and need for a permanent external appliance
- *The role of ileostomy alone in CD*
Ileostomy alone has a limited role in CD
 1. Temporary diversion as with repair of anal sphincters or advancement flaps in repair of peri-anal or recto-vaginal fistulae
 2. End or loop ileostomy is done for patients destined for procto-colectomy because of severe perianal sepsis to avoid later unhealed perineal wound complication

3. In cases with toxic colitis and patients who cannot tolerate subtotal colectomy or segmental colectomy or proctectomy.
- *Treatment of colonic fistulas in CD*
 1. Internal fistulas occur in CD in 25-40% of cases depending on the site of the disease, highest in the ileo-colic CD.
 2. Ileo-sigmoid fistulas are the most common. The procedure of choice is resection of the ileo-cecal area involved + with ileo-colic anastomosis with separate segmental resection of the sigmoid with colo-colonic anastomosis. A covering temporary loop ileostomy is a must if there is associated pelvic sepsis or small bowel obstruction
 3. Bladder fistula is treated with closure after dealing with the gut CD

Perineal Crohn's Disease

- The perianal abnormalities associated with CD include edematous skin tags or hemorrhoids, cyanotic, discoloration, fissures or canal ulceration, abscesses, fistula and anorectal stricture.
- Patients with colonic disease are more likely to have perianal manifestations than those with ileocolic or small bowel disease patterns.
- Surgery is reserved for symptomatic problems. If the patient is too uncomfortable for an adequate evaluation, a well-conducted examination under anesthesia (EUA) is mandated.
- *Simple incision and drainage* of the abscess adequately relieves acute symptoms and allows resolution of inflammation. Controlling the proximal disease improves perianal disease.

Duodenal Crohn's Disease (1-2% of cases)

- The most common problem is *stricture* leading to obstructive symptoms.
- Most patients can be managed successfully with *medical therapy*.
- Surgical approaches are used when obstructive symptoms remain disabling despite medical treatment. *Gastro-jejunostomy* is the simplest and most widely used operation for relief of obstruction.

Main Differences between UC and CD (Table 5)

Table 5: Main differences between ulcerative colitis and Crohn's disease

Main Differences between UC and CD	
-	UC affects the colon; CD can affect any part of the GIT, but particularly the small and large bowel
-	UC is a <i>mucosal</i> disease whereas CD affects the <i>full thickness</i> of the bowel wall
-	UC produces confluent disease in the colon and rectum, whereas CD is characterized by <i>skip</i> lesions
-	CD more commonly causes <i>stricturing</i> and <i>fistulation</i>
-	<i>Granulomas</i> may be found on histology in CD but <i>not</i> in UC
-	CD is often associated with <i>peri-anal disease</i> , whereas this is unusual in UC
-	CD affecting the <i>terminal ileum</i> may produce symptoms mimicking acute <i>appendicitis</i> , but this does <i>not</i> occur in UC
-	Resection of the colon and rectum (procto-colectomy) cures the patient with UC, whereas <i>recurrence</i> is common after resection in CD

Rectal Prolapse and Intussusception

RECTAL PROLAPSE

Definition

- Prolapse means descent of part of the wall or full thickness wall of the rectum into the anal canal (during straining). It is associated with *fecal incontinence* in 50-70% of cases.

Predisposing Factors

- Prolonged straining
- Vaginal delivery
- Neurological
- Dementia
- Chronic constipation
- Hyper motile sigmoid colon
- Bilharziasis.

Classification

Partial (Incomplete or Mucosal) Prolapse

- ***Definition:***
Partial prolapse means a “1-4 cm *protrusion* of rectal mucosa and submucosa outside the anus” (Figures 1 and 2).
- ***Age***
It occurs in the extremes of age (children and elderly): In children, the prolapse is usually mucosal and should be treated conservatively.
- ***Predisposing factors***
 1. *In infants:* Underdeveloped sacral curve - low anal store
 2. *In children:* diarrhea – whooping cough – loss of weight.
 3. *In adults:* Hemorrhoids - prolonged straining - perineal tears (more among females) - secondary to surgery (due to damage of sphincters).



Figure 1. Patient with partial rectal prolapse showing prolapse of rectal mucosa

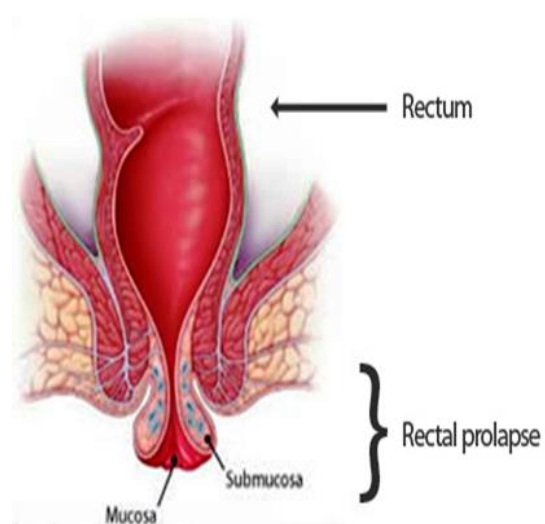


Figure 2. Diagram illustrating mucosal (partial) rectal prolapse.

Complete Prolapse

- **Definition:** Full thickness prolapse of the rectum. It contains the two layers of the rectal wall and has an intervening peritoneal sac.
- **Age:** It occurs in older adults and is often associated with incontinence.
- **Gender:** It occurs in females > males (6:1) worldwide; however, in Egypt, this ratio is reversed with a high incidence in *young males*.
- **Associated** with weak pelvic and anal musculature. Sigmoid & rectum are floppy & redundant.
- **Surgery** is necessary. The operation is performed either via the perineum or abdomen
- **Grades of complete rectal prolapse:**
 - Grade 1:** *Occult (internal) prolapse* It is internal intussusceptions of upper into lower rectum. (Figure 3)
 - Grade 2:** *Concealed prolapse:* Prolapse to but not through the anus i.e. does not emerge from the anus (Figure 4 A,B). May be associated with *solitary rectal ulcer* (SRU).
 - Grade 3:** *Complete rectal prolapse through the anus* (Figure 5A-C).

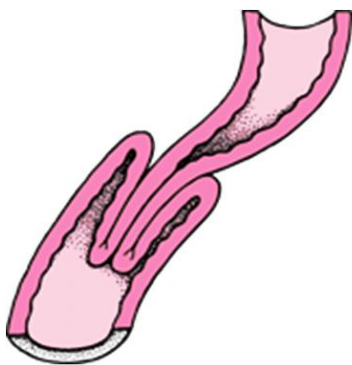


Figure 3. Grade 1 prolapse

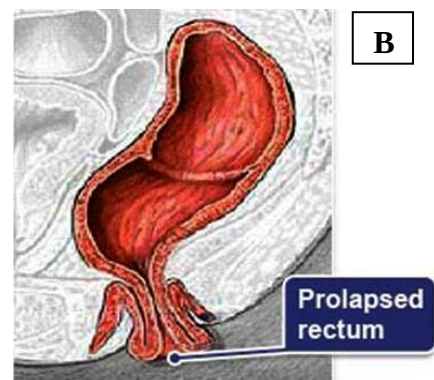
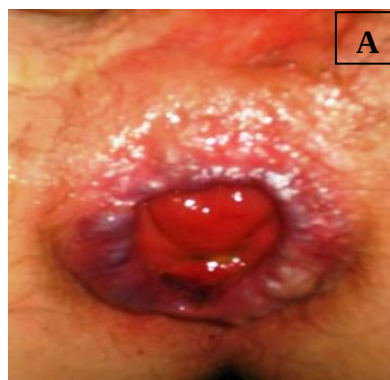


Figure 4. Grade 2 prolapse (does not pass through the anus)

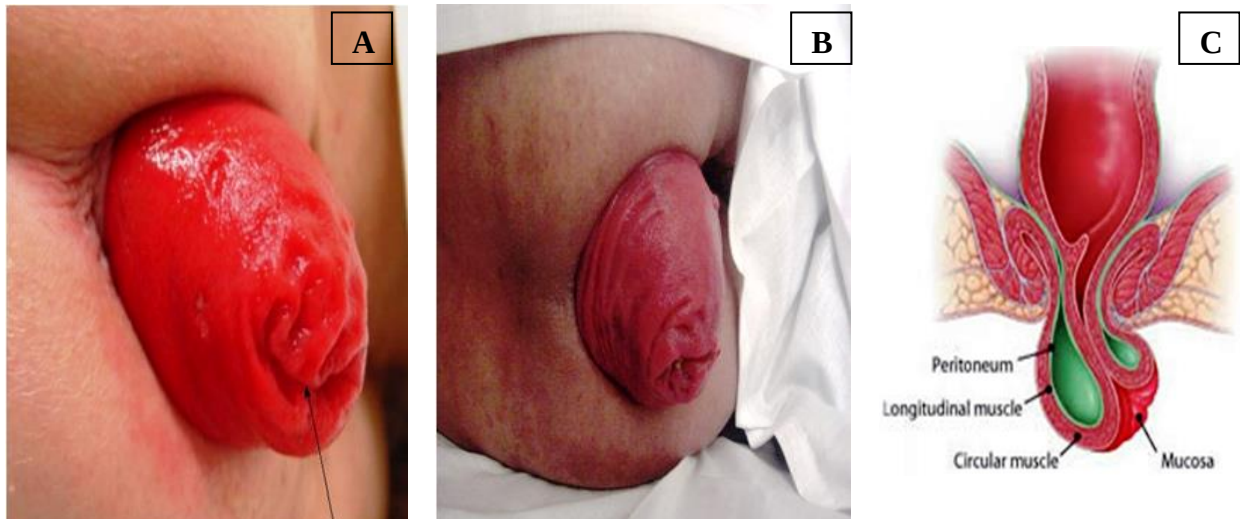


Figure 5. Grade 4 prolapse (passes through the anus)

The 6 typical anatomical features are:

1. The intussusception itself
2. Deep cul-de-sac or pouch of Douglas
3. Absent fixation of the rectum to the sacrum
4. Redundant rectum and sigmoid colon
5. Weakness of the pelvic floor and/or the anal sphincters
6. Possibly, the presence of a *rectocele*

Presentation

- Usually occurs after defecation & may reduce spontaneously or be reduced manually
- Bleeding, mucus discharge or incontinence may be troublesome
- PR Examination usually shows poor anal tone
- Prolapse may be visible on straining. Most prolapses that are > 5 cm are complete (Figure 6)
- Differential diagnosis in adults includes (1) Large hemorrhoids (Figure 7), (2) Prolapsing rectal tumors and (3) Prolapsing anal polyp.

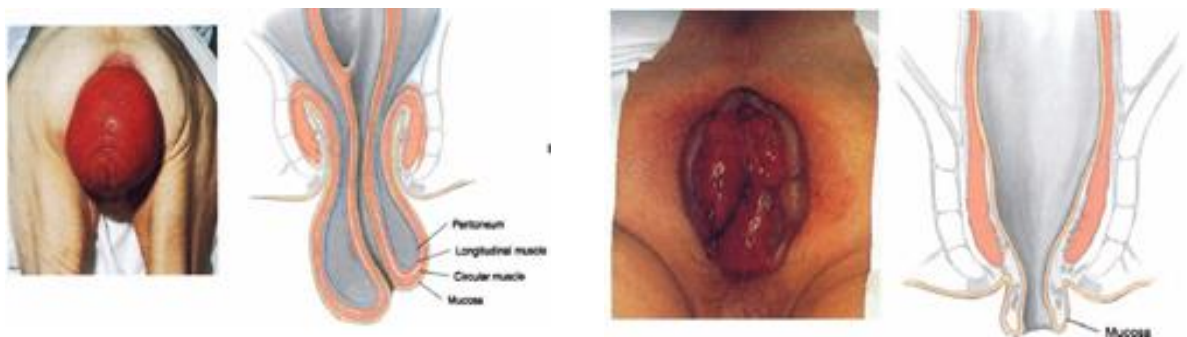


Figure 6. Folds in prolapsed of the rectal wall. (A) Concentric folds of rectal procidentia. (B) Bowel wall layers seen with complete rectal procidentia.

Figure 7. (C) Radial folds of prolapsing internal hemorrhoids. (D) Bowel wall with only mucosal prolapse or prolapsing internal hemorrhoids.

In adults

Internal or occult prolapse

- This may cause complete prolapse
- There are symptoms of outlet obstruction with urgency, straining, sense of incomplete evacuation and digitated defecation
- This may lead also to *solitary rectal ulcer syndrome*, which is a precursor of procidentia (prolapse) and is caused by injury & ischemia caused by the internal prolapse and straining, which force the anterior wall of the upper rectum into the anal canal (serpiginous or polypoidal ulcer) with passage of excess mucus and blood-tinged mucus per rectum

Complete prolapse

- The prolapse itself is the main complaint
- There is soiling of the underwear by blood, mucus and feces
- Some degree of fecal incontinence may be present (can be the start of the intussusception or the effect due to the stretch of the sphincter).
- Generalized *perineal descent* with stretch of the pudendal nerves can also contribute in long-standing cases to the incontinence of the patient
- On PR examination, the anus is patulous, with deficient sphincter tone and little discomfort.
- On straining, the prolapse is seen (>5 cm is definite complete and < 5 cm can be complete or partial) and needs bidigital examination.

In children

- It is the feature of the first 2 years being more common in boys
- Parents complain of the passage of red area around the anus of the child after defecation.
- Excessive mucus, and traces of blood is a very common presentation
- Inserting a suppository & putting the child on a pot may be necessary to precipitate the prolapse because usually examination is negative.

Complications of Prolapse

1. Ulceration (Figure 8)
2. Strangulation (Figure 9)
3. Spontaneous rupture with evisceration (Figure 10 and 11)
4. Incontinence (fecal and urinary)

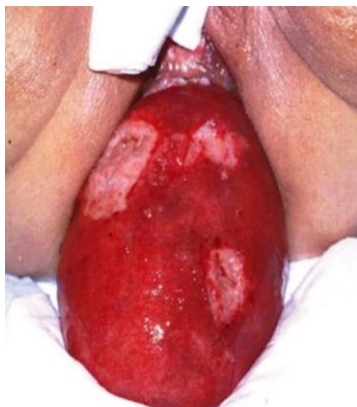


Figure 8.
Ulcers on
top of rectal
prolapse



Figure 9.
Rectal prolapse
with
strangulation of
bowel



Figure 10. Rectal prolapse with spontaneous rupture and evisceration of normal bowel



Figure 11. Rectal prolapse with spontaneous rupture and evisceration of strangulated bowel (note the dark color)

Investigations (in Elective Cases)

1. To detect the precipitating factor → At least a *flexible sigmoidoscopy*
2. Assessment of surgical risk (no effective non-operative treatment)
3. Anorectal manometry, pudendal nerve test (predicts functional outcome after surgery).

Treatment

In Children (Partial or Complete)

- Alleviation of straining (constipation, diarrhea/tenesmus) and training of regular bowel habits.
- Buttocks are strapped together after defecation after spontaneous or manual reduction
- The body of the child and fat reservoirs should be built-up
- *Sclerosant* injection (phenol in almond oil for *submucous* injection in partial prolapse and alcohol for *retro-rectal* injection in complete prolapse)
- In case of failure, one of the operations described below is resorted to

In Adults

- In *partial prolapse* mucosal hemorrhoidectomy will often suffice to deal with the condition. Recently, *Longo's procedure* (PPH stapler is used to induce anal lift and refixation of the prolapsed mucosa back to the rectum and anal canal).
- In *complete prolapse*, the final functional result is dependent on the degree of continence rather than simple control of the prolapse.

Operations of Prolapse

- The choice of the operation depends on:
 1. Degree of prolapse present
 2. Associated disorders (cystocele, rectocele, incontinence or constipation)
 3. Co-morbid conditions (spinal cord lesion and mental, psychic or vital system problems)
 4. The main symptoms of presentation

- *Goals of surgery are*
 1. Resection of redundant colon
 2. Fixation of the rectum to the sacrum
 3. Improving symptoms of fecal incontinence and constipation

Perineal Procedures

1. Thiersch Procedure: *Obsolete nowadays!*
2. Altemeier Operation: (Perineal Procto-sigmoidectomy) (Figure 12)
 - Mortality rate (MR) = 0-5%
 - Complications: pelvic sepsis, leakage
 - Recurrence rare (RR) = 0-16%
 - Best if combined with posterior levatorplasty
 - Ideal for incarcerated or strangulated prolapse
 - Difficult to perform for small prolapse
3. Delorme Procedure
 - It is the minimum procedure that should be done!!! (Figure 13)
 - Mortality rate = 0-4%
 - Recurrence rate is high = 4-38% (At St. Marks Hospital-UK = 12.5%)
 - Good for short prolapse
 - Can be repeated if necessary



Figure 12. Altemeier operation
(Perineal procto-sigmoidectomy)

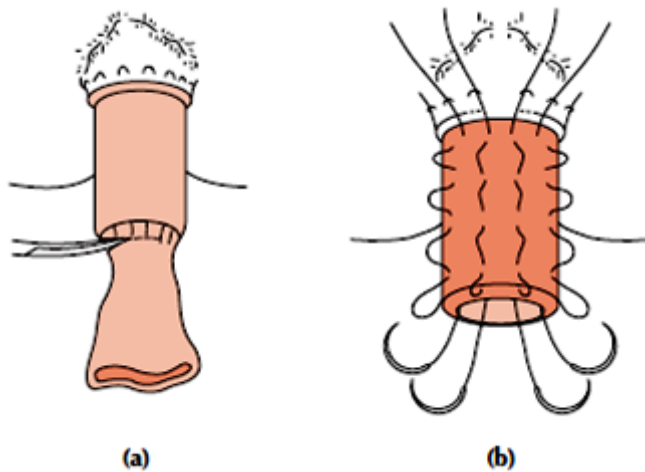


Figure 13. Delorme's operation for rectal prolapse. **(a)** The mucosa has been removed from the prolapse. **(b)** Interrupted sutures have been placed in the underlying muscle so that, when tied, the muscle is *plicated*.

Abdominal Procedures

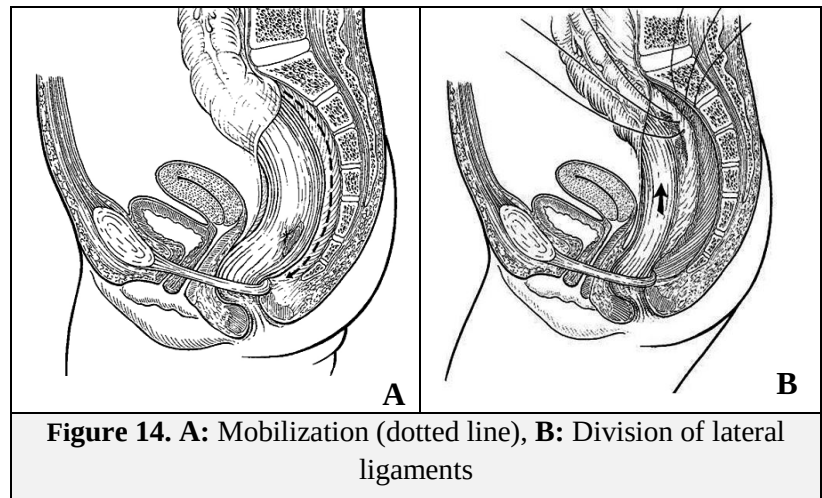
Rectopexy

1. *Sutured Posterior Rectopexy (Ripstein Procedure)*
2. *Prosthesis or Mesh Rectopexy*
 - a. Anterior ventral rectopexy
 - b. Posterior rectopexy (*Wells operation*)
3. *Resection rectopexy (Frykman-Goldbery procedure)*

Laparoscopic versus open procedure

1. Sutured Rectopexy (Ripstein Procedure)

- The procedure entails posterior mobilization of the rectum to the tip of the coccyx and division of the lateral ligaments on either side (Figure 14).
- No reported mortality
- RR = majority 0-8% (range: 0-28%)
- There is a variable response to constipation



2. Prosthesis or Mesh Rectopexy

- a. Anterior ventral rectopexy (Figure 15)
- b. Posterior rectopexy (*Wells operation*) (Figure 16).

Disadvantages: *Constipation* following *rectopexy* is a real problem that may be caused by:

- Redundant sigmoid kinked in the Douglas pouch
- Division of the parasympathetic nerves during rectal mobilization

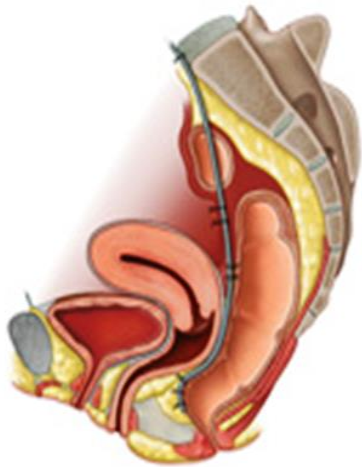


Figure 15.
Anterior
ventral
rectopexy
(using
mesh)

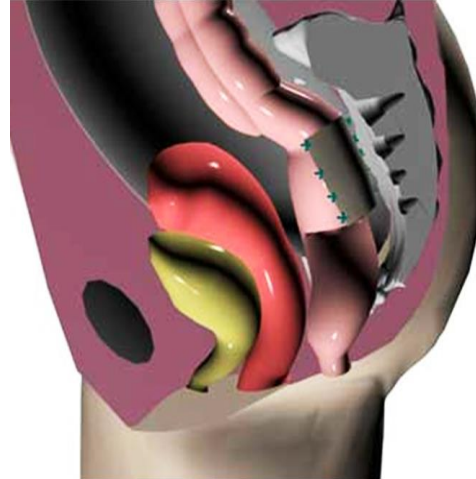


Figure 16.
Posterior
rectopexy
(using
mesh)
(Wells
operation).

3. Resection-rectopexy (*Frekman Repair*)

- It is the best-used operation for adults now.
- It entails, *sigmoidectomy* & *posterior rectopexy* (Figure 17)
- The operation added 1% to MR
- Recurrence rate = 0-5%
- *Improved constipation*. Other forms of posterior rectopexy cause severe constipation.

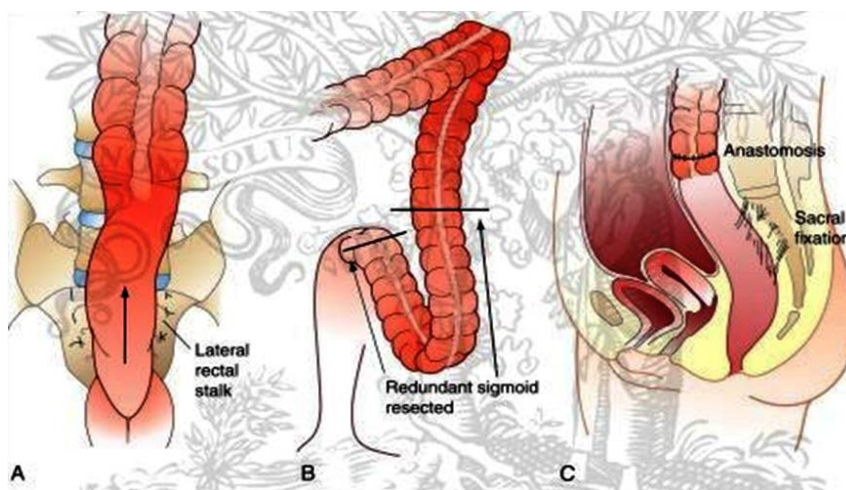


Figure 17. *Frekman repair* **A.** Rectal mobilization **B.** sigmoid resection **C.** Colorectal anastomosis and rectopexy

Comparison between Operations for Rectal Prolapse (Table 1 below)

Criteria	Sutured Rectopexy	Mesh Rectopexy	Resection-Rectopexy
- Mortality	0%	0-2.8%	1-4%
- Recurrence	0-8%	0-13%	0-5%
- Complications	Rare	8-52%	Up to 30%
- Continence	Improved	Improved	Improved
- Constipation	Variable	Up to 42%	Improved

Laparoscopic Approach for Rectopexy

- It includes the following procedures for rectopexy (sutured, stapled, posterior mesh, resection)
- It is as effective as the open approach (no long-term difference)
- Recurrence 0-10%
- Benefits include:
 1. Shorter hospitalization
 2. Overall decrease in cost
 3. Less morbidity and earlier recovery
 4. Earlier return to work.
- Laparoscopic approach is desirable because of:
 1. The benign nature of the condition
 2. Patients are often at high surgical risk for laparotomy.

The Choice of Operation should be Individualized

- Abdominal procedures are ideal for young fit patients & provide the best chance of cure
- Sutured rectopexy gives good results
- Combination of a resection reduces constipation
- Laparoscopic approach provides similar results with less morbidity
- Perineal procedures are indicated for frail patients with extensive co-morbidity, not fit for major abdominal surgery
- Perineal recto-sigmoidectomy, combined with levatorplasty gives better result than Delorme's operation.

INTUSSUSCEPTION

Definition

- Intussusception is a process in which a segment of intestine invaginates into the adjoining intestinal lumen (Figure 18) (causing bowel obstruction)
- As a common cause of abdominal pain in children, intussusception is suggested readily in pediatric practice based on a *classic triad of signs and symptoms* (vomiting, abdominal pain, and passage of *bloody red currant stools* per rectum)

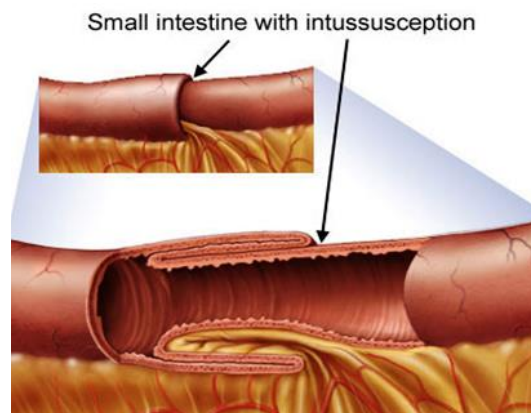


Figure 18. Intussusception of the intestine

Classification

According to Location

- Ileo-colic (*the most common*) (Figure 19)
- Colo-colic
- Ileo-ileo-colic
- Ileo-ileal.
- Jejun-jejunal

According to Etiology

- Primary (idiopathic) Intussusception
- Secondary Intussusception

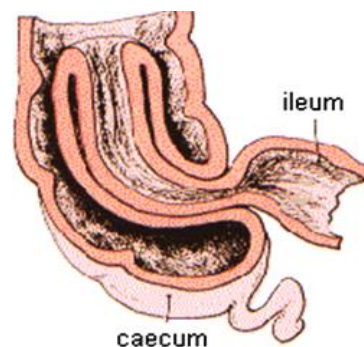


Figure 19. Primary ileo-colic intussusception

Etiology

Primary Intussusception

- In most infants and toddlers with intussusception, etiology is unclear. This group is believed to have *idiopathic intussusception*.
- One theory about the etiology of idiopathic intussusception is that it occurs because of *enlarged Peyer's patches* (following viral upper respiratory tract infection or gastro-enteritis) starting the process of intussusception.

Secondary Intussusception

- In approximately 2-8% of children with intussusception, a surgical lead point is found.
- Occurrence of surgical lead points increases with age and indicates that the probability of non-operative reduction is highly unlikely.
- Examples of lead points are:
 1. *Meckel's diverticulum* or appendix
 2. *Benign tumor of malignant tumor* of the mesentery or intestine e.g. lymphoma, polyps (Figure 20) and hamartomas associated with *Peutz-Jeghers syndrome*

3. Mesenteric or duplication cysts.
4. Cystic fibrosis
5. Ectopic pancreatic and gastric rests
6. Foreign body (FB)
7. Following abdominal operations and trauma.

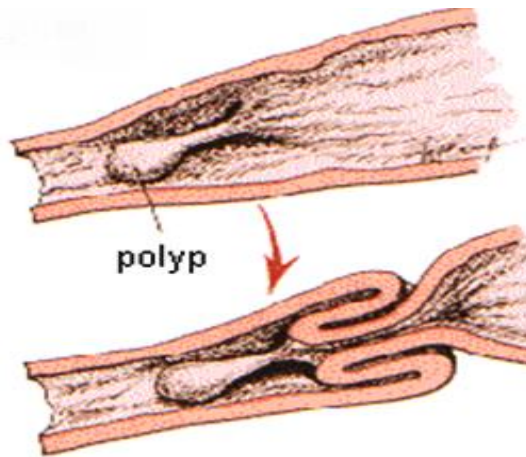


Figure 20: Secondary (polyp) ileo-ileal intussusception

Pathophysiology

- The pathogenesis of intussusception is believed to be 2^{ty} to an imbalance in the longitudinal forces along the intestinal wall.
- This imbalance can be caused by a mass acting as **a lead point** or by a disorganized pattern of peristalsis (eg, an ileus in the post-operative period). As a result of the imbalance, an area of the intestinal wall invaginates into the lumen, with the rest of intestine following.
- The *invaginating* portion of the intestine (**intussusceptum**) completely invaginates into the *receiving* portion (**intussusciens**) (Figure 21). This process continues & more proximal areas follow, allowing the intussusceptum to proceed along the lumen of the intussusciens.
- Early in this process, *lymphatic return* is impeded, then, with the rise in the pressure within the wall of the intussusceptum, *venous drainage* is impaired. Finally, the pressure reaches a point at which **arterial inflow** is inhibited & infarction ensues (Figure 22).
- The *mucosa* is most sensitive to ischemia because it is furthest away from the arterial supply. Ischemic mucosa sloughs off, which initially → the classic "**red currant jelly stool**" (a mixture of sloughed mucosa, blood & mucus – Figure 23). If untreated, the process → transmural gangrene & perforation of the leading edge of the intussusceptum.

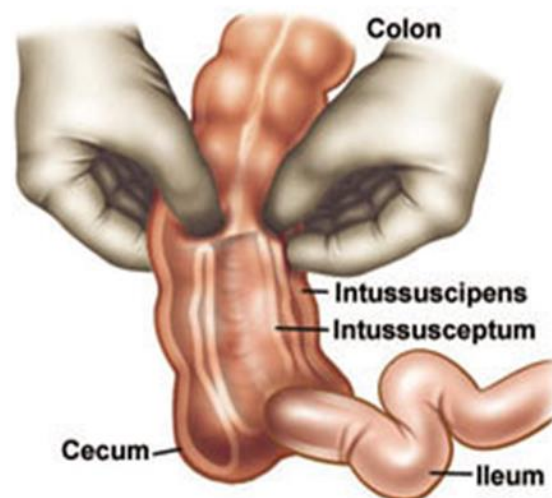


Figure 21. Ileo-colic intussusception showing the *intussusceptum* and *intussusciens*

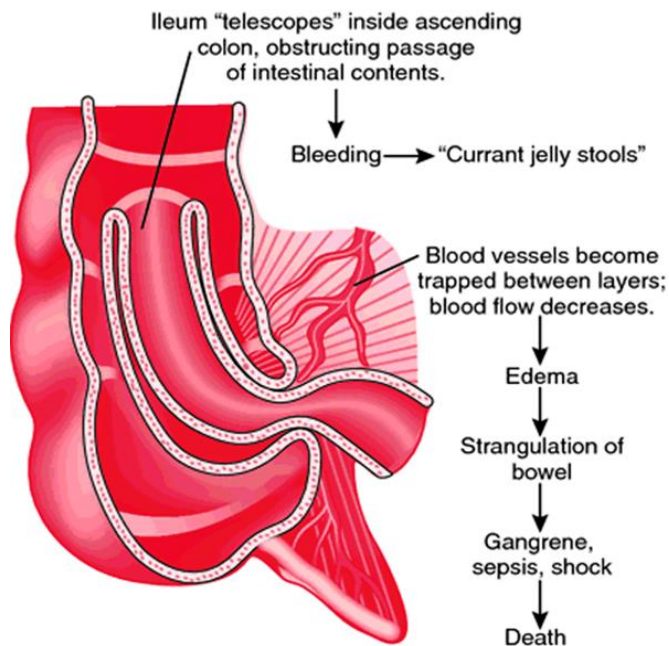


Figure 22. Pathophysiology of intussusception



Figure 23. Red current jelly

Incidence

- General incidence: 1-4 cases / 1000 live births.
- *Gender*: Slight preponderance of males (3:2).
- *Age*:
 - 2/3 of children with intussusception are <1 year; most commonly, in *infants aged 5-10 months*. Intussusception has been very rarely reported in the neonatal period.
 - Intussusception can account for as many as 25% of abdominal surgical emergencies in children <5 years, exceeding the incidence of appendicitis.
 - *Intussusception is the commonest cause of intestinal obstruction between 5 months-3 years.*

Clinical Features

History (Symptoms)

- History of preceding upper respiratory tract or GIT infection.
- *Pain* is *colicky*, severe and intermittent, and occurs in 100% of cases. The parents or caregivers describe the child as *drawing the legs up to the abdomen* and kicking the legs in the air.
- Initially, *vomiting* is non-bilious and reflexive, but when intestinal obstruction occurs, it becomes bilious. It occurs in 80% of cases. *Any child with bilious vomiting is assumed to have a condition that must be treated surgically until proven otherwise.*
- Parents also report the passage of stools that look like *redcurrant jelly*. This is a mixture of mucus, sloughed mucosa & shed blood, and is seen in 90% of cases.
- *Lethargy* is relatively common with intussusception, especially after attack of colic.

Physical Signs

- On clinical examination, the patient is *usually plump & in good health*. Intussusception is uncommon in children who are malnourished.
- The child is found to have periods of lethargy alternating with crying spells, and this cycle repeats every 15-30 minutes. The infant can be pale, sweaty and hypotensive if shock has occurred.
- The *hallmark findings* are a right hypochondrium sausage-shaped *mass* (Figure 24) and emptiness in the right lower quadrant (*Dance sign*). This is hard to detect and is palpated best when the infant is quiet between spasms of colic.

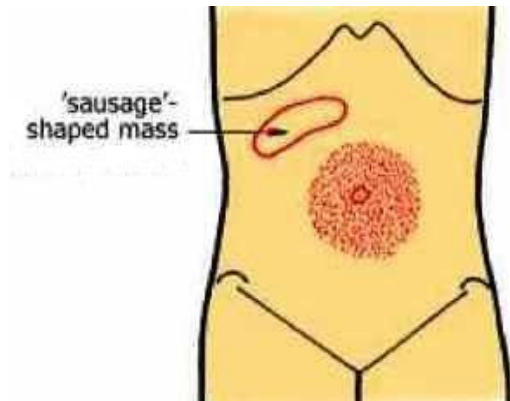


Figure 24. The site of sausage shaped intussusception in the abdomen, shown schematically in and clinically in a female child.

- Abdominal *distention* is frequently found if obstruction is complete.
- If intestinal gangrene and infarction have occurred, *peritonitis* can be suggested on the basis of rigidity and involuntary guarding.
- Classic *red currant jelly* stools appear.
- *Fever and leukocytosis* are late signs and can indicate transmural gangrene and infarction.
- A rare presentation is *prolapse of the intussusceptum* through the anus: This prolapse can be confused with *rectal prolapse*, in which the anal crypts are everted with prolapse (not with intussusception). Moreover, an examining finger can pass between the prolapse and the anus in patients with intussusception but not in patients with rectal prolapse.
- *Complications*
 - o These include *gangrene and perforation*.
 - o In complicated cases, vomiting becomes bilious and physical findings include lethargy, increased abdominal distention, tender and rigid abdomen.

Investigations - Imaging Studies

1. Plain Radiograph

- It may be *normal* early in the course of intussusception.
- As the disease progresses, earliest radiographic evidence includes an absence of air in the right lower and upper quadrants (*radiological Dance's sign*) (Figure 25)
- Visible abdominal mass
- Meniscus sign (Figure 26)
- Late cases show an obvious pattern of small bowel obstruction, with small bowel dilatation and *air-fluid levels* (Figure 27) or free intra-abdominal air



Figure 25. Plain X-ray showing paucity of air in the right side

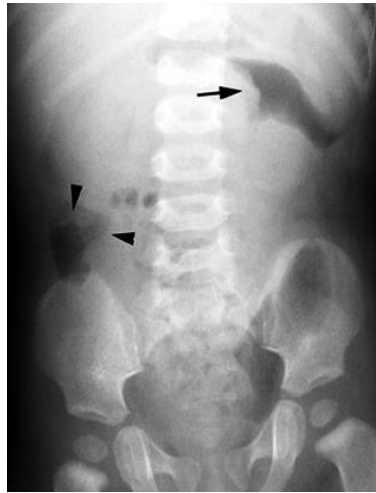


Figure 26. Plain X-ray showing meniscus sign



Figure 27. Plain X-ray showing multiple fluid levels (obstruction)

2. Contrast Enema (Barium, Saline or Air)

- Contrast enema is a quick and reliable, and has the potential to be therapeutic.
- Exercise caution when performing contrast enema in patients >3 years because most of them have a surgical lead point in the small bowel and the diagnostic and therapeutic yield of the enema is lower in these patients.
- The classic appearance of the head of intussusception in contrast enema is *Cobra head sign* or *Napoleon cap sign* (Figure 28). It results from arrest of the barium in the colon with crescentic outline (*cobra-head appearance*) around the apex of intussusception.
- Barium flowing between 2 layers of intussusception (*spring coil appearance*) (Figure 29)
- Appearance of the rounded apex of the intussusceptum protruding into a column of contrast material (*meniscus sign*) (Figure 30).



Figure 28. Barium enema showing *Cobra-head* appearance



Figure 29. Barium enema showing *Spring coil* appearance

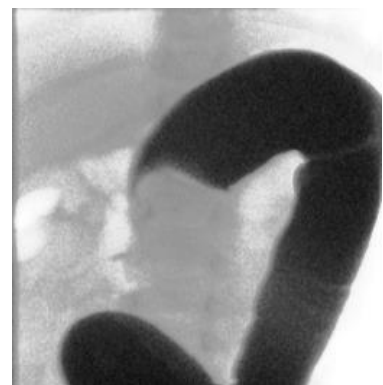


Figure 30. Barium enema showing meniscus sign

3. Ultrasonography(US)

- It is a very accurate and non-invasive method of diagnosis.
- The characteristic *target sign* (Figure 31) represents transverse section through the mass while the *pseudo-kidney* sign represents a more oblique cut.

4. CT Scan

- *Target sign* (Figure 32) and to detect the cause in 2^{ry} intussusception

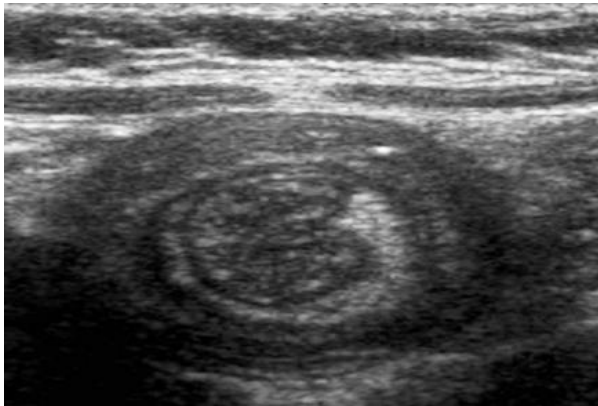


Figure 31. Ultrasound showing target sign



Figure 32. CT scan showing target sign (arrow)

Principles of Treatment

- *Early cases:* Reduction of the invaginating segment, either non-operatively or operatively.
- *Neglected cases:* Resection of the strangulated bowel & restoring continuity of the bowel.
- *For all children,* start IV fluid resuscitation and naso-gastric (NG) decompression as soon as possible.
- *The presence of peritonitis and any evidence of perforation* revealed on plain X-ray are the only 2 absolute *contra-indications* to an attempt at non-operative reduction with a therapeutic enema.

Medical Treatment

1. Hospitalization
2. Nasogastric tube
3. IV fluids.
4. Antibiotics

Non-surgical Reduction (Therapeutic enemas)

- Can be *hydrostatic*, with either (1) *barium* or (2) *water-soluble contrast*, or (3) *pneumatic*, with air insufflation. Therapeutic enemas must be performed *under fluoroscopic or ultrasonic guidance*.
- *Criteria of successful reduction* = the flow reaches the small bowel + disappearance of the target sign
- A systematic approach to hydrostatic reduction of intussusception, the reported success rate of this non-operative intervention varied widely from <40% to >90%.

- When performing a therapeutic enema, the recommended pressure of air insufflation should *not* exceed 120 cmH₂O. When using barium or water-soluble contrast, the column of contrast should *not* exceed 100 cm above the level of the buttocks.
- An attempt is *not* considered successful until the reducing agent, whether air, barium, or water-soluble contrast, is observed refluxing back into the terminal ileum.

Surgical Reduction

- *Indications*
 1. Failure or contra-indications of hydrostatic reduction.
 2. Above the age of 2 years (2^{ry} cases).
 3. Previous abdominal operations.
 4. Recurrent intussusception.
- Surgical options include simple manual reduction or *resection and anastomosis* according to severity of presentation

- Simple manual reduction: Traditional entry into the abdomen is through a transverse right upper abdominal muscle cutting incision (Figure 33). Deliver the intussusception into the wound & attempt simple reduction. *Milking* the intussusceptum out of the intussusciens is important. Sustain gentle manual pressure rather than pulling out the intussusceptum to avoid risk of iatrogenic perforation (Figure 34).

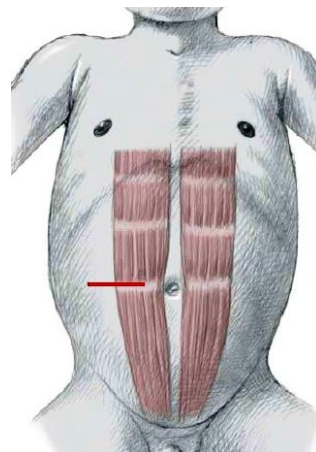


Figure 33. Transverse right upper abdominal muscle cutting incision (red line)

- Resection with an end-to-end anastomosis (R-EEA)

○ *Indications of R-EEA*

1. Failed simple reduction.
2. Gangrenous bowel loops.
3. Perforation.
4. Secondary causes (eg. Meckel's)

Diligent search for any lead points is advised especially if the patient is >2-3 years.

- *Post-operative course*

- After manual reduction (or hydrostatic reduction) the child is observed and starts oral feeding after passage of stools, usually after 6 hours and is usually discharged after 24 hours. A sure sign of complete release of intestinal obstruction after manual or hydrostatic reduction is the passage of a normal motion.
- After R-EEA, the child receives IV fluids and antibiotics & oral feeds are withheld for 4-5 days.

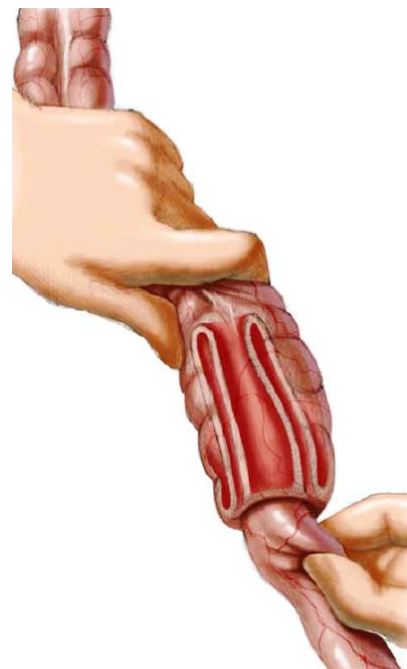


Figure 34. Manual reduction

Prognosis

- Early diagnosis, appropriate fluid resuscitation and treatment reduced the *mortality rate* from intussusception to <1%. If left untreated, this condition is uniformly *fatal* in 2-5 days.
- The *morbidity rate* is very low after treatment of intussusception.
- *Recurrence*
 - o Risk of *recurrence* of the intussusception after operative reduction is 5-10 %.
 - o Most of recurrences occur within the first 72 hours.
 - o Multiple recurrences indicate a pathological lead point that necessitates surgery

SUMMARY OF INTUSSUSCEPTION

Age (3-12m)

Etiology(1^{ry} or 2^{ry})

Clinical (pain, vomiting, red current jelly stools)

Test (US target sign)

Treatment (Hydrostatic or surgical reduction)

Hemorrhoids

Definition

Anal fibrovascular cushions (or hemorrhoids) are part of the *normal* anatomy within the anal canal and are believed to be important in maintaining continence. The term *hemorrhoidal disease* means engorgement or enlargement of the normal fibrovascular cushions lining the anal canal.

SURGICAL ANATOMY AND CLASSIFICATION

- **External hemorrhoids** arise from the *inferior* hemorrhoidal plexus & are covered by modified squamous epithelium *distal* to the dentate line (Figure 1).
- **Internal hemorrhoids** originate from the *superior* hemorrhoidal plexus & are covered by mucosa *proximal* to the dentate line (Figure 1).
- Internal hemorrhoids are classified by history and *not* by physical examination.
- They are graded as shown in Table 1 as follows:



Figure 1. External and internal hemorrhoids

Table 1: Grading of hemorrhoids into 4 degrees (1-4)

Degree	Description
First degree	Bleeding per rectum only, no prolapse.
Second degree	Prolapse but reduce spontaneously.
Third degree	Prolapse & have to be manually reduced
Fourth degree	Permanently prolapsed

PATHOPHYSIOLOGY

1. Mechanism of Bleeding (Venous Obstruction Theory)

- Repeated straining to evacuate a fecal bolus impedes the venous drainage of the anal cushions. Consequently, the anal cushions become engorged the overlying mucosa becomes thin & more friable with subsequent bleeding at defecation.

2. Mechanism of Prolapse (Cushions Sliding Theory)

- Repeated straining & passage of hard fecal bolus cause stretching & disruption of the elastic fibers holding the cushion to the underlying internal sphincter.

CLINICAL PICTURE AND DIAGNOSIS

Clinical Features

- *External hemorrhoids* may swell or become thrombosed causing pain or they may ulcerate with subsequent bleeding. Thrombosis can ultimately resolve or a skin tag may remain which affords the possibility of itching, burning and soiling.
- *Internal hemorrhoids* may cause burning, swelling, itching, pain, protrusion, discharge, bleeding &/or soiling.
- *Hemorrhoidal Bleeding*: occurs with, or following, bowel movements, is almost universally bright red, very commonly drips into the toilet water and *not* mixed with the stool, dark, or melanotic in nature. Rarely, may present with anemia 2^{ry} to chronic blood loss.
- *Hemorrhoidal Prolapse* usually occurs in association with a bowel movement, particularly when straining is present. Hemorrhoids may also prolapse during walking or heavy lifting as a result of increased intra-abdominal pressure. The prolapse is associated with a full, uncomfortable feeling, which resolves when the prolapse reduces.

Summary of Symptoms of Hemorrhoids

- Hemorrhoids or piles are *symptomatic* anal cushions. They are more common when intra-abdominal pressure is increased e.g. in obesity, constipation and pregnancy
- Classically, they occur in the 3, 7 and 11 O'clock positions with the patient in the lithotomy position
- Symptoms may include:
 1. Bright red bleeding.
 2. Prolapse that may cause a protruding mass (Figure 2).
 3. Mucoïd discharge with prolapse.
 4. Itching (if hygiene is poor).
 5. No pain unless external and thrombosed or prolapsed



Figure 2. Prolapsed hemorrhoids (protruding mass)

Evaluation of Per-anal Bleeding

Rigid Sigmoidoscopy

- It is indicated in young individuals with bleeding associated with hemorrhoids and no other systemic symptoms and no family history of colorectal cancer (CRC).

Colonoscopy

- It is indicated in older patients with a family Hx of CRC or change in bowel habits

TREATMENT

- Treatment of symptomatic internal hemorrhoids varies from simple reassurance to operative hemorrhoidectomy.
- Treatments are classified into three categories: (1) Dietary and lifestyle modification, (2) Non-operative/office procedures and (3) Operative hemorrhoidectomy.

Dietary and Lifestyle Modifications

- Avoid prolonged attempts at defecation
- Increasing fluid and fiber in the diet
- Exercise
- Stool modifiers
- Warm sitz baths and topical creams (prolonged use of topical steroids is harmful)

Non-operative/Office Procedures (Grade I or Grade II)

1. Rubber Band Ligation (RBL)

- It is useful for Grade I or Grade II hemorrhoids and, in some circumstances, Grade III hemorrhoids.
- It involves strangulation of the redundant mucosa proximal to the internal anal cushions by using tight rubber bands, the resulting ulcer heals with fixation of the mucosa to the underlying muscle wall (90% success rate) (Figure 3)



Figure 3. Rubber band ligation (RBL)

2. Infrared Coagulation

- It is useful for Grade I or Grade II hemorrhoids; however, it may need repetitive treatment sessions, thus, alternative methods may be more efficacious (i.e. RBL).

3. Sclerotherapy

- It involves the injection of an irritating material into the submucosa in order to reduce vascularity and increase fibrosis.
- This, in theory, reduces blood entering the hemorrhoidal bundle, also fixes the hemorrhoidal bundle to the underlying internal sphincter and $\therefore$ r bleeding a prolapse.
- Injecting agents have traditionally been Phenol in oil or Sodium Morrhuate.

Operative Procedures (Grade III or Grade IV)

Hemorrhoidectomy

Principle

- Various types of hemorrhoidectomy have developed throughout the years, but they are all associated with the same basic principle. As hemorrhoids are thought to be redundant fibrovascular cushions, most treatments reduce blood flow and remove redundant tissue. Additionally, mucosal and submucosal blood flow is interrupted by a circular staple line

Techniques

1. Open (*Milligan and Morgan*) hemorrhoidectomy
 - It is most commonly used. This technique involves ligation and excision of the hemorrhoids while leaving the wound open.
 - One, two or three hemorrhoids may be treated in this manner.
2. Closed (*Ferguson*) hemorrhoidectomy
 - This technique involves excision of the hemorrhoid and the pedicle is then sutured with a suture ligature. The wound is then closed with a running suture beginning at the apex and extending at the anoderm.
 - As with open hemorrhoidectomy, one, two or three hemorrhoidal bundles may be excised in this fashion. It is essential in this procedure that only minimal amounts of anoderm are excised, otherwise, closing the anal wounds can result in significant post-operative pain and perhaps long-term anal stenosis.

Complications of open & closed hemorrhoidectomy

- Severe post-operative pain (avoid intra-anal packing after surgery)
 - Acute urine retention (conservative treatment, if failed a urinary catheter should be inserted)
 - Reactionary or 2^{ry} hemorrhage (needs re-admission for surgical control of bleeding)
 - Anal fissure (associated with stenosis due to excessive excision of anal verge skin and pectin mucosa)
 - Abscess or fistula
 - Skin tags
 - Anal stenosis (avoidable by leaving adequate islands of mucosa between hemorrhoidectomy wounds)
 - Pseudo polyps
 - Incontinence
3. Diathermy Hemorrhoidectomy / LigaSure
 - Both open and closed techniques can be done using diathermy electro-coagulation.
 - This technique results in less post-operative bleeding and pain as no intra-anal pack is needed. Recently, vascular-sealing devices are used e.g. LigaSure
 4. Stapled Hemorrhoidectomy
 - This technique involves trans-anal, circular stapling of redundant anorectal mucosa with a standard circular stapling instrument.
 - No incisions are made in the somatically innervated, highly sensitive anoderm theoretically resulting in significantly less post-operative pain, short hospital stay and early return to normal activity.

- It is a promising new technique available to patients otherwise requiring surgical hemorrhoidectomy. It is associated with significantly less pain and similar complication rates when compared to conventional surgery.

Less Invasive Operative Techniques

1. Pile Sutures

- This technique involves three levels suture ligation of perforating veins at the base of vascular cushions

2. Ligation Anopexy

- This technique allows reduction of the redundant mucosa, reduction of the blood flow to the hemorrhoidal cushions, re-setting of hemorrhoidal cushions in their normal position, and fixation of hemorrhoidal cushions to the underlying internal sphincter

COMPLICATIONS OF INTERNAL HEMORRHOIDAL DISEASE

1. Thrombosis / Strangulation / Ulceration/ Gangrene

Pathogenesis

- Prolapse of internal hemorrhoid, which becomes gripped by the sphincter thus interfering with venous return (causing strangulation) and so it becomes tense, very painful and if strangulation is not relieved in few hours, the pile becomes dark purple or black and feels solid, with considerable edema around the anus and persistence of tenderness rather than pain (thus become thrombosed).
- *Superficial ulceration* is the 3rd step in the events after *strangulation and thrombosis*, and this occurs in the exposed mucosa. Ulceration is usually associated with bleeding that may not stop except with urgent surgery.
- Thrombosis with bacterial invasion and/or *gangrene* of the cushion or the whole anal canal can occur. This is due to occlusion of the arterial blood supply or superadded infections with anaerobic bacteria. This is rare now and responds to antibiotic. *Portal pyemia* is history.

Conservative Treatment

- Bed rest
- Stool modifiers
- Warm sitz baths
- NSAIDs and analgesics
- Venotonic such as Diosimin
- Antibiotics: Metronidazol & broad spectrum antibiotics
- Local decongestant applications (e.g. jelly, ointment, or lead sub-acetate compresses)

Indications of early surgery (open hemorrhoidectomy under antibiotic cover).

- In case of very early diagnosis (rare) with no extensive edema & a single prolapsed, strangulated, thrombosed pile
- Ulceration with partial extrusion causing bleeding or infection.

2. Anemia

- Anemia may develop secondary to chronic blood loss.

3. Perianal dermatitis

- It results from mucus leakage, skin tags, bad hygiene, and sensitivity to drugs and additives to sitz baths.
- It is treated by taking care of the perianal skin hygiene, excision of prolapsed piles, skin tags, and stoppage of use of unnecessary local applications

4. Fibrosis

- Fibrous tissue polyps of the anal lining may develop as a result of resolution of thrombosis.

5. Suppuration

- Suppuration may occur due to bacterial invasion of ulcerated hemorrhoids (anaerobic or portal pyemia)

6. Partial rectal mucosal prolapse

COMPLICATIONS OF EXTERNAL HEMORRHOIDS

Thrombosis of external vascular channels

- It is *incorrectly* termed *peri-anal hematoma*.
- It presents as a *tense* swelling that feels indurated and painful, smooth and tender, with blue coloration seen below the skin with variable amount of edema.
- Self-resolution is the rule, but, it may persist and need surgical excision for extrusion with continuous bleeding or for persistence

Ano-rectal Abscess and Fistula-in-Ano

ANO-RECTAL ABSCESS

Etiology

- Ano-rectal abscesses start as an infection in the anal glands at the base of the anal crypt. The abscess develops into the plane between the IAS & EAS (i.e. *inter-sphincteric*), and then extends to adjacent areas as the abscess expands as shown in Figure 1.

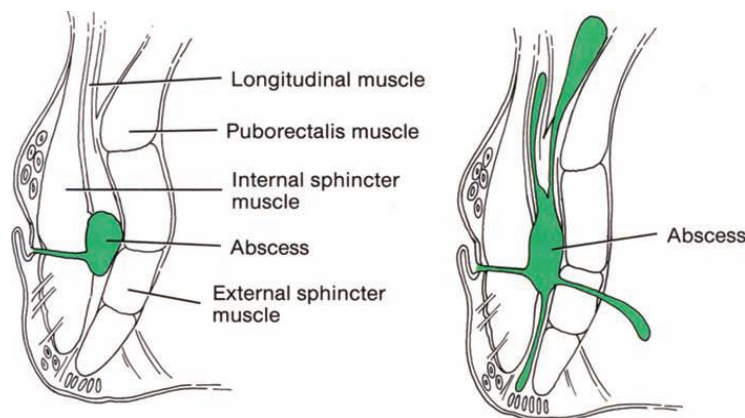


Figure 1. Spread of ano-rectal sepsis from infected anal glands in the inter-sphincteric space (1) downwards toward the perineum, (2) upwards to supra-levator space, (3) through the EAS either peri-anal or ischio-rectal, and (4) internally into the anal canal along the duct to dentate line.

Classification

- The classification is based on the predominant area of expansion: perianal, ischio-rectal, low inter-sphincteric, high inter-sphincteric, submucous, or pelvirectal, in addition to the posterior space of Courtney, which is located posterior to the rectum in the midline loose areolar tissue joining the 2 ischio-rectal fossa (Figure 2)
- The spaces, which allow the pus to spread in a circumferential pattern, are spaces connected posteriorly to each other with no septae in between. These are mainly 3 spaces: (1) the ischio-rectal spaces, (2) the pelvi-rectal spaces, and (3) the low inter-sphincteric spaces.

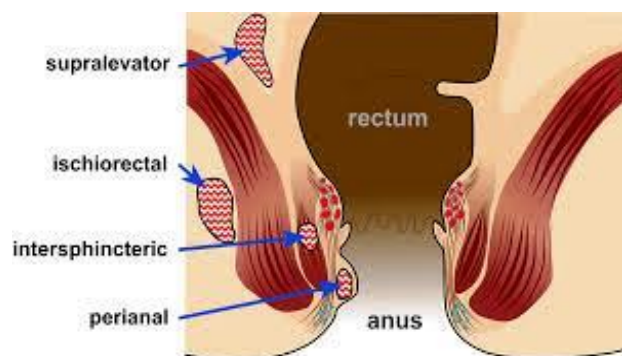


Figure 2. Different sites of ano-rectal abscess

Clinical Picture

- Severe throbbing pain (worse on sitting and coughing) and dyschasia (painful defecation).
- Systemic symptoms (fever, malaise, etc) (especially in ischio-rectal and pelvi-rectal collections)
- Local area of tenderness and induration (around the anus, or felt by PR examination).
- Quite often pus is seen (discharging from the anus or a spontaneous external opening).
- Determine if the patient is *immuno-compromised* (diabetic or on steroids or immuno-suppressants).

Differential Diagnosis

- An ano-rectal (or peri-anal) abscess should be differentiated from other *painful peri-anal conditions*, which include:
 1. Anal fissure
 2. Peri-anal hematoma
 3. Strangulated or thrombosed internal piles
 4. Peri-anal fistula
 5. Anal carcinoma
 6. Pruritis ani
 7. Proctalgia fugax.

Treatment

Conservative Treatment

- Very early in the course of the disease, where only a localized area of tenderness is found, massive doses of antibiotics with gram negative spectrum and anaerobes efficiency can be used (Quinolones and Metronidazole) with analgesics, sit baths and close follow-up.

Surgical Treatment (Incision and Drainage - I&D)

- The sign for incision drainage is the presence of induration (*do not wait for fluctuation*), and this can be easily detected in suppuration of spaces close to the perianal skin e.g. low inter-sphincteric, perianal, and ischio-rectal spaces suppuration.
- I&D is done under general anesthesia to ensure complete evacuation of pus.

Summary: Ano-rectal abscess

- They usually produce a painful, throbbing swelling in the anal region.
- The patient often has swinging pyrexia.
- They are subdivided according to anatomical site into perianal, ischiorectal, submucous and pelvirectal
- Underlying conditions include fistula-in-ano (most common), Crohn's disease, diabetes, and immunosuppression.
- Treatment is drainage of pus in first instance, together with appropriate antibiotics
- Always look for a potential underlying problem

FISTULA-IN-ANO

Definition

- A fistula is a tract between 2 epithelial-lined surfaces AND has 2 openings, an internal and an external.
- Fistula-in-ano opens deeply in the anal canal (or rectum) and superficially on the skin around the anus.

Etiology

- All fistulae arise from *pyogenic abscesses*, which have been allowed to develop to the point of spontaneous discharge or surgical drainage especially if inadequate. The point of debate thereafter is to explain why the condition has persisted.
- This may be due to:
 1. Persistence of the internal opening (recurrent activation of infection). This is the most accepted explanation nowadays.
 2. The presence of anal glands, which act as a reservoir for infection or as foreign bodies
 3. Inflammatory bowel disease (IBD) is very important association: Crohn's disease (CD) in 20% of cases, ulcerative colitis (UC) to a lesser extent and indeterminate colitis in between.
 4. Active TB is present in 8-16 % (one half shows evidence of healed pulmonary TB).
 5. Very rarely, higher carcinoma is coexistent.

Classification

It is necessary to consider the outline of the track in 2 planes: *horizontal & vertical*.

Horizontal Plane (Goodsall's Rule)

- Fistulae with their external opening in the anterior half of the anus have their internal opening opposite the external one, while those with the external opening in the posterior half have their internal opening in the midline i.e. have curved tracks.
- Lateral extension on both sides may take place leading to the characteristic horse-shoe fistula (Figure 3).

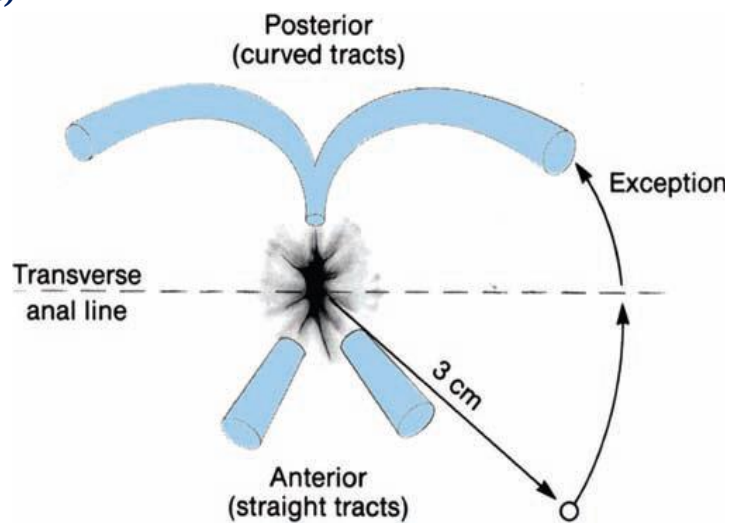


Figure 3. Goodsall's rule of fistula-in-ano

Vertical Plane (Park's Classification)

Alan Park's classification of anorectal fistulae is demonstrated in Figure 4 and includes the following types:

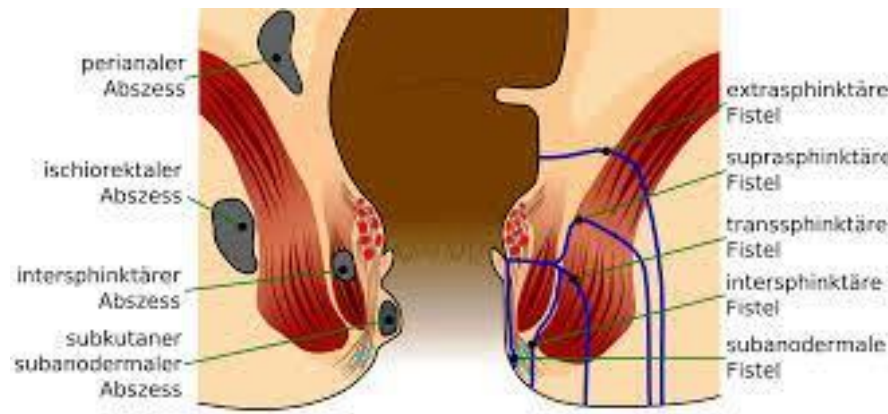


Figure 4. Alan Park's classification of ano-rectal fistulae.

1. Subcutaneous (SC)

- Rare, just below the perianal skin (superficial to the SC part of the EAS) near the pectinate line or at the base of a fissure

2. Inter-sphincteric

- Low inter-sphincteric: The track runs from the region of the pectinate line through the lower part of the IAS and skirts underneath the SC part of the external anal sphincter (EAS).
- Inter-sphincteric with an internal opening located at the dentate line or higher a track is extending in the inter-muscular plane.
- Inter-sphincteric with upward blind extension.
- Inter-sphincteric with blind upward extension and a supralelevator pocket.
- Inter-sphincteric with no external opening, and only an internal opening located at the dentate line with blind upward inter-sphincteric tract (a sinus rather than a fistula).

3. Trans-sphincteric

- Trans-sphincteric: the internal opening may lie at or higher than the pectinate line but below the ano-rectal ring (ARR). The tract extends through the EAS.
- Trans-sphincteric with occasional side track extending blindly upwards between the two sphincters or in the ischio-rectal fossa.
- Trans-sphincteric with a blind upward extension in the ischio-rectal spaces with a supra-levator pocket.

4. Extra-sphincteric

- Iatrogenic or due to pelvic pathology of the rectum.

5. Supra-sphincteric

- It is originally an inter-sphincteric fistula with a blind upward extension & supralelevator pocket, which has penetrated through the levator to present as an ischio-rectal abscess with an exit to the ischio-rectal skin.

Fistulae may thus be classified as:

1. Inter-sphincteric (70%)
2. Trans-sphincteric (25%)
3. Supra-sphincteric (5%)
4. Extra-sphincteric (<1%)

Extra-sphincteric fistulae are usually *not* associated with inter-sphincteric sepsis. In such cases, IBD or neoplasia should be considered.

Clinical Features and Differential Diagnosis

History of abscess (spontaneous or surgical drainage) or recurrent abscesses or discharge

A. Symptoms

- Discharge: Continuous or intermittent sero-purulent discharge with pruritis & soiling. Sometimes it stops due to obstruction of the external opening.
- Pain: due to abscess formation or obstructed internal opening.

B. Examination

- *Inspection:* The following should be defined: (1) number, (2) site (anterior or posterior), (3) distance from the anal canal (the nearer the external opening from the anus, the lower is the fistula), (4) discharge, and (5) granulation tissue or temporary healing.
- *Palpation:* The fistulous tract is felt and its relation to the ARMIR is determined.
- *PR Exam:* the internal opening may be felt, induration of the track may be identified and other pathologies excluded e.g. cancer.
- *Proctoscopy or sigmoidoscopy* is a *must* before embarking on the treatment.
- Hidradenitis suppurativa, pilonidal sinus, dermal suppurative sinuses, infected sebaceous cysts and urethra-perineal fistulas must be *differentiated* from cryptogenic fistulas.

Investigations

Endo-anal/Endo-rectal Ultrasound (US)

- Passage of a 7 to 10-MHz transducer into the anal canal helps define the muscular anatomy to differentiate inter-sphincteric from trans-sphincteric lesions. A standard water-filled balloon transducer can help evaluate the rectal wall for any supra-sphincteric extension.
- This modality has *not* been used widely for routine clinical fistula evaluation.

Anal Manometry

- Pressure evaluation of the sphincter mechanism is helpful in certain patients.
- Decreased tone observed during preoperative evaluation may be due to:
 1. Previous fistulotomy or obstetrical trauma
 2. High trans-sphincteric or supra-sphincteric fistula (if known)
 3. Very elderly patients
- If reduced, surgical division of any portion of the sphincter mechanism should be avoided

Magnetic Resonance Imaging (MRI)

- Magnetic resonance imaging may be used in difficult cases.

Examination under Anesthesia (EUA)

- An examination of the perineum, digital PR and anoscopy are performed under anesthesia.
- This EUA is necessary before surgical intervention, especially if outpatient evaluation causes discomfort or has not helped delineate the course of the fistulous process.
- Several techniques have been described to help locate the course of the fistula and, more importantly, identify the internal opening.
 1. Hydrogen peroxide, milk, or dilute methylene blue (MB) is injected into the external opening & its egress at the dentate line is observed (The MB often obscures the field rather than helping to identify the opening).
 2. Traction (pulling or pushing) on the external opening may also cause a dimpling or protrusion of the involved crypt.
 3. Insertion of a blunt-tipped crypt probe via the external opening may help outline the direction of the tract. If it approaches the dentate line within a few millimeters, a direct extension likely existed. Avoid using excessive force and creating false passages.

Treatment

- *The Puborectalis is the key to future continence*
- *Low fistulas:* Lay open with either fistulotomy or fistulectomy
- *High fistulas:* Require two stage surgery
 1. Setons - loose or tight.
 2. Anorectal advancement flap (may be considered).

Fistula Lay-Open

- The main target of the operation is to lay open the tract from its external opening to the internal opening in order to allow granulation of the wound to close the defect created.
- The main problem is cut through a significant part of the EAS especially if the deep part is involved or more seriously the puborectalis, thus compromising continence.
- In these cases, the use of a *Seton* is an acceptable technique to avoid the occurrence of this dreadful complication

Seton Placement

- A Seton can be placed alone, combined with fistulotomy, or in a staged fashion. This technique is useful in patients with the following conditions:
 1. Complex fistulae (high Trans-sphincteric, supra-sphincteric, extra-sphincteric) or multiple fistulae.
 2. Recurrent fistulae after previous fistulotomy.
 3. Anterior fistulae in female patients.
 4. Poor preoperative sphincter pressures.
 5. Patients with Crohn's disease (CD) or patients who are immunosuppressed.

- *Setons* can be made from large silk suture, Silastic vessel markers, or rubber bands that are threaded through the fistula tract. They have 2 purposes beyond giving a visual identification of the amount of sphincter muscle involved; namely: (1) To drain and promote fibrosis and (2) To cut through the fistula.
- *Single-stage seton (cutting)*
 - o The Seton is passed through the fistula tract around the deep EAS after opening the skin, SC tissue, IAS muscle and subcutaneous EAS muscle.
 - o The Seton is tightened down and secured with a separate silk tie.
 - o With time, fibrosis occurs above the Seton as it gradually cuts through the sphincter muscles and essentially exteriorizes the tract.
 - o The Seton is tightened on subsequent office visits until it is pulled through over 6-8 weeks.
 - o A cutting seton can also be used without associated fistulotomy
- *Two-stage Seton (draining/fibrosing)*
 - o The Seton is passed around the deep portion of the EAS after opening the skin, SC tissue, IAS muscle & subcutaneous EAS muscle.
 - o Unlike the cutting Seton, the Seton is left loose to drain the inter-sphincteric space and to promote fibrosis in the deep sphincter muscle.
 - o Once the superficial wound is healed completely (2-3 m later), the Seton-bound sphincter muscle is divided.
 - o Two studies (74 patients combined) support the 2-stage approach with a 0-nylon Seton.
 - o Once wound healing is complete, the Seton is removed without division of the remaining encircled deep EAS muscle. Eradication of the fistula tract occurs in 60-78% of cases.

Mucosal Advancement Flap (MAF)

- The MAF is reserved for patients with chronic high fistula, but is indicated for the same disease process as Seton use.
- Advantages include a one-stage procedure with no additional sphincter damage.
- A disadvantage is poor success in patients with Crohn's disease or acute infection.
- This procedure includes total fistulectomy with removal of the 1st and 2nd tracts, in addition to complete excision of the internal opening.
- A rectal mucomuscular flap with a wide proximal base (2 times the apex width) is raised.
- The internal muscle defect is closed with an absorbable suture, and the flap is sewn down over the internal opening so that its suture line does not overlap the muscular repair.

Post-operative Complications

Early Post-Operative Complications

- Urinary retention
- Bleeding
- Fecal impaction
- Thrombosed hemorrhoids

Delayed post-Operative Complications

- Recurrence
- Incontinence (stool)
- Anal stenosis: The healing process causes fibrosis of the anal canal. Bulking agents for stool help prevent narrowing.
- Delayed wound healing: Complete healing occurs by 12 weeks unless an underlying disease process is present (ie, recurrence, Crohn's disease).

Complication	Fistulotomy	Seton	Mucosal Advancement Flap
Recurrence rate	0-18%	0-17%	1-10%
Fecal incontinence	3-7%	0-17%	6-8%.

Causes of recurrent fistula

1. Inadequate excision or missing side tracks
2. Specific infection e.g. TB.
3. Presence of active disease e.g. Crohn's disease.
4. Malignancy.
5. Epithelization of the track (with incomplete excision).

Future Perspectives

- Recent *biotechnology* has led to the development of many new tissue-adhesive materials.
- Reported series exist of *fibrin glue treatment* of fistula-in-ano, with one-year follow-up success rates approaching 60%. By its less invasive nature, this therapy leads to decreased postoperative morbidity. However, information on long-term success rates and modification of techniques will have to await further studies.

Summary: Ano-rectal Fistulae

- Anorectal fistulae are common and may be simple or complex
- Anorectal fistulae are classified according to their relationship to the anal sphincters
- They may be associated with underlying disease such as TB or Crohn's disease
- Laying-open is the surest method of eradication, but sphincter division may result in incontinence

Pilonidal Disease and Fissure-in-Ano

PILONIDAL DISEASE

Definition

- Common infection around hair containing sinuses in the natal cleft.
- *Pilus* means 'hair or tuft of hair', *nidus* means 'nest'.

Etiology

- *Congenital*: sinus of ano-coccygeal area
- *Acquired*: Most common, supported by :
 1. Inter-digital pilonidal sinus (PNS) in hair dressers
 2. The age incidence is between 20-29 years
 3. Hair follicles are demonstrated in the wall of the sinus
 4. The disease usually hairy men
 5. Recurrence is common

Pathology

- Abscess formation
- Sinus formation: the sinus extends into the subcutaneous planes. Branching channels are not infrequent with stratified squamous epithelial lining. Hair shafts are found lying loose in the sinus and embedded in granulation tissue. In 3/4 of the cases, foreign body giant cells are common.

Clinical Picture

1. *Acute presentation*
 - The patient presents with an abscess in the natal cleft.
2. *Chronic presentation*
 - Midline sinus at the first piece of coccyx
 - Tuft of hair projecting from its mouth
 - Discharge, mostly blood-stained, and contains foul sebum and sometimes hair.
 - Secondary openings may be present on either side of the midline.
 - Age: third decade
 - Gender: male: female = 4:1

Treatment

Conservative Treatment

- Cleaning out of the openings
- Shaving of hair
- Frequent washing by detergents and water
- Avoiding long sittings

Treatment of Acute Exacerbation (Abscess)

- Incision-drainage
- Broad spectrum antibiotics
- Instillation of pure phenol solution

Treatment of Chronic Sinus

- Local excision and healing by secondary intention (lay-open)
- Lateral drainage and midline pit excision (Buscom)
- Excision and closure
- Rhomboid excision and flap repair
- Karydakis procedure: sound closure to the side of midline.

Causes of Recurrence

- Over-looked diverticulum in the primary operation
- New hair enters the skin or the scar
- Deformity of natal fold by scarring (the least trauma causes tearing of the scar).

FISSURE-IN-ANO

Definition

- An anal fissure is an elongated ulcer in the long axis of the anal canal.

Location

- Midline posterior
- Midline anterior
- Lateral

Etiology

- Lack of support of the anal mucous membrane
- High anal pressure
- Trauma and ischemia
- Inflammatory bowel disease (IBD)
- Anal stenosis (post-operative)

Pathology

Acute Fissure

- It is a deep tear of anal margin skin extending to the dentate line
- The edges are inflammatory and edematous
- There is spasm of anal sphincter

Chronic Fissure

- It is characterized by more inflamed, indurated margins and a base consisting of scar tissue.
- A skin tag can be present at the inferior extremity of the ulcer called “sentinel pile”
- The muscle becomes contracted by infiltration of fibrous tissue.
- Chronic fissure-in-ano may have a specific cause e.g. Crohn’s disease.

Clinical Presentation

- Pain during defecation
- Bright red bleeding
- Constipation
- Puckered anus

Treatment

Conservative Treatment

- Glyceryl trinitrates --- nitric acid
- Laxatives
- Xylocaine ointment

Operative Treatment

- It is rarely needed for acute anal fissure
- For chronic anal fissure, anal sphincterotomy is done (lateral or dorsal sphincterotomy)

Malignancy of Colon, Rectum and Anus

COLO-RECTAL CANCER (CRC)

Incidence

- In Egypt, colorectal cancer (CRC) is the 4th most commonly diagnosed cancer in both men and women, with a median age of 48 years and a male to female ratio of 1.1.

Risk Factors for CRC

1. Hereditary CRC Syndromes
 - a. Adenomatous polyposis syndromes
 - Familial adenomatous polyposis (FAP)
 - MYH-associated polyposis (MAP)
 - b. Non-polyposis syndrome
 - Hereditary non-polyposis CRC (HNPCC)
 - c. Hamartomatous polyp syndromes
 - Peutz-Jeghers syndrome (PJS)
 - Juvenile polyposis syndrome (JPS)
 - Cowden disease (Bannayan-Ruvalcaba-Riley)
2. Inflammatory Bowel Disease (IBD): Ulcerative Colitis
3. Personal History (Hx) of CRC
4. Family Hx of CRC
5. Ethnic background: Ashkenazi Jews
6. Age: above 50 y in the Western community. In Egypt the median age of CRC is <50 y.
7. Environmental Factors
 - a. Increased Risk of CRC ↑ with a diet high in red meat & animal fat, and low-fiber diet (low overall intake of fruits & vegetables).
 - b. Lifestyle choices that are associated with ↑ risk for CRC include alcohol & tobacco consumption, obesity & sedentary habits

Factors Associated with Lower Risk Include:

1. Folate intake
2. Calcium intake
3. Estrogen replacement therapy

Genetics of CRC

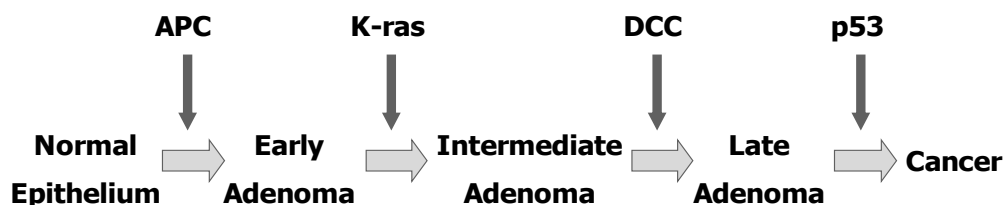
- Genetics does *not* mean inherited, genetics means *pathogenesis*. Genetically, CRC represents a complex disease, and accumulation of genetic alterations is associated with progression from normal epithelial cells or premalignant lesion (adenoma) to invasive adenocarcinoma. Colorectal cancers have several gene expression profiles with different pathogenic (genetic) pathways (Table 1).
- The major known mechanisms of genetic instability are *chromosomal* instability (CIN), which constitutes 85% of colon cancer and the other 15% is due to *microsatellite* instability (MSI) mismatch repair (MMR) pathway.

Table 1. Types of Colorectal Cancer (CRC)

Type	%
Sporadic	75
Family History of CRC	18
HNPCC	5
FAP	1
MAP	1
PJS & JP	<1
IBD	1

Pathway I CIN (Chromosomal Instability)

- Acquired (somatic) or inherited (germline) macrogenetic alterations at chromosomal level either in the form of abnormal number (aneuploidy) or gross changes i.e. loss of heterozygosity (LOH).
- The important genes involved in these chromosome losses are APC (5q), DCC (18q) and TP53 (17p). This pathway is seen in FAP and adenoma-carcinoma model.



Pathway II Microsatellite Instability (MSI) Mismatch Repair Pathway (MMR)

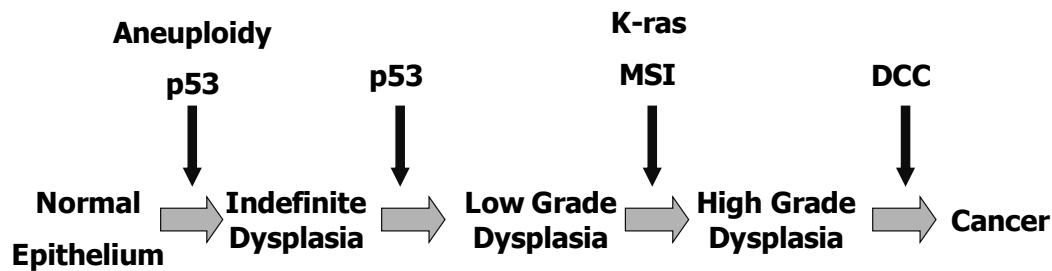
- Germline or somatic microgenetic (intragenic) mutations of DNA damage-repair genes result in microsatellite instability (MSI) i.e. insertion or deletion of nucleotides in one or more alleles. MSI means alterations in repeating units of DNA that occur normally throughout the genome.
- The important genes involved in this MSI are MMR genes (hMSH2, hMSH6, hMLH1, hMLH3, hPMS1, hPMS2)
- This pathway is seen in 95% of HNPCC and 18% of sporadic CRC.

Alternative Pathways

- Alternative pathways are involved in up to 30% of CRC, precursors not easily seen with rapid evolution following genetic instability. Alternative pathways include:
 1. Flat and depressed adenoma
 2. De Novo CRC
 3. Colitis induced CRC
 4. Serrated Pathway

Colitis-induced CRC

- This pathway is seen in ulcerative colitis and schistosomiasis Japonicum. Chronic inflammation results in free O₂ radicals that cause DNA damage (dysplasia-carcinoma model)



Serrated Pathway: CpG Island Methylator Phenotype (CIMP)

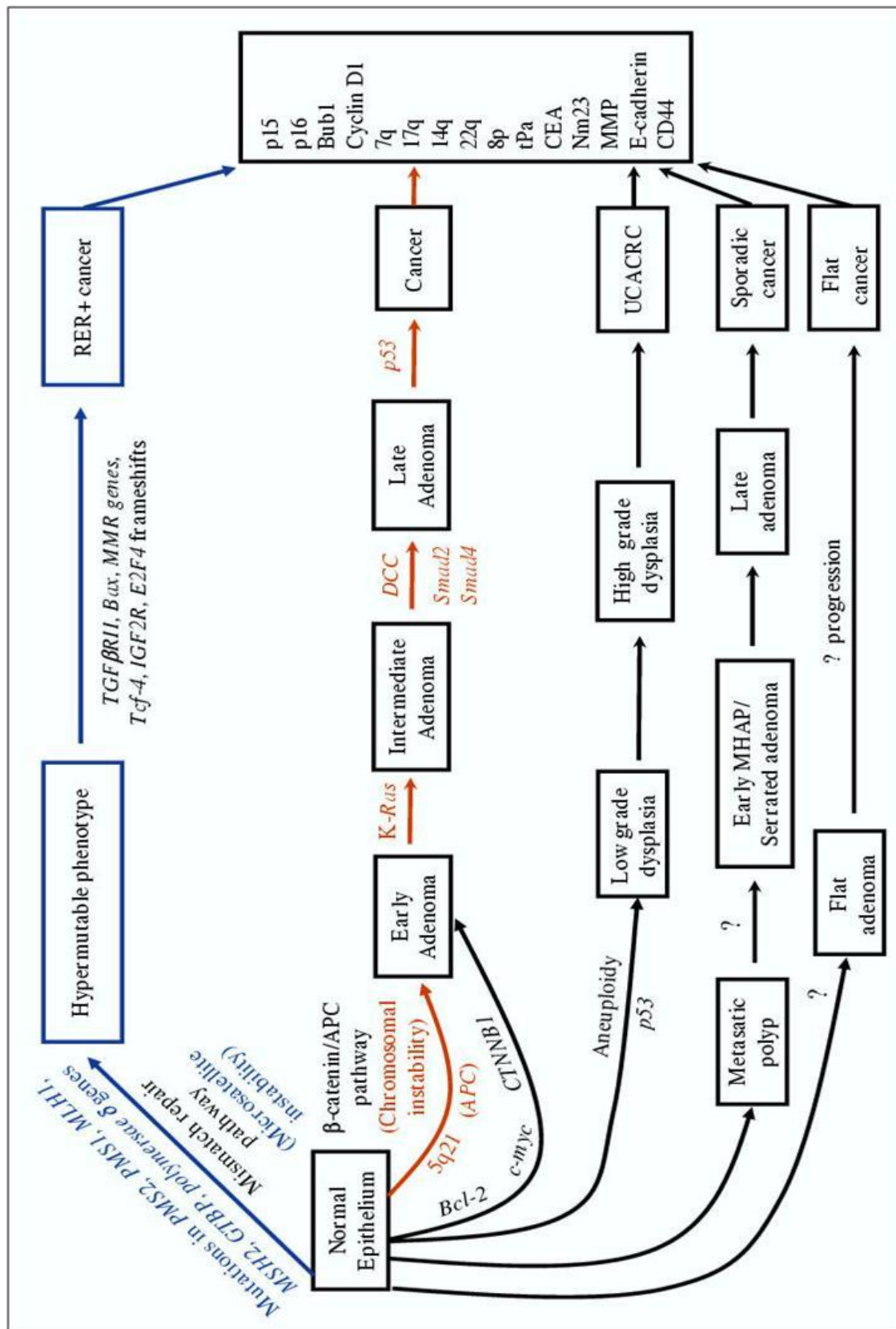
- Epigenetic factors are mechanisms outside the gene such as a cell's exposure to carcinogens or hormones, or genetic variations that modify a gene or its protein by methylation, demethylation, phosphorylation, or dephosphorylation. Genes that must be expressed in all tissues have unmethylated regions, called CpG islands. On the other hand, genes that must be turned off in differentiated tissues have these islands methylated.
- In this pathway type C tumor suppressor genes silenced by promoter region CpG methylation.
- This pathway is seen in juvenile polyposis, Peutz-Jegher's syndrome & serrated adenomas

Screening for CRC

- The life-time risk of CRC is summarized in Table 2.

Table 2: Life-Time Risk of CRC

Group	Life-time Risk of CRC
General Population	5%
Ashkenazi Jews	6-10%
Family Hx of CRC	10-15%
Personal Hx of CRC	15-20%
IBD	15-40%
PJS	50%
JP	50%
HNPCC	70-80%
FAP	>95%



High-Risk Group

- Patients with a 10-100% lifetime risk according to family history criteria, pathological criteria and pathogenic gene mutation.
- Screening Method: in Genetics Centre for formal counseling & mutation analysis

Moderate-Risk Group

- First-degree relative (FDR) with CRC aged <45 years or two FDRs:
 1. Single colonoscopy at age 55yrs.
 2. Polyps snared: If adenomatous, then adenoma surveillance
 3. Colon is clear of neoplasia: Reassured & shift to population risk fecal occult blood test (FOBT) screening.

Low-Risk Group

- No family history of CRC
- Population-based screening
 1. Annual (FOBT)
 2. Flexible Sigmoidoscopy / 5 years
 3. Colonoscopy / 10 years

Clinical Picture of CRC

- Patients with CRC may be asymptomatic (detected by screening programs mentioned above) or present with manifestations of the primary lesion, metastases or complications of the disease.
 1. Asymptomatic (detected by screening)
 2. Primary Lesion
 3. Metastases
 4. Complications

Clinical Picture (C/P) of Primary CRC

- The sites of primary CRC are demonstrated in Figure 1. As may be seen, the most common site is the rectum (38%) followed by the sigmoid colon (20%). The least commonly affected site is the appendix (0.5%) followed by the splenic flexure (2%)
- The most common presenting symptoms associated with CRC are abdominal pain, followed by change in bowel habits, rectal bleeding and occult blood in the stool. Table 3 and Figure 2 summarize the patient's symptoms according to the location of the tumor. Differences are shown in Table 4.

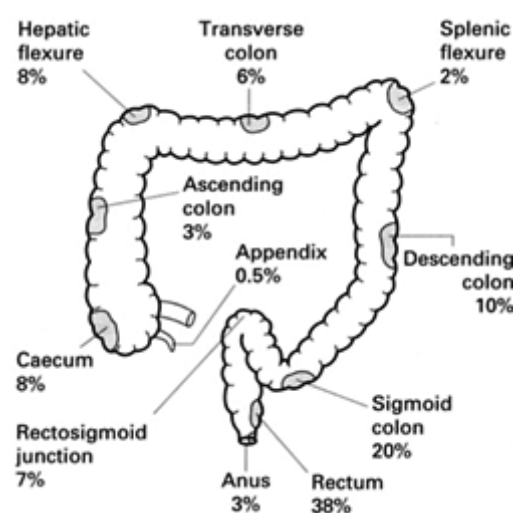
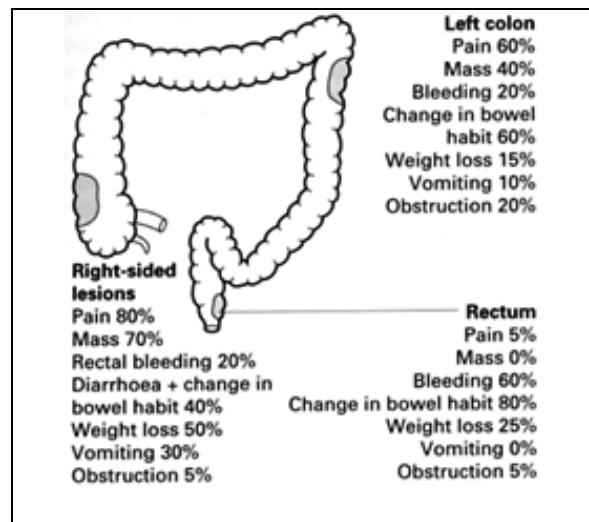


Figure 1. Sites of primary CRC

Table 3: Clinical presentation according to tumor location

Cancer Right Colon	Cancer Left Colon	Cancer Rectum
1. Change in bowel habits: Diarrhea 2. Vague upper abdominal pain 3. Iron deficiency anemia 4. Weight loss 5. Mass (advanced cases) 6. Obstruction: Rare	1. Change in bowel habits: Increasing constipation 2. Colicky pain with distension 3. Weight loss 4. Mass (advanced or fecal matter) 5. Obstruction: Common or 1 st presentation 6. Bleeding /rectum 7. Mucous/rectum	1. Bleeding /rectum 2. Mucous/ rectum 3. Tenesmus (sense of incomplete evacuation) 4. Change in bowel habits: Spurious morning diarrhea 5. Pain: Colicky pain in recto-sigmoid lesions. Severe pelvic pain in advanced cases due to infiltration. 6. Weight Loss 7. Obstruction: Rare, in recto-sigmoid lesions.

**Figure 2.** Clinical presentation of primary CRC according to location of the tumor**Table 4:** Comparison between symptoms according to tumor site

Symptom	Right Side	Left Side	Rectum
Change in bowel habits	Diarrhea	Constipation	Constipation spurious morning diarrhea
Change in bowel habits	Diarrhea	Constipation	Constipation Spurious morning diarrhea
Bleeding/rectum	Altered	Dark	Fresh
Colicky pain	+	++++	upper 1/3
Mass	+++	++	-
Mucous/rectum	-	+	++++
Tenesmus Incomplete evacuation	-	-	+++++
Weight loss	+++	+	++
Iron deficiency anemia	++++	+	+

Spread of CRC

1. Direct spread: (a) intramural (circumferential – longitudinal), and (b) transmural
2. Lymphatic spread
3. Hematologic spread: liver and lung metastases
4. Trans-peritoneal spread: Malignant ascites, Krukenberg's tumor, etc.

Complications of CRC

1. Intestinal obstruction
2. Penetration (fistula)
3. Bleeding (anemia or hypovolemia)
4. Intussusception: incomplete obstruction
5. Perforation: tumor perforation or cecal perforation caused by a left-sided colon cancer

Diagnosis (Dx)

An individualized approach to diagnosis considers the following:

1. Patient's symptoms
2. Age
3. Personal history of IBD, colon polyps, or CRC.
4. Family history of CRC or predisposing genetic syndromes (e.g., FAP or HNPCC).

Diagnostic Evaluation

- 1- *Colonoscopy*: Evaluation of the patient's entire colon and rectum for the presence of synchronous neoplasms. Colonoscopy allows biopsy (Bx) and histological confirmation of diagnosis. It also allows for identification and endoscopic removal of synchronous polyps. *Pre-operative histology should be done in all rectal tumors.*
- 2- Depending on the patient's age and health status, a variety of laboratory, radiologic, and cardiorespiratory tests may be appropriate to assess the patient's operative risk.

Pre-operative Assessment

- 1- *Pre-operative, carcinoembryonic antigen (CEA) level*. Beneficial for 2 reasons:
 - a. Post-operative return to normal of an elevated pre-operative CEA is associated with complete tumor resection, whereas persistently elevated values indicate the presence of visible or occult residual disease.
 - b. Elevated preoperative CEA level is an independent prognosticator of poor outcome. CEA is *not* useful as a screening tool as it increases in many conditions eg, CRC, proximal GIT cancers, lung & breast cancers, benign inflammatory conditions of the GIT, and smoking.
- 2- *Pre-operative CT scanning*: CT scans can be used to evaluate local extension of the tumor & regional lymphadenopathy, as well as for the presence of hepatic metastases.
- 3- *Preoperative chest x-rays or CT scan* to evaluate the lungs for evidence of metastatic disease.
- 4- PET–CT scan is *not* routinely indicated.

Staging of CRC

TNM Staging

T

Tx: Incomplete information

Tis: It involves only the mucosa

T1: Extends into the submucosa

T2: Extends into the muscularis propria

T3: Extends into subserosa

T4a: Penetrates visceral peritoneum

T4b: Tumor invades other organs or structures

N

Nx: Incomplete information

N0: No LN involvement

N1: In 1- 3 regional LNs.

- N1a: in 1 regional LNs

- N1b: in 2- 3 regional LNs

- N1c: Tumor Deposits in subserosa, mesentery, nonperitonealized pericolic or perirectal tissues without regional LNs metastasis

N2: In 4 or more regional LNs

- N2a: in 4-6 regional LNs

- N2b: in 7 or more regional LNs

M

M0: No distant Metastasis

M1: Distant Metastasis

- M1a: Metastasis confined to one organ or site (Liver, lung, ovary, non-regional LNs)

- M1b: Metastasis in more than one organ/site or peritoneum

Stage grouping of CRC and the 5-y SR according to stage are shown in Tables 5 and 6, respectively

Table 5. Stage grouping of CRC

Stage	TNM			Duke's
0	Tis	N0	M0	-
I	T1	N0	M0	A
	T2	N0	M0	A
IIA	T3	N0	M0	B
IIB	T4a	N0	M0	B
IIC	T4b	N0	M0	B
IIIA	T1-2	N1/N1c	M0	C
	T1	N2a	M0	C
IIIB	T3-4	N1/N1c	M0	C
	T2-T3	N2a	M0	C
	T1-T2	N2b	M0	C
IIIC	T4a	N2a	M0	C
	T3-T4a	N2b	M0	C
	T4b	N1-N2	M0	C
IVA	Any T	Any N	M1a	-
IVB	Any T	Any N	M1b	-

Table 6. 5-y-Survival Rate (SR) of CRC

Stage	5-Y-SR	
I	92%	
II	73%	
IIA		85%
IIB		72%
III	56%	
IIIA		83%
IIIB		64%
IIIC		44%
IV	8%	

Dukes Staging of Rectal Cancer

Dukes classified carcinoma of the rectum into three stages (Table 7)

Table 7: Dukes Staging (Figure 3)

Stage	Description
A	The growth is limited to the rectal wall (15%): Prognosis excellent
B	The growth is extended to the extra-rectal tissues, but no metastasis to the regional LNs (35%): Prognosis reasonable.
C	There are secondary deposits in the regional LNs (50%). These tumors are further subdivided into: - C1, in which the local para-rectal LNs alone are involved. - C2, in which the LNS accompanying the supplying blood vessels are implicated up to the point of division. This does not take into account cases that have metastasized beyond the regional LNs or by way of the venous system: Prognosis is poor.
D	<i>It is often included, which was not described by Dukes. This stage signifies the presence of widespread metastases, usually hepatic.</i>

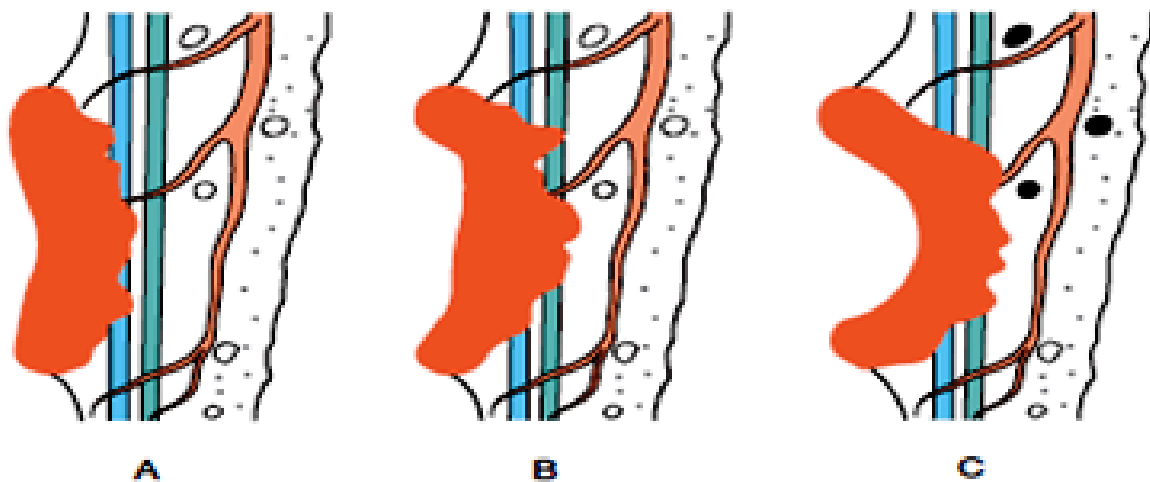


Figure 3. The three cardinal stages of progression of the neoplasm (After C. Dukes, FRCS).

Treatment of CRC

Preparation for Operation

A. Informed consent

All patients who are to undergo surgery for colon cancer need to be clearly informed of the reasons for and the extent of the proposed resection, the likely outcome of the surgery, the pertinent complications and their likelihood of occurring, expected length of hospitalization and recovery, alternatives to the proposed surgery and prognosis.

B. Mechanical bowel preparation

Despite its nearly universal use, the literature does *not* support a defined benefit for preoperative mechanical bowel preparation. The persistence in using it may be justified by the advantages it affords in ease of handling the prepared colon, the proven safety of the methods used for bowel cleansing and the relatively low cost. Outpatient bowel preparation (the day before surgery) is generally safe and cost effective.

C. Prophylactic antibiotics

Prophylactic antibiotics have proven effectiveness in decreasing the rate of infection, mortality and cost of hospitalization after colonic resection. Regardless which parenteral antibiotic regimen is selected, it must be given before the start of the operation.

D. Blood cross match & transfusion

The need for transfusion is based on the starting hemoglobin (Hgb), patient's physiologic status & extent of intra-operative blood loss. The immunosuppressive effect of transfusion increases the incidence of infection, but *not* the rate of cancer recurrence. Other factors (extent of resection, tumor location and experience of surgeon) in patients requiring transfusion may actually be the cause for the increased recurrence rate (RR).

E. Thrombo-embolism prophylaxis

All patients undergoing surgery for colon cancer should receive prophylaxis against thrombo-embolic disease as they have a high incidence of venous thrombo-embolism, including deep venous thrombosis (DVT) and pulmonary embolism (PE).

Operative Issues

A. Operative Technique

- The determinant of adequate bowel resection for colon cancer is resection of the 1st feeding arterial vessel and its corresponding lymphatics. Tumors located in border zones should be resected with the neighboring lymphatic regions to encompass both possible directions of spread. The length of bowel resected is usually governed by the blood supply to that segment. Ligation of the origin of the 1st feeding (Figure 4) vessel ensures the inclusion of the apical LNs, which may affect prognosis.
- There is much concern regarding intra-operative manipulation of the tumor with shedding of cancer cells into the portal circulation. However, the value of the “no touch” technique has *not* been proven, although there is a theoretic basis for its use.
- Operative procedures according to tumor location are summarized in Table 7

Table 7: Operative procedures according to site of the lesion

Site of the Lesion	Operation	Vessels	Excised Parts
Appendix, cecum, or ascending colon	Right hemicolectomy with ileo-transverse anastomoses (Figure 5)	- Ileocolic - Right colic - Right branch of middle colic	Cecum, appendix, ascending colon and right 1/3 of transverse colon
Hepatic flexure	Extended right hemicolectomy with ileo-transverse anastomosis	- Ileocolic - Right colic - Middle colic	Cecum, appendix, ascending colon and right 2/3 of transverse colon
Transverse colon	Transverse colectomy with anastomosis between the ascending and descending colons	Middle colic	Transverse colon including both hepatic and splenic flexures ± greater omentum in T4 lesions
Splenic flexure	Extended left Hemicolectomy with anastomosis between transverse and sigmoid colons	- Inferior mesenteric - Left branch of Middle colic	Descending colon and left half of the transverse colon
Descending colon	Left hemicolectomy with anastomosis between transverse and sigmoid colons or (rectum) (Figure 6)	- Inferior mesenteric	Descending colon and left third of the transverse colon ± sigmoid colon
Sigmoid colon	Sigmoidectomy with anastomosis between left colon and upper rectum (Figure 7)	- Inferior mesenteric	Sigmoid colon

Figure 4: Arterial supply of the colon and rectum

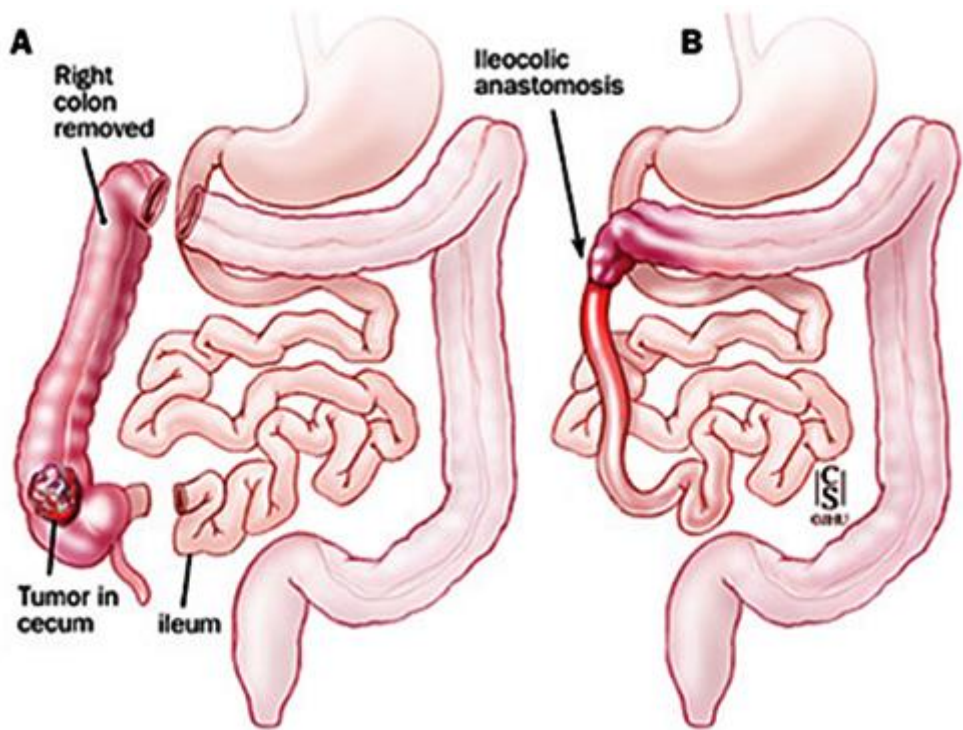
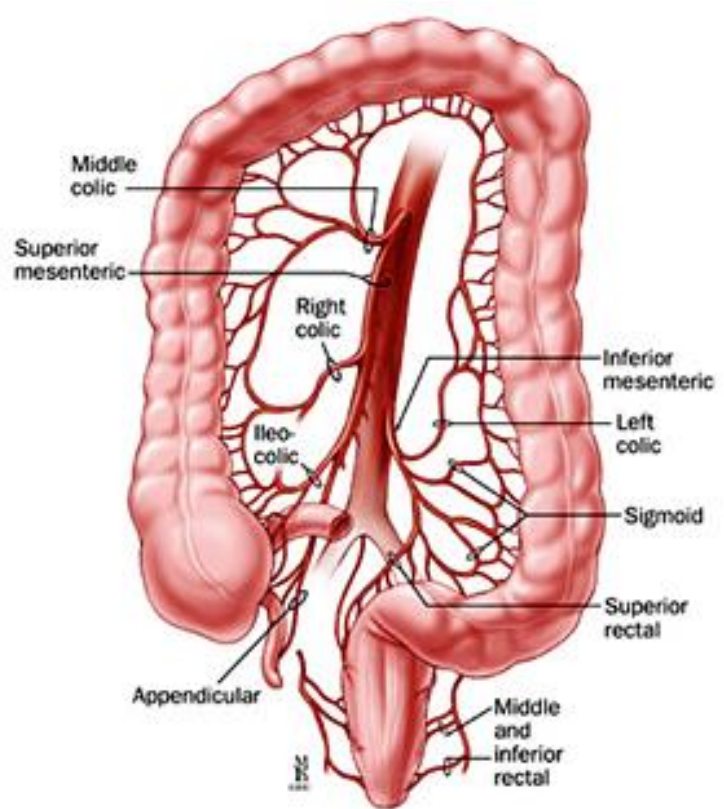


Figure 5. Right hemicolectomy with ileo-colic anastomoses for a tumor in the cecum

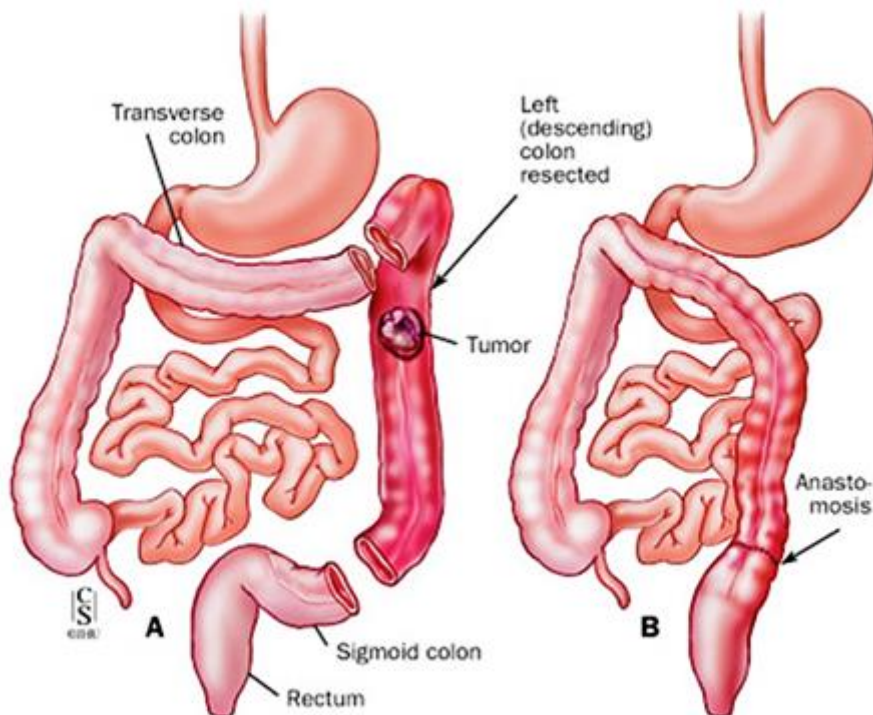


Figure 6: Left hemicolectomy with anastomosis between transverse and sigmoid colons for a tumor in the descending colon.

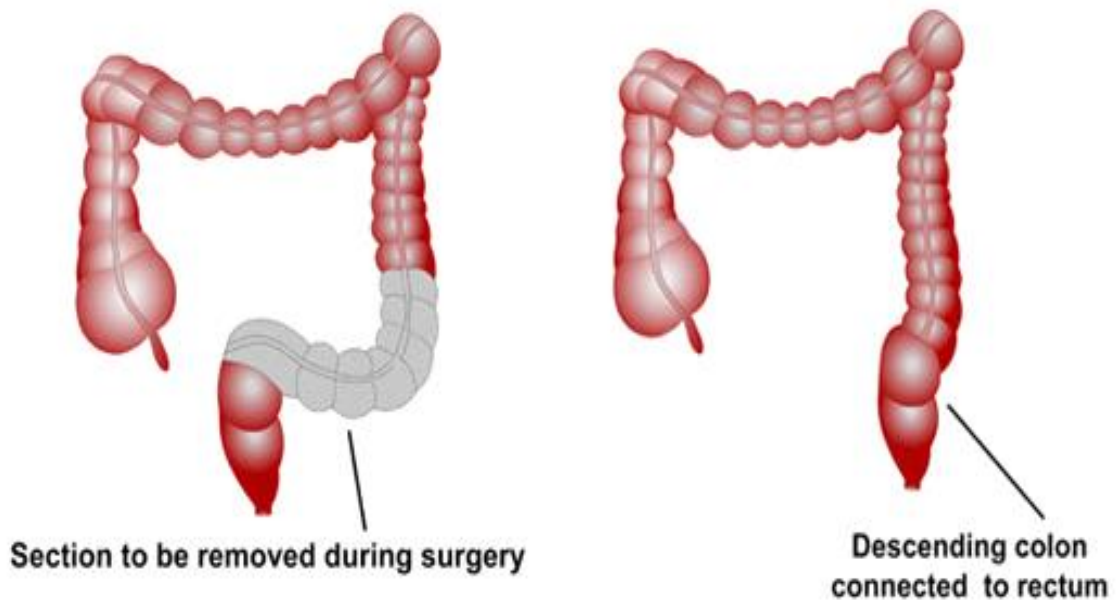


Figure 7: Sigmoidectomy with colo-rectal anastomoses for a tumor in the sigmoid colon.

B. Synchronous colon cancer

- Synchronous colon cancers can be treated by 2 separate resections or subtotal colectomy.

C. Contiguous organ attachment (15% of cases)

- Colon cancers adherent to adjacent structures should be resected en bloc. At the time of surgery, it often is impossible to distinguish between malignant & inflammatory adhesions.
- Because these adhesions harbor malignant cells in 40% of the time, an en bloc excision is necessary to achieve a tumor-free resection.

D. Synchronous resection of liver metastases (10-20% of cases)

- Resection (or ablation) of synchronous liver metastases should be performed at the time of the initial colon resection because it remains only means of obtaining long-term survival in this group of patients.
- Removal of the metastasis (*metastatectomy*) can proceed if the following conditions are met:
 1. Colon resection has proceeded with minimal blood loss or contamination
 2. The medical condition of the patient permits combining both procedures.
 3. Resection can be done with at least 1-cm margin
 4. The incision is appropriate for hepatic resection.
 5. The surgeon is comfortable with performing the hepatic resection.

E. Role of oophorectomy

- Bilateral oophorectomy is advised when one or both ovaries are grossly abnormal or involved with contiguous extension of the colon cancer. However, *prophylactic* oophorectomy is *not* recommended.

F. Role of laparoscopic resection

- Laparoscopic resection of colon cancer is feasible but requires specific surgical expertise. Adherence to oncologic principles is possible and adequate lymphadenectomy with disease-free margins can be achieved comparable to open surgery.

Operative Issues - Emergency

A. Obstructing colon cancer

- a. *Obstructing right or transverse colon cancer*: Right or extended right colectomy with a 1^{ry} ileo-colic anastomoses.
- b. *Left-sided colonic obstruction*: The procedure should be individualized.
 - Resection with end colostomy and Hartmann's pouch
 - Resection with on-table colonic lavage and 1^{ry} anastomosis
 - Subtotal colectomy with ileo-rectal anastomosis.
 - Insertion of colonic Stent to relieve the acute obstruction thereby permitting an elective colonic oral lavage, colonoscopy and then resection with 1^{ry} anastomosis.

The 3-stage approach of performing "proximal diversion, then resection, then colostomy closure" is less advantageous because of its high mortality and morbidity rates.

B. Colonic Perforation

The site of colonic perforation caused by colon cancer should be resected if at all possible.

- a. *Right-sided colon perforation* from a right colon cancer should be resected. If there is a free perforation with peritonitis, an anastomosis may be unwise & the patient is best left with an end ileostomy. The distal end may be brought out as a mucous fistula or stapled off as a Hartmann's pouch. Alternatively, if there is limited fecal spillage, the surgeon may choose to reanastomose the bowel $\pm$ fecal diversion.
- b. When a *left colon cancer perforates* resulting in peritonitis, a Hartmann's resection is mostly indicated. If there is massive proximal colonic distention and/or ischemia, a subtotal colectomy may be the best choice. If peritoneal contamination is limited, an ileo-rectal or ileo-sigmoid anastomosis with a diverting loop ileostomy may be done.
- c. *In the case of a right colon perforation caused by a left-sided colon cancer*, most experts advocate a subtotal colectomy. Whether an anastomosis or a loop ileostomy to protect the anastomosis is performed, it depends on the surgeon's judgment.

C. Massive Colonic Bleeding

- a. Acutely bleeding colon cancers that require emergent resection should be removed following the same principles as in elective resection. Because of the cathartic effect of the bleeding, the bowel has been effectively cleansed of the bulk of fecal matter and a 1^{ry} anastomosis can be considered. Whether to proceed with an anastomosis or an end stoma & mucous fistula (or Hartmann's pouch) is based on the surgeon's judgment.
- b. If the site of the bleeding cannot be identified, a subtotal colectomy is preferred.

Post-operative Staging of Colon Cancer

- It is important that accurate pathologic evaluation of the *radial margin of resection* be performed. Each operation is given a resection code to denote completeness of resection:
 - R0:** Complete tumor resection with all margins negative
 - R1:** Incomplete tumor resection with microscopic involvement of the margin
 - R2:** Incomplete tumor resection with gross residual tumor that was not resected.
- *Other factors* that have an impact on the patient's risk of recurrence and survival.
 1. Microscopic venous or lymphatic invasion within the specimen worsen the prognosis for every stage.
 2. Histologic grade & histologic type
 3. Serum CEA
- *Lymph node (LN) number*
 - To be properly evaluated, one should strive to have a minimum of 15 LNs examined microscopically.
 - The accuracy of colon cancer staging improves with increasing the number of LNs evaluated microscopically. Ten or more LNs can be found in 98% of colon specimens and 13 or more LNs can be found in 91% of specimens without using fat-clearing techniques.

Adjuvant Therapy

A. Chemotherapy

- Administration of 5-Fluorouracil (5-FU)/Leucovorin for 6 m post-operatively, in patients with stage III, may ↓ RR and ↑ SR.
- Patients with Stage II who are considered at higher risk for recurrence include those with one or more of the following: tumor perforation, adherence, or invasion of adjacent organs; non-diploidy by flow cytometry; poorly differentiated tumor; or venous, lymphatic and peri-neural invasion. These patients may receive adjuvant chemotherapy using *Capecitabine*, which is an *oral* Fluoropyrimidine Carbamate preferentially converted to 5-FU in tumor cells.

B. Immunotherapy

- Its value is undetermined and its use is recommended in the setting of clinical trials.

C. Intra-peritoneal/Intra-portal chemotherapy

- Intra-peritoneal and intra-portal infusion of chemotherapy is limited to clinical trials.

D. Radiation Therapy (RT)

- The role for RT in colon cancer is limited. It is rarely used in the treatment of colon cancer. Radiation's potential for injury to the abdominal viscera limits its usefulness.

Surveillance

1. History and Physical Examination: Every 3-6 months for 2 years, then every 6 month for a total of 5 years.
2. Serum CEA: Every 3-6 months for 2 years, then every 6 months for a total of 5 years.
3. CT scan of abdomen and pelvis: Annually for 3 years.
4. Colonoscopy: At one year, then as clinically indicated.

Management of Rectal Cancer

- Anatomically, the rectum is the distal 15-cm of the large bowel leading to the anal canal. *Thus, any cancer whose distal margin is 15 cm or less from the anal verge using a rigid sigmoidoscope should be classified as rectal.*
- Rectal cancer comprises about 25% of the malignancies arising in the large bowel.
- Cancers of the *intra-peritoneal* rectum behave like colon cancers regarding recurrence and prognosis.
- The *extra-peritoneal* rectum resides within the confines of the bony pelvis; it is this distal 10 to 12 cm that constitutes the rectum from the oncologic standpoint.
- Rectal cancer is a challenging disease because:
 1. It is a common malignancy
 2. Difficult dissection
 3. Treatment not infrequently entails a permanent stoma
 4. Sexual and urinary morbidity
 5. Local recurrence

Pre-operative Assessment

An individualized approach to the diagnosis considers:

- 1- Patient's symptoms
- 2- Age
- 3- Personal history of IBD, colon polyps, or CRC
- 4- Family history of CRC or predisposing genetic syndromes (eg, FAP and HNPCC).

Diagnostic evaluation

- 1- *Colonoscopy*: Evaluation of the patient's entire colon and rectum for the presence of synchronous neoplasms. Colonoscopy allows biopsy (Bx) and histological confirmation of Dx. It also allows for identification & endoscopic removal of synchronous polyps.
- 2- Depending on the patient's age & health status, a variety of laboratory, radiological and cardio-respiratory tests may be appropriate to assess the patient's operative risk.
- 3- When an *ostomy* is a consideration, preoperative counseling with an entero-stomal therapist should be offered when available.
- 4- *Digital rectal examination and rigid procto-sigmoidoscopy*
 - a. Digital rectal examination enables detection and assessment of the size and degree of fixation of mid and low rectal tumors.
 - b. Rigid procto-sigmoidoscopy and digital rectal examination allow the most precise assessment of tumor location and the distance of the lesions from the anal verge.
- 5- *Abdominal and pelvic CT scans* often used to evaluate local extension of the tumor, regional LNs and liver metastases. However, its role in local staging is limited.
- 6- *Trans-rectal ultrasound (TRUS)* is the *diagnostic modality of choice* for preoperative local staging of mid and distal rectal cancers. It may be more accurate in defining T1, T2.
- 7- *Pelvic MRI* is more accurate in assessing T3 and T4 lesions, offers a multiplanar & larger field of view of the mesorectal fascia and more accurately predicts obtaining a tumor-free circumferential resection margin.
- 8- *Preoperative routine chest X-ray or chest CT scanning*: Rectal cancer is more likely than colon cancer to be associated with lung metastases without liver metastases.
- 9- CEA level is most useful when it is elevated pre-operatively and then normalizes after tumor resection. Subsequent increase suggests recurrence or metastases. CEA is *not* used as a screening test.
- 10- PET-CT scan is *not* routinely indicated

Treatment Considerations

- Informed opinion of the multidisciplinary team (MDT) (radiologist, pathologist, medical oncologist and colorectal surgeon) should be obtained by the patient and family. The patient should have an access to treatment within 31 days of discussion with MDT.
- Surgery is the mainstay of treatment for rectal cancer. The risk of recurrence depends on TNM stage. Early stage cancer can be treated by surgical resection alone. More advanced lesions require adjuvant therapy to increase the probability of cure.
- The surgeon is a critical variable regarding morbidity, sphincter preservation as well as local recurrence.

Surgical Therapy

Resection Margin

- The proximal resection margin is determined by blood supply considerations. For upper third rectal cancers both the rectum and meso-rectum are divided *not* <5 cm below the lower margin of the tumor.
- A 2-cm distal margin is adequate for most low rectal cancers. In smaller cancers of the low rectum without adverse histologic features, a 1-cm distal margin is acceptable.

Operative Procedures

- Operative procedures advocated according to rectal tumor locations are listed in Table 8.

Table 8: Operative procedures according to tumor location

Site of the Lesion	Operation	Vessels	Excised Parts
Upper 1/3 rectum	Tumor specific meso-rectal excision (Figure 8) with anastomosis between descending colon & middle 1/3 rectum. Anastomosis above the peritoneal reflection = anterior resection, while below it = low anterior resection (LAR)	Inferior mesenteric	Sigmoid colon and upper third rectum with distal safety margin of 5 cm
Middle and lower 1/3, Rectum down to > 2 cm from the dentate line	Total meso-rectal excision (TME) (Figure 9) with anastomosis between descending colon & anal canal (colo-anal anastomoses = ultra-LAR)	- Inferior mesenteric - Middle rectal	Sigmoid colon & total proctectomy with distal safety margin at least 2 cm
< 2 cm from the dentate line or lesion infiltrating the anal sphincter or the pelvic floor muscles	TME with abdomino-perineal excision of the rectum (APER) with permanent colostomy (Figure 10).	- Inferior mesenteric - Middle rectal - Inferior rectal	Sigmoid Colon, total proctectomy, anal canal, anal sphincters and pelvic floor muscles

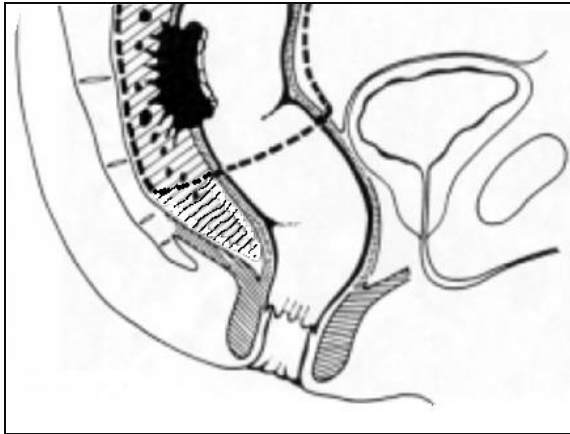


Figure 8. Tumor Specific Meso-rectal Excision (TSME)

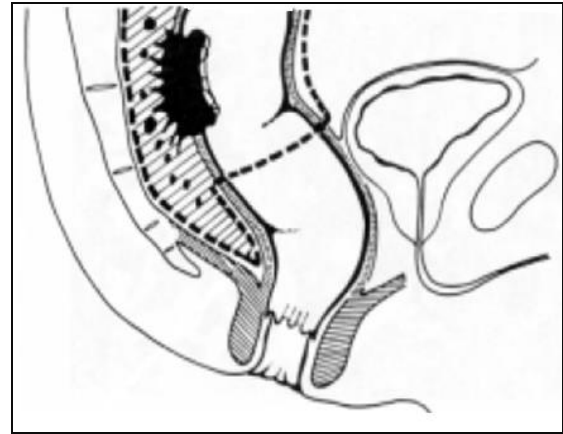


Figure 9. Tumor Meso-rectal Excision (TME)

Level of proximal vascular ligation

- Proximal lympho-vascular ligation at the origin of the superior rectal artery is adequate for most rectal cancers. There is no demonstrable survival advantage for a high ligation of the inferior mesenteric artery (IMA) at its origin.
- In patients with LNs thought to be involved clinically, removal of all suspicious nodal disease up to the origin of IMA is recommended.
- High ligation of the inferior mesenteric vessels may be helpful to provide additional mobility of the left colon, as often is required for a low colorectal anastomosis.



Figure 10. TME with APER

Circumferential resection margin

- For distal rectal cancers, TME is recommended. For upper rectal cancers, TSME is adequate.
- The meso-rectum is the fatty tissue that encompasses the rectum. It contains lympho-vascular and neural elements. Surgical excision of the meso-rectum is accomplished by sharp dissection in the plane between the fascia propria of the rectum and the pre-sacral fascia. Radial clearance of meso-rectal tissue enables the en-bloc removal of the 1st rectal cancer with any associated lymphatic, vascular, or perineural tumor deposits. TME is associated with the lowest local recurrence rates.
- Pathologic assessment of rectal cancer specimens suggests that distal meso-rectal spread may occur up to 4 cm away from the 1st tumor. Thus, a cancer in the distal rectum should be treated with a TME in most cases. Upper rectal cancers may be treated with a TSME with distal safety margin of 5 cm.

- Circumferential margin involvement in the presence of an intact mesorectal specimen is a strong predictor for local recurrence. A margin of ≤ 2 mm between tumor and the mesorectal fascia was considered positive and is associated with a higher local recurrence rate. Furthermore, patients who had a margin ≤ 1 mm had an increased risk of distant metastases

En Bloc Resection of Adherent (T4) Tumors

- Rectal cancers with adjacent organ involvement should be treated by en bloc resection.
- Tumors may be adherent to adjacent organs by malignant invasion or inflammatory adhesions. Locally invasive rectal cancer (T4) is removed by an en-bloc resection to include any adherent tissues. If a tumor is transected at the site of local adherence, resection is deemed incomplete, because it is associated with a higher incidence of local recurrence rate.

Inadvertent Perforation

- It worsens oncologic outcome and should be documented.
- It must be considered in postoperative adjuvant treatment decisions and outcome measurements.

Other Operative Considerations

1. Role of oophorectomy

Bilateral oophorectomy is advised when one or both ovaries are grossly abnormal or involved with contiguous extension of the colon cancer. However, prophylactic oophorectomy is *not* recommended.

2. Laparoscopic-assisted resection of rectal cancer

It is feasible but requires specific surgical expertise.

3. Emergency intervention:

Hemorrhage, obstruction and bowel perforation are the most common indications for emergency intervention for rectal cancer. Appropriate treatment must be individualized with options, including resection with anastomosis and proximal diversion, or diversion alone followed by radiation. Self-expandable metallic stents can be used to relieve obstruction by a proximal rectal cancer.

Pre-operative Clinical Staging

1. T1-2, N0
2. T3, N0 OR T any, N1-2
3. T4 &/or locally unresectable primary
4. T any, N any, M1 resectable metastases
5. T any, N any, M1 unresectable metastases or medically inoperable

Treatment of T1-2, N0, M0

Surgery

A. Trans-anal Resection (conventional resection or trans-anal microsurgery)

Full thickness excision is performed through the bowel wall into the peri-rectal fat.

Indications include:

1. T1N0
2. Small (<3 cm)
3. Grade 1-2
4. within 8 cm of AV
5. < 30% of rectal circumference
6. Negative (> 3 mm) deep and mucosal margins
7. No Tumor fragmentation

According to histopathology of the resected specimen, either Surveillance would be sufficient or further radical surgery might be indicated

B. *Trans-abdominal radical resection* is indicated in unfavorable histological features:

1. LVI + PNI+
2. Grade 3-4
3. Positive Margins
4. T2

C. *Trans-abdominal radical resection*:

1. Mid to upper rectum: LAR 5 cm below distal edge of tumor + colorectal anastomosis
2. Low rectal lesions
 - a. Tumor within 2 cm above the dentate line: APER or ISRR if Ta and T2
 - b. Tumor above 2 cm from the dentate line:
 - i. TME with colo-anal anastomosis (CAA)
 - ii. Spare autonomic nerves

Adjuvant Chemotherapy

- If the pathology review after trans-abdominal resection revealed T3N0 or T1-3/N1-2
- Regimens over 6 months
 1. 5-FU +/- LV or FOLFOX or Capecitabine (1 Cycle)
 2. Concurrent 5-FU/RT or Capecitabine/RT
 3. 5-FU +/-LV or FOLFOX or Capecitabine (2 Cycles)

Treatment of T3, N0 / T any, N1-2, M0

- *Pre-operative (Neoadjuvant) therapy is a MUST:* Continuous 5-FU/RT or bolus 5-FU/LV/RT or Capecitabine/RT
- *Operative treatment:* Trans-abdominal approach *only*, performed 5-10 weeks after completion of neoadjuvant therapy
- *Postoperative adjuvant therapy:* 5-FU +/- LV or FOLFOX or Capecitabine. All patients receiving preoperative (Neoadjuvant) therapy should receive postoperative adjuvant therapy for 6 months.

Treatment of T4 and/or Locally Unresectable Primary

Resectable case = anticipated negative (R0) microscopic circumferential resection margin

- *Pre-operative (Neoadjuvant) therapy:* Continuous 5-FU/RT or bolus 5-FU/LV/RT or Capecitabine/RT.
- *Operative treatment:* More radical (multi-organ) approach, complete removal of tumor & involved viscera with R0 margins.

- *Post-operative adjuvant therapy:* 5-FU +/- LV or FOLFOX or Capecitabine. All patients receiving pre-operative (Neoadjuvant) therapy should receive postoperative adjuvant therapy for 6 months.

Treatment of T any, N any, M1 Resectable Metastases

1. Combination chemotherapy:
 - a. FOLFOX/FOLFIRI/capecOX +/- Bevacizumab
 - b. FOLFIRI/FOLFOX +/- Cetuximab (KRAS wild-type gene only)
2. Staged or synchronous resection of metastases and rectal lesion
3. Adjuvant Therapy:
 - a. Continuous IV 5FU/ RT
 - b. Bolus 5FU+leucovorin/ RT
 - c. Capecitabine/RT

Treatment of T any, N any, M1 Unresectable Metastases or Medically Inoperable

1. Role of palliative surgery: Only if obstruction, perforation or bleeding
2. Symptomatic R/
 - a. Chemotherapy alone
 - b. Combined modality: 5-FU/RT or Capecitabine / RT.
 - c. Resection of involved rectal segment + CT
 - d. Intra-luminal Stent + CT
 - e. Diverting colostomy + CT

Pathological Evaluation of Rectal Cancer

1. Gross: Tumor and Specimen
2. Grade
3. LVI and PNI
4. T Stage
5. Number of regional LNs evaluated
6. N Stage: Number of +ve LNs
7. M Stage: Other organs – peritoneum – nonregional LNs.
8. Proximal, distal, and CRM
9. “p” = pathologic staging
10. “yp” = pathologic staging following neoadjuvant therapy

Surveillance

1. History and physical examination: Every 3-6 months for 2 years, then every 6 months for a total of 5 year.
2. CT scan of abdomen and pelvis Annually for 3 years
3. Colonoscopy At one year then as clinically indicated

Adjuvant Therapy

Neoadjuvant Therapy (pre-operative chemoradiation) should be offered to patients with stage II and III rectal cancers.

- It is given as long course (45-50 Gy in 25-28 fractions)
- 5-FU based chemotherapy should delivered concurrently with radiation
- All patients receiving preoperative (Neoadjuvant) therapy should receive postoperative adjuvant therapy for 6 m

Rationale for neoadjuvant therapy

1. It avoids irradiation of neo-rectum & small bowel
2. It provides early systemic therapy i.e. less burden of disease and better drug perfusion (non-surgically manipulated - well vascularized oxygenated tissue).
3. It acts as a Radio-sensitizer: It thus enhances rates of down-staging and pathological complete response (ypCR)

Treatment according to site is summarized in Table 9.

Table 9: Treatment of rectal cancer according to "T" and 'tumor location'.

Site	At DL	< 2cm DL	> 2 cm DL	Mid-Rectum	Upper Rectum
Pedunculated Polyp LVI-, G1-2	Polypectomy and observe				
Sessile Polyp or LVI+, G3-4	Radical Surgery				
T1, LVI-, G1-2	SCRT +APR	APR/ CRT+ISRR	Local Excision	Local Excision/ TME	TSME
T1, LVI+, G3-4, or T2	SCRT+ APR	APR/ CRT+ISRR	TME	TME	TSME
Good T3	CRT + APR	CRT+APR	TME	TME	TSME
Bad T3	CRT+ APR	CRT+APR	CRT+TME	CRT+TME	CRT+TSME
T4a	CRT+APR	CRT+APR	CRT+TME	CRT+TME	CRT+TSME
T4b	CRT+APR	CRT+APR	CRT+TME	CRT+TME	CRT+TSME

NEOPLASTIC DISEASE OF THE ANAL CANAL

Pathology

- These tumors are classified into squamous (SCC), basaloid, or muco-epidermoid types. It is almost agreed that the dentate line separates the anal margin (the area below) from the anal canal (the area above up to the ano-rectal ring).
- SCC (30%) (keratinizing or not) tend to arise from skin of anal margin. On the other hand *basaloid* cell tumor (50%) (cloacogenic cell tumor) arise from the anal canal (transitional zone above the dentate line).
- The *muco-epidermoid* type is rare, and it produces both keratin and mucus.

Spread

- *Direct spread*, usually towards the rectum in an upward direction in the submucous layer.
- *Lymphatics* can spread to both inguinal & towards lymphatics of the rectum
- *Hematogenous*: Very rarely it can give blood borne metastasis

Clinical Picture

- The presence of the ulcer, with all its consequent symptoms of pain, discharge, itching etc is complained of.
- Inspection and PR examination will reveal the diagnosis.

Management

- It usually start with *biopsy of the lesion*, which when proved to be malignant is treated recently with NEGRO regimen; a pre-operative course of chemo-radiotherapy, which can be followed by either, local excision, no surgery or abdomino-perineal resection according to the response to the initial treatment.
- *Inguinal LNs* are only dissected as a block excision 4-6 weeks after anal lesion treatment if they proved to be involved

Colo-rectal Emergencies

VOLVULUS

- The word volvulus derives from the Latin 'to twist'. A volvulus is the twisting of a loop of intestine around its mesenteric attachment, resulting in a closed loop bowel obstruction. The affected bowel can become ischemic due to a compromised blood supply, rapidly leading to bowel necrosis and perforation.
- Most volvuli occur at the sigmoid colon and are a common cause of large bowel obstruction in many countries (after malignancy and diverticular disease). They can also occur at the stomach, cecum, small intestine, and transverse colon, but are much rarer.
- The long mesentery of the sigmoid colon (which increases with age) means that this segment bowel is more prone to twisting on its mesenteric base (Figure 1) to form a volvulus than any other region.

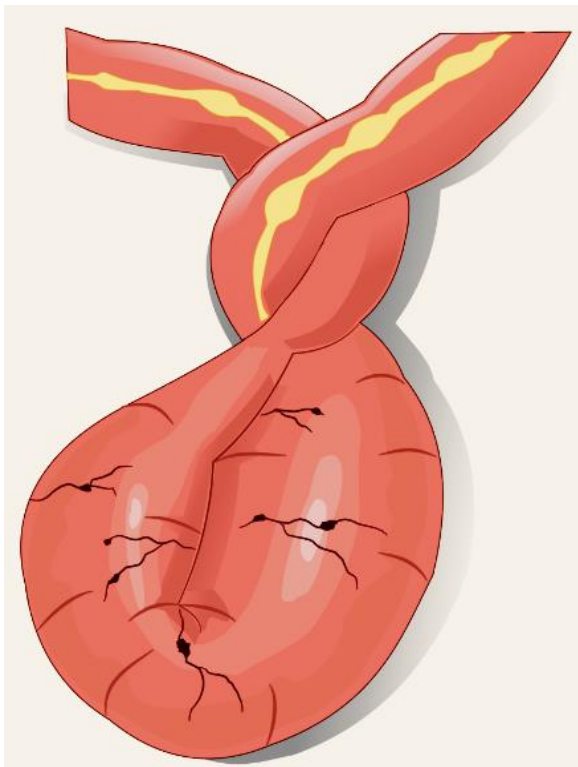


Figure 1: A volvulus occurs when bowel twists on its own mesentery

Etiology

A long narrow based mesentery predisposes to volvulus.

1. Congenital anomaly of long and mobile mesentery (neonatal midgut volvulus).
2. Acquired:
 - Mesenteric adhesions producing narrowing of the base, or causing fixation of the loop at its top
 - Overloading of one limb of the gut as in sigmoid colon

Pathogenesis of Volvulus Sigmoid

- Anti-clockwise torsion of 180° is physiological and can be reverted spontaneously. If the posterior loop gradually fills with gas and stools, and become heavier, it then changes its position and becomes anterior, thereby completing the 360° anti-clockwise torsion as shown in Figure 2.

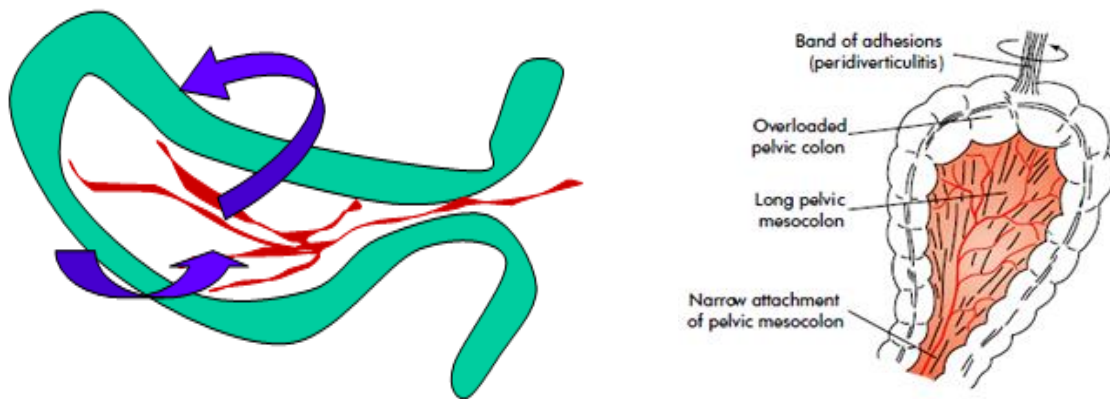


Figure 2: Pathogenesis of volvulus of the sigmoid

Risk Factors

The risk factors for developing a volvulus include:

- Increasing age
- Neuropsychiatric disorders
- Chronic constipation or laxative use
- Male gender
- Previous abdominal operations

Clinical Features

- Patients with a volvulus will present with the clinical features of bowel obstruction.
- As the sigmoid colon is located distally in the gastro-intestinal tract (GIT), vomiting is usually a late sign, whilst the colicky pain, abdominal distension, and absolute constipation occur earlier in the clinical course.

- Particularly noteworthy in cases of volvulus, compared to other causes of bowel obstruction, is the rapidity of onset (over a few hours) and degree of abdominal distension.
- On physical examination, the abdomen is often very tympanic to percussion. Ensure to examine for signs of perforation or peritonism, as this indicates ischemia or perforation and is a surgical emergency.

Differential Diagnosis (DD)

The main DD to consider are the alternative causes of bowel obstruction, as well as severe constipation, pseudo-obstruction, and severe sigmoid diverticular disease.

Investigations

All patients with clinical features of bowel obstruction should be investigated accordingly.

- *Routine blood tests* should be taken, including electrolytes, calcium, and thyroid function tests (TFTs) to exclude any potential pseudo-obstruction.
- *CT scan abdomen-pelvis* with contrast, as this is much more sensitive and specific for bowel obstruction and identifies the site and cause. CT imaging classically will demonstrate a severely dilated sigmoid colon with a ‘*whirl sign*’, from the twisting mesentery around its base (Figure 3).

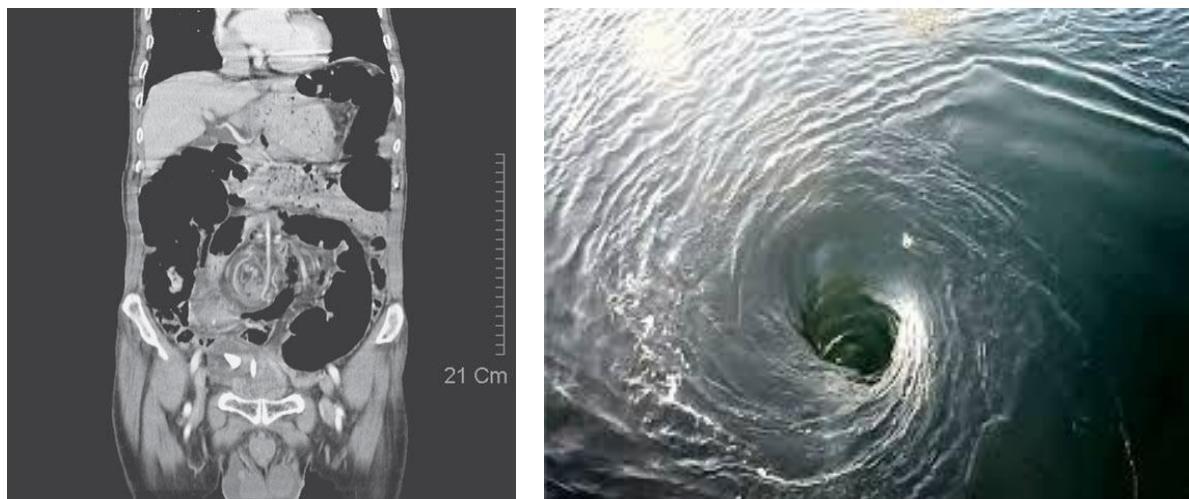


Figure 3: CT scan reveals characteristic mesenteric whirl sign.

- *Abdominal radiograph* (plain X-ray) will classically show (in around 60-75% cases) a “*coffee-bean sign*” arising from the left iliac fossa; if the ileocecal valve is incompetent, the abdominal X-ray will also show signs of small bowel dilatation (Figure 4).



Figure 4. Plain X-ray abdomen: characteristic coffee bean appearance, with the convexity of the loop lying in the right upper quadrant (opposite the site of obstruction).

- *Gastrografin enema* shows a narrowing at the site of the volvulus and a pathognomonic ‘bird’s beak’ sign (Figure 5)



Figure 5. Gastrografin enema shows ‘bird’s beak’ sign.

Management

All patients admitted with suspected sigmoid volvulus should be managed initially as per any patient with bowel obstruction. They should be examined for any signs of ischemia and given fluid resuscitation.

Conservative Treatment

- Most patients with sigmoid volvulus are treated conservatively initially with decompression by sigmoidoscope and insertion of a rectal tube (Figure 6).

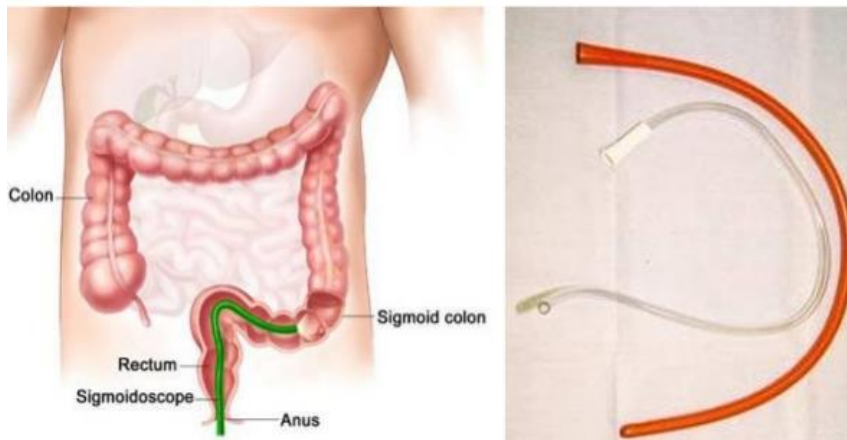


Figure 6.
sigmoidoscope (left)
and rectal tube (right)

- In *sigmoidoscope decompression*, the patient is placed in the left lateral position and a lubricated sigmoidoscope gently guided into the rectum. It is maneuvered to locate the twisted bowel and once the sigmoidoscope is in the correct position, there will be a rush of air and liquid feces as the obstruction is relieved.
- A *flatus tube* is often left in-situ for a period (up to 24 hours) after initial decompression to allow for the continued passage of contents and aid recovery of the affected area. In 25-50% of patients, this approach is unsuccessful and a formal decompression with a flexible sigmoidoscope is required.

Surgical Management

- The indications for surgery (which is usually a laparotomy for a Hartmann's procedure) are:
 1. Colonic ischemia or perforation
 2. Repeated failed attempts at decompression
 3. Necrotic bowel noted at endoscopy.
- The decision on which operation to perform will depend on the patient's nutritional status, adequacy of blood supply, hemodynamic stability, and the presence of any perforation or peritonitis.
- Patients with recurrent volvulus who are otherwise healthy may choose to have an elective procedure (most commonly sigmoidectomy with primary anastomosis) to prevent further recurrence.

Complications

- The main immediate complication of a sigmoid volvulus is bowel ischemia and perforation. Longer term complications are mainly the risk of recurrence (occurring in up to 90% of patients) and complications from a stoma if placed.

- Overall mortality from surgery in these patients is high as they are generally old, frail and with multiple co-morbidities. There is also often a delay in getting these patients to theatre which contributes to their physiological compromise and high mortality.

Cecal Volvulus

- The second most common site for a volvulus to occur is at the cecum (Figure 7), accounting for around 25-40% of all colonic volvulus. There is a bimodal age of onset, in the 10–29-year group and then again in the 60-79-year group.
- Those in the younger group may have intestinal malformation or excessive exercise as the predisposing cause, whilst in older patients it is more associated with chronic constipation, distal obstruction, or dementia.

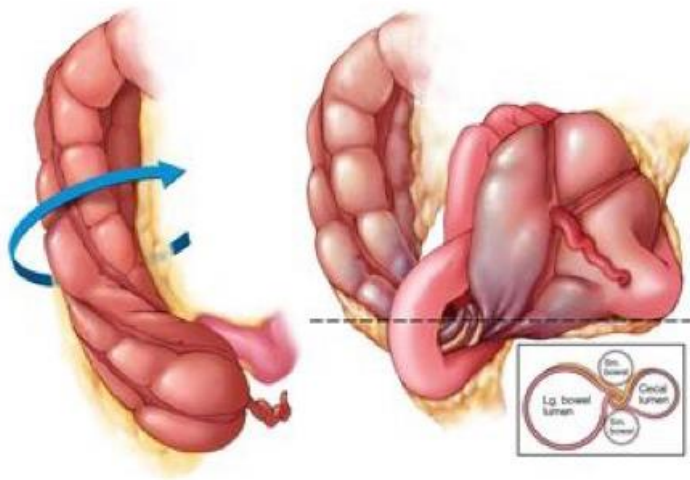


Figure 7. Cecal volvulus is second most common site of volvulus

- Diagnosis is once again made on CT imaging, which shows a distended cecum, a mesenteric “swirl” and small bowel obstruction.
- The management of a cecal volvulus is always laparotomy and ileo-cecal resection (assuming the patient is well enough to tolerate the surgical procedure)).

ISCHEMIC COLITIS

Ischemic colitis (colonic ischemia): hypoperfusion of the large bowel, which is mostly transient and self-limiting (non-gangrenous form), but can also lead to severe acute ischemia with bowel infarction (gangrenous form)

Types

1. *Acute mesenteric ischemia*: acute inadequate blood flow to the intestine (arterial or venous) that can result in bowel infarction.
2. *Chronic mesenteric ischemia*: constant or episodic hypoperfusion of the small intestine, usually due to atherosclerosis.

Epidemiology

- Ischemic colitis is the most common form of intestinal ischemia.
- Mainly occurs in adults > 60 years.
- Mild, non-gangrenous form occurs in 80–85%.

Etiology

- It is usually caused by transient hypoperfusion
- Thrombo-embolism
- Hypotension, hypovolemia (e.g., sepsis, dehydration, hemorrhage)
- Cardiovascular surgery (especially aortic repairs or cardiac bypass)
- Vaso-constrictive drugs
- Thrombophilia (e.g., anti-phospholipid syndrome)
- Colonic obstruction from, e.g., tumors, adhesions

Pathophysiology

- Intestinal blood flow of the superior mesenteric artery (SMA) and/or inferior mesenteric artery (IMA) is suddenly compromised → intestinal hypoxia → intestinal wall damage → mucosal inflammation and possibly bleeding → may progress to infarction and necrosis (gangrenous type) → disruption of mucosal barrier and perforation → release of bacteria, toxins, vasoactive substances → life-threatening sepsis.
- The splenic flexure and the recto-sigmoid junction are at high risk for colonic ischemia because they are 'watershed areas'.

Clinical Features

Typically presents with 3 clinical stages.

1. *Hyperactive phase:*

- Sudden onset of crampy abdominal pain (usually left lower quadrant)
- Bloody, loose stools
- Most patients recover and do not progress beyond this phase.

2. *Paralytic phase:*

- Pain more diffuse
- Bowel sounds become absent.
- Bloating
- Bloody stools.

3. *Shock phase:*

- Acute abdomen with abdominal guarding and rebound tenderness.
- Signs of septic shock.

Diagnosis

Laboratory Studies

1. No specific laboratory findings in mild ischemic colitis
2. In severe ischemic colitis:
 - ↑ Lactate, ↑ LDH, ↑ creatine kinase
 - Leukocytosis
 - Metabolic acidosis

Imaging

- *Plain abdominal radiograph*: insensitive, unspecific (air-filled, distended bowel), but helps exclude other disorders (Figure 8)

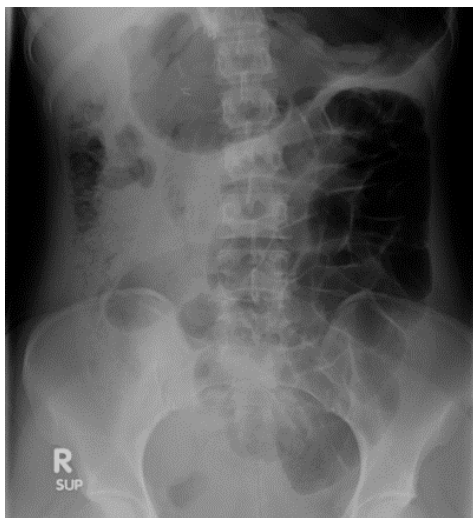


Figure 8. Plain X-ray abdomen air-filled, and distended left colon

- *Selective angiography* can demonstrate blood flow in celiac trunk and mesenteric arteries (Figure 9)



Figure 9. Mesenteric Angiogram. Note complete lack of contrast in mesenteric vessels in AP view (left). The occluded origins of the celiac axis and superior mesenteric artery are demonstrated in the lateral view (right).

- CT scan with oral contrast may show increase in wall thickness, pneumatosis intestinalis, and edematous thickening of the mucosa causes indentations in the large bowel wall (*Thumbprint sign*). (Figures 10 and 11)



Figure 10. CT scan: increase wall thickening, pneumatosis intestinalis

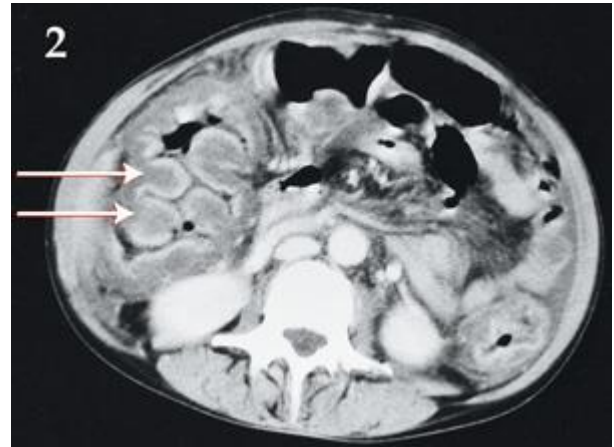


Figure 11. Thumbprint sign

Colonoscopy

- A pale mucous membrane and a loss of vascular pattern can be seen.

Treatment

Mild to Medium-Severe Forms

- Supportive care (IV fluids, bowel rest, nasogastric tube in case of an ileus)
- Anti-platelet drugs
- Reduce risk of atherosclerosis (anti-coagulation therapy)

Severe Forms (Signs of Peritonitis, Sepsis):

Surgical intervention

- Pre-operative preparation
- Assure adequate resuscitation.
- Monitoring
- Foley Catheter
- Urgent exploration
 1. Surgical embolectomy.
 2. Bowel resection.

INJURY (TRAUMA)

Etiology

Penetrating Trauma

- This is the most common type of trauma seen and is usually due to:
 1. High-velocity missiles (75%) - stab wounds (20%).
 2. Iatrogenic injuries due to uterine perforation during curettage of the uterus, use of endoscopies whether diagnostic or during polypectomies, or other rectal instrumentations and/or during surgical operations for urological or gynecological operations.
 3. Rectal impalement injuries that result when a patient falls on a penetrating object or using foreign bodies for sexual satisfaction.

Blunt Trauma

- The commonest cause is motor vehicle accidents followed by falls and crush injuries.
- This is rare due to the protected situation of the anus and rectum.
- It is usually associated with fracture pelvis.

How to Suspect Large Bowel Injuries? How to Investigate?

- *Intra-abdominal injuries* are diagnosed as any abdominal trauma by the presence of manifestations of peritoneal irritation, free intra-peritoneal fluid, or air.
- *Pain x-ray* may show air under diaphragm (Figure 12).
- *CT scan* indicates free intra-peritoneal fluids and pneumo-peritoneum (Figure 13).

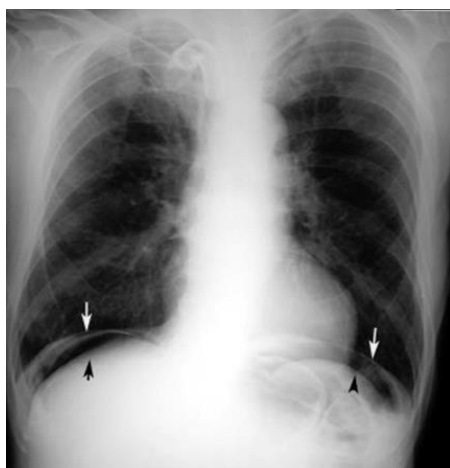


Figure 12. Plain X-ray abdomen showing air under diaphragm (arrows)

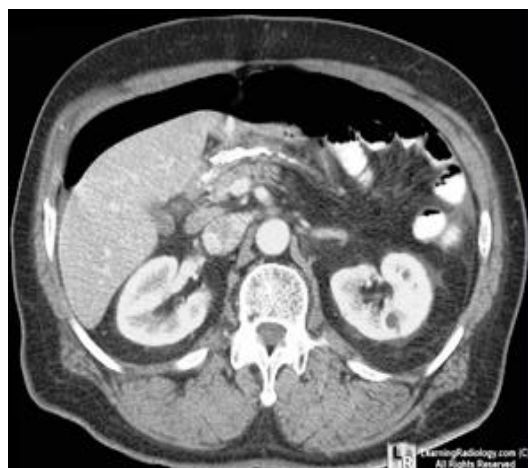


Figure 13. CT scan of the abdomen showing pneumo-peritoneum and free fluid

- *Diagnostic peritoneal lavage (DPL)*: It will not evaluate the retro-peritoneum. Bacteria / vegetable matter suggestive
- The *focused assessment with sonography in trauma (FAST)*
- *Rigid procto-sigmoidoscopy*

- However, *rectal, anal canal and perineal trauma* are diagnosed by proper inspection and PR examination. Presence of bleeding PR is a very important sign. The presence of different types of urethral injuries or fracture pelvis should stimulate the surgeon to properly examine and even sigmoidoscope (the preferred method of investigation) the rectum and the pelvic colon.
- If enema is to be used, water-soluble contrast (gastrographin) is a must and barium should never be used. Again, a CT scan abdomen and pelvis with double or at least IV-contrast is indispensable for proper diagnosis.

Grading

According to AAST (American Association for Surgery of Trauma Organ Injury Scale)

Colon Injury

- Grade I – Contusion or hematoma; partial thickness laceration
- Grade II – Full-thickness laceration <50 percent of circumference
- Grade III – Full-thickness laceration ≥50 percent of circumference
- Grade IV – Transection
- Grade V – Transection with tissue loss; DE vascularized segment

Rectum and Recto-Sigmoid Injury

- Grade I – Contusion or hematoma; partial thickness laceration
- Grade II – Full-thickness laceration <50 percent of circumference
- Grade III – Full-thickness laceration ≥50 percent of circumference
- Grade IV – Full-thickness laceration with perineal extension
- Grade V – DE vascularized segment

Treatment Options

Direct laceration sutures closure (Figure14)

- Criteria and contraindications of primary closure are summarized in Table1.
- In practice, direct laceration sutures closure this is only valid in situations where the colon and/or rectum are injured in a patient whose large bowel is prepared as in operative or endoscopic iatrogenic injuries.

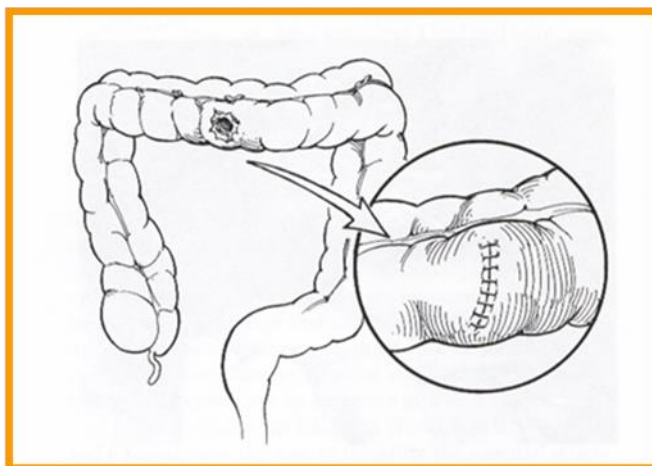


Figure 14. Sutured closure of a laceration in the colon

Table 1: Direct laceration sutures closure.

Criteria for Primary Closure	Contraindications of Primary Closure
No hemodynamic shock.	Patient has been in shock (systolic < 80 mm Hg)
Interference in a time < 8 hours from wound injury.	The interval between injuries and closure is > 8 hours.
Small tear < 2 cm	large tear > 2 cm
Minor spillage reaching to a distance < 5 cm with unloaded colon.	Massive contamination with loaded colon.
No other large bowel injuries	Injuries at two different locations of large bowel
No other organ injuries	More than one organ injured.
Absent of prosthetic material or the necessity of its insertion	Presence of prosthetic material or the necessity of its insertion

Resection + Ileo-transverse Anastomosis

- Resection of the injured area with direct anastomosis of the small bowel to the transverse colon (Figure 15) is only valid in right-sided lesions where direct closure is contraindicated.

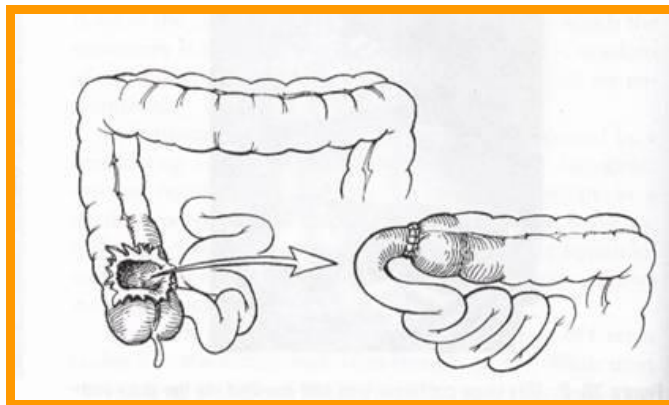


Figure 15. Resection with ileo-transverse anastomosis.

Double-Barrel Colostomy

- Double-barrel colostomy at the site of injury (instead of exteriorization of the repaired injured colon) is done in the transverse or sigmoid colon (mobile area with long mesentery) (Figure16).

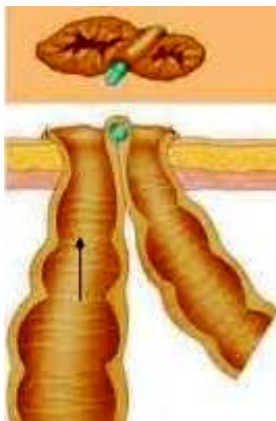


Figure 16. In mobile segments of the colon, the mesentery allows the injured segment to be exteriorized as a double barrel colostomy.

Resection with End-Colostomy and Mucosal Fistula or Hartmann Pouch (Figure 17)

- This is done if the injury is at a site where mobility of the distal limb is limited while mobility of the proximal limb is free.
- One condition is a must. This is to ensure evacuation and emptiness of the distal limb of the large bowel. This is especially applicable in the following situations:
 1. Injury near the splenic flexure of the colon
 2. Injury at the distal region of the sigmoid colon.

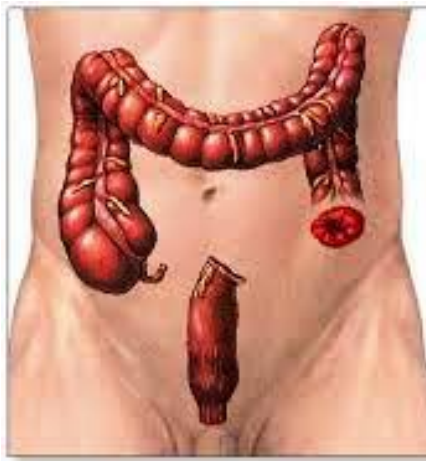


Figure 17. An injury of the distal sigmoid is managed by proximal colostomy & Hartmann closure of the distal rectum.

Suture Closure with Proximal Diversion (Figure18)

- It means 1st closure of the laceration (even if some conditions prohibit that closure) with protection of the 1st repair with proximal fecal diversion, by ileostomy or colostomy with the following conditions:
 1. Diversion should be complete with no chance of any fecal matter passage to the distal limb.
 2. Assurance of removal of all fecal residue in the distal limb
- This is suitable in (1) descending colon injuries, (2) distal sigmoid lesions and (3) proximal intra-peritoneal rectal injuries.

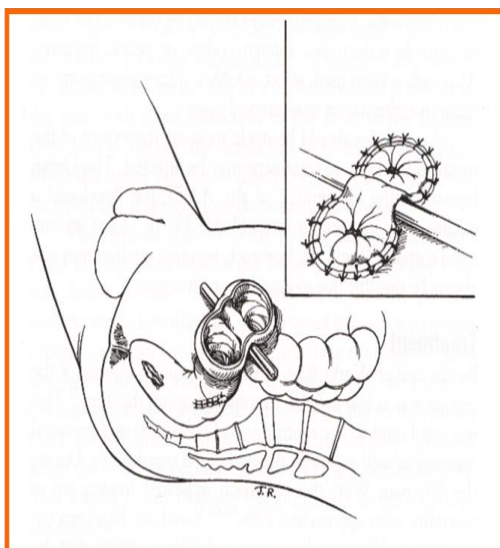


Figure 18. Suture closure with protective colostomy

Principles of Management of Rectal Injuries

1. The injury in the rectum is detected by a through endoscopic examination preferably done by a rigid sigmoidoscope in the left lateral or lithotomy position to determine the injury's location whether in the intra-peritoneal segment or extra-peritoneal one. The extra-peritoneal space for the rectum is divided into retro-peritoneal high up in the abdomen & sub-peritoneal low in the pre-sacral space. Also using the scope, removal of the retained feces with irrigation is done.
2. Direct per rectal repair is done in low injuries (sub-peritoneal spaces) with possible drainage of the pre-sacral space through an incision situated midway between the coccyx and the anus. This is specially indicated in posterior injuries.
3. Direct repair through abdominal exploration is done in injuries of the intra-peritoneal segment or in the retro-peritoneal segment, with possible use of suction drainage of the pre-sacral space if the injury is posteriorly located, and drainage of the Douglas pouch if it is anteriorly located as is usually the case.
4. A proximal complete fecal diversion is a must in all situations.
5. Removal of all retained feces should be ensured by irrigation through either the distal limb of the colostomy, or better still through the rectum.

Principles of Management of Perineal and Anal Canal Injuries

1. In perineal and anal canal injuries, no attempt should be done for 1^{ry} sphincteric or tissue repair. Only debridement and hemostasis are done.
2. It is however, mandatory to divert the fecal stream totally from the wound in the perineum if the lesion is even moderately extensive and specially if involving the sphincteric complex of the anal canal.
3. After healing of the wound and before any colostomy closure, sphincteric repair can be done under cover of the diversion usually with satisfactory results. Treatment options of sphincter injury repair are summarized in Table 2.

Table 2: Treatment options of sphincter injury repair according to presentation

Presentation	Treatment
No associated injuries – no delay in treatment	Primary repair
Significant associated injuries - minor sphincter injury	Delayed repair
Significant associated injury - major sphincter injury	Fecal diversion followed by delayed repair
No significant associated injuries - Delayed presentation (inflammation and edema)	Fecal diversion followed by delayed repair

BLEEDING

Definitions

- *Bleeding per rectum (hematochezia)* is the passage of altered maroon-colored or reddish-dark, brown-colored offensive fluidly stool motions with possible clots. The nature of blood is evident to the patient.
- *Fresh bleeding per rectum* is the passage of fresh blood during passing a stool motion, and it takes several forms as splashing, dripping, streaking of stools or as terminal drops. Sometimes it stains the underwear after defecation. It may be associated with mucus &/or pus.

Massive (Serious) Lower GIT Bleeding

- Passage of a large volume of red or maroon blood through the rectum
- Hemodynamic instability and shock (systolic blood pressure of < 90 mmHg)
- Initial decrease in hematocrit level of 6 g/dL or less
- Transfusion of at least 2 units of packed red blood cells (RBCs)
- Bleeding that continues for 3 days
- Significant rebleeding in 1 week

Causes of Lower GIT Bleeding:

Causes of lower GIT bleeding by *source* are summarized in Table 3, whereas common causes of lower GIT bleeding by *age* are listed in Table 4,

Table 3: Causes of Gastrointestinal Bleeding by Source

Etiology	Upper GIT	Small Bowel Sources	Colonic Sources
<i>Common causes</i>	- Esophageal varices - Ulcers (DU & GU) - Gastritis - Gastric tumors	- Meckel's diverticulum - IBD and vasculitis - Lymphoma	- Angiodysplasia - Diverticula - Neoplasms - IBD - Polyps - Hemorrhoids
<i>Less common causes</i>	- Aorto-enteric fistula - Deulafoy's lesion - GERD - Mallory Weiss syndrome	- Mesenteric ischemia - Radiation enteritis - Intussusceptions	- Solitary rectal ulcer syndrome - Radiation proctitis - Endometriosis - Rectal varices

Table 4: Common Causes of Lower Gastrointestinal Bleeding by Age

Adolescents and young adults	Adults to 60 years of age	Adults > 60 years
- Meckel's diverticulum - IBD - Polyps	- Diverticula - IBD - Neoplasms	- Angiodysplasia - Diverticula - Neoplasms

IBD: Inflammatory bowel disease

Differential Diagnosis

Diagnosis of anal conditions, which present with the following:

Pain and Bleeding

- Fissures

Pain, Lump and Bleeding

- Prolapsed hemorrhoids
- Carcinoma of the anal canal
- Prolapsed rectal polyp or carcinoma.
- Prolapsed rectum

Diagnosis of Conditions Presenting with Rectal Bleeding but No Pain

- Blood mixed with stool → colon carcinoma.
- Blood streak on stool → anal fissure & rectal carcinoma
- Blood after defecation → hemorrhoids
- Blood and mucus → colitis
- Blood alone → diverticular disease
- Melena → peptic ulcer

Investigations

Laboratory Investigations

- Complete blood count (CBC) count
- Serum electrolytes
- Blood urea nitrogen, creatinine
- Blood grouping and cross matching
- Coagulation profile including: Partial thromboplastin time (PTT), platelet count, bleeding time

Colonoscopy

Colonoscopy has diagnostic and therapeutic capabilities. The patient must be in stable condition.

- It can be used even with ongoing massive bleeding
- Timing ideally 6-24 hours post-presentation
- Increased risk of bowel perforation (2% risk of perforation)
- Should not be attempted with ischemia or severe mucosal inflammation.
- Endoscopic coagulation treatment of choice for bleeding AVMs (Arterio-venous malformations)
- Diverticular bleeding cannot usually be endoscopically coagulated (technically challenging)

Nuclear Scintigraphy

It can detect bleeding at a rate of ≥ 0.05 -0.1mL/minute.

Technetium 99m (^{99m}Tc) Sulfur-Colloid

- Requires no preparation → inject immediately
- Short half-life
- Enhances liver and spleen → can obscure bleeding sites
- Able to detect slower rates of bleeding

^{99m}Tc-labeled RBCs

- Preferred method (Figure 19)

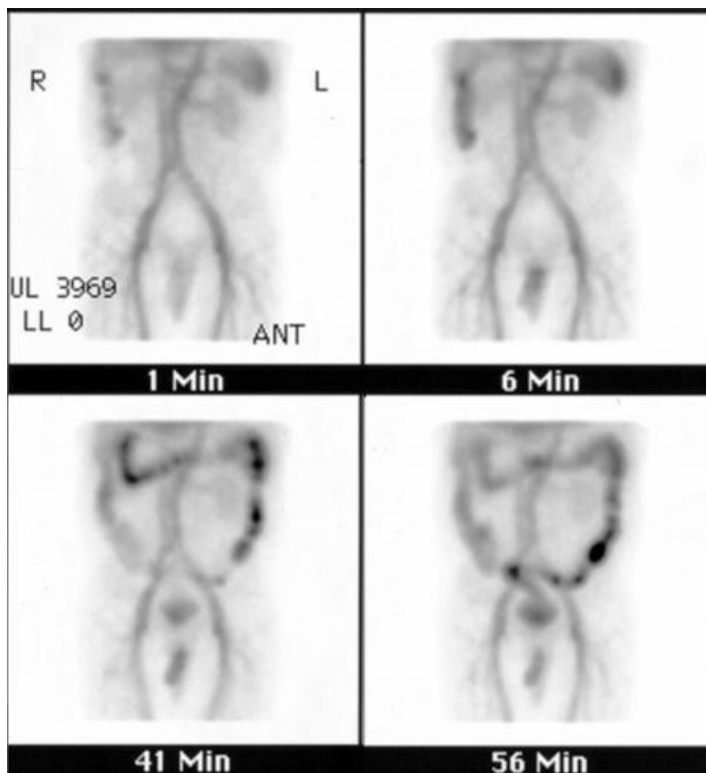


Figure 19. Active bleeding from ascending colon

- Good for intermittent bleeds
- Able to visualize bleeds near liver/spleen
- 73-98% sensitive
- False localization rate ~20% (3-59%)
- Blood does not remain stationary within bowel lumen: Reflux into small bowel - rapid transit through bowel with pooling in other areas
- Institution dependent: Not available in all hospitals - Variability in quality

Angiography

- Selective mesenteric angiography (Figure 20)
 - o Femoral artery punctured
 - o Evaluates superior mesenteric (SMA), then inferior mesenteric artery (IMA), then the celiac axis
 - o Positive test if extravasation of contrast into bowel lumen
- Can detect hemorrhage rates $\geq 0.5\text{mL/min}$
- Diagnostic and potentially therapeutic: Adrenaline infusion - embolization
- Sensitivity 40-86%
- Complication rate ~ 2%



Figure 20. Normal selective inferior mesenteric artery (IMA) angiogram

CT Angiography

- New and evolving technology
- Requires modern CT scanner
- Fast (< 15 minutes)
- Non-invasive
- Identifies large and small bowel hemorrhage
- Sensitivity and specificity 72-80%

Treatment

An algorithm for management of lower GIT bleeding is depicted in Figure 21.

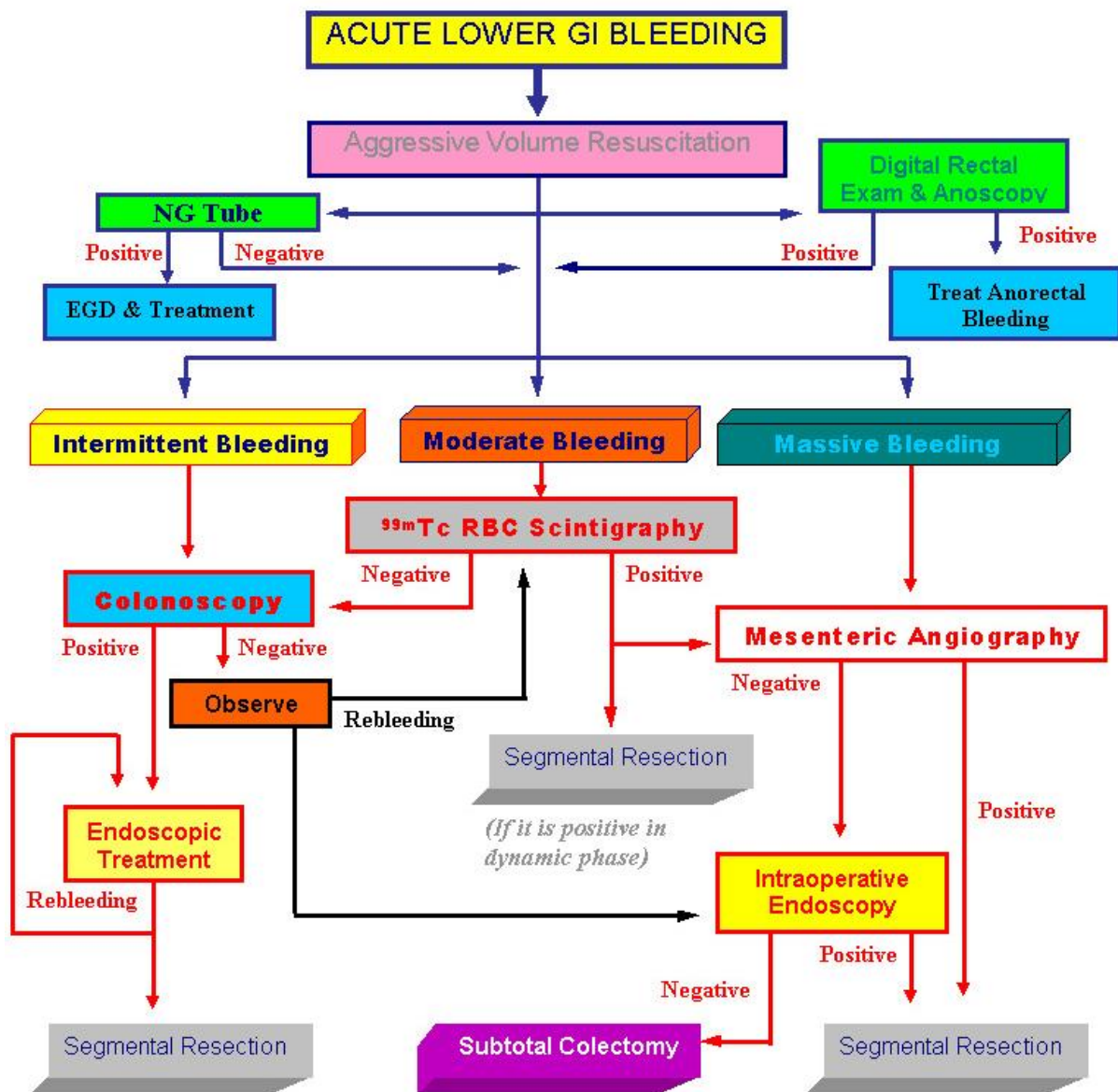


Figure 21: Algorithm for management of lower GIT bleeding

- Resuscitation with crystalloid solutions, preparing for blood transfusion & diagnostic nasogastric (NG) intubations are important steps. The presence of bile in the aspirate is the only sure sign that there is no active upper GIT source of bleeding. Upper endoscopy can diagnose more accurately the upper GIT source.
- It should be recognized that about 85% of bleeding from lower GI sources stops spontaneously; however, the presence of large amounts of blood in the colon can give the impression of a continuous bleeding process. That is why in monitoring bleeding, one should depend more on the hemodynamic figures rather than frequent passage of bloody motions.

- PR examination, anoscopy and possibly rigid sigmoidoscopy are imperative to rule out palpable neoplasms and other anal and rectal conditions. It is important, even if detection of a lesion is not possible due to bad preparation, to ascertain a healthy lower 15-20 cm for a possible use of subtotal colectomy. Treatment of a simple visible cause like piles can be done at the same setting by injection or ligation.
- Colonoscopy offers the chance of proper diagnosis and sometimes-non-surgical stoppage of bleeding. Mechanical preparation can be done. It should be done as soon as the patient is stabilized and better under general anesthesia. Diverticulosis, IBD, polyps & neoplasms are confidently diagnosed; however, AVMs are sometimes difficult to diagnose, but if seen can be treated with injection of diluted adrenaline or photocoagulated by laser therapy. Colonoscopy can also diagnose the source of bleeding as small bowel if the cecum is reached & more fresh blood is coming out of the ileo-cecal valve. This will indicate small bowel site (refer to the Table above for causes).
- Arteriography (by selective catheterization of all mesenteric vessels) and ^{99m}Tc-labeled RBC scanning are only of use if active bleeding is present with a high rate (2ml/minute or higher), which is not usually the case on admission of the patient. Their use is more or less of academic interest only. Selective vasopressin infusion can help, as well as peripherally used vasopressin.

Vermiform Appendix

Epidemiology

- Acute appendicitis affects 6-7 % of the population.
- Peak incidence in adolescents and young adults, with a slight male predominance in this age group.
- Infants, elderly, pregnant women and immuno-compromised patients tend to have atypical presentations and have higher morbidity and mortality

Anatomical Positions (Figure 1)

- Retro-cecal: 64%
- Pelvic position: 32%
- Other positions: 5%

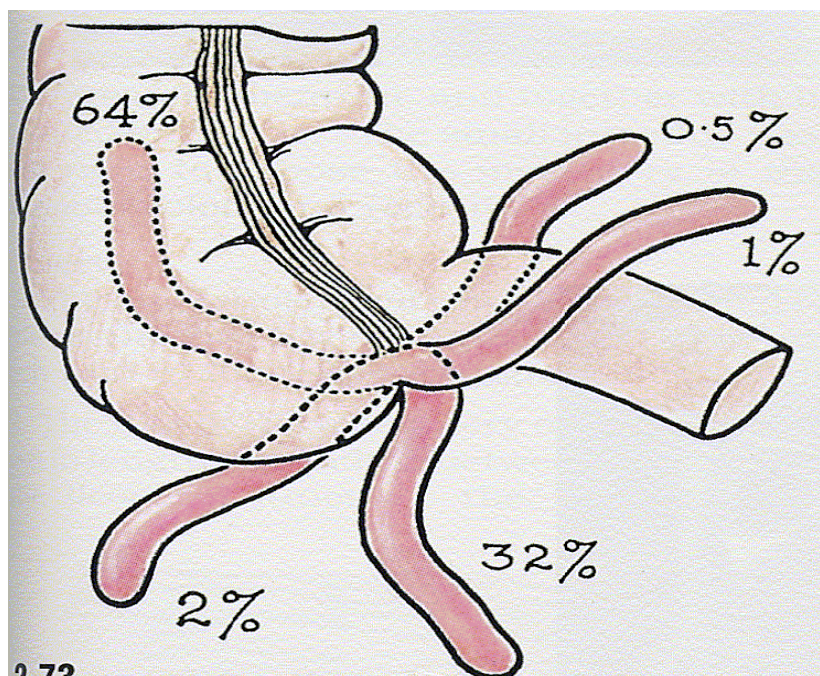


Figure 1: Different anatomical positions of the appendix. The retro-cecal is the commonest.

Position of the appendix also changes according to the position of the cecum, which can be sub-hepatic, or pushed up by pregnancy

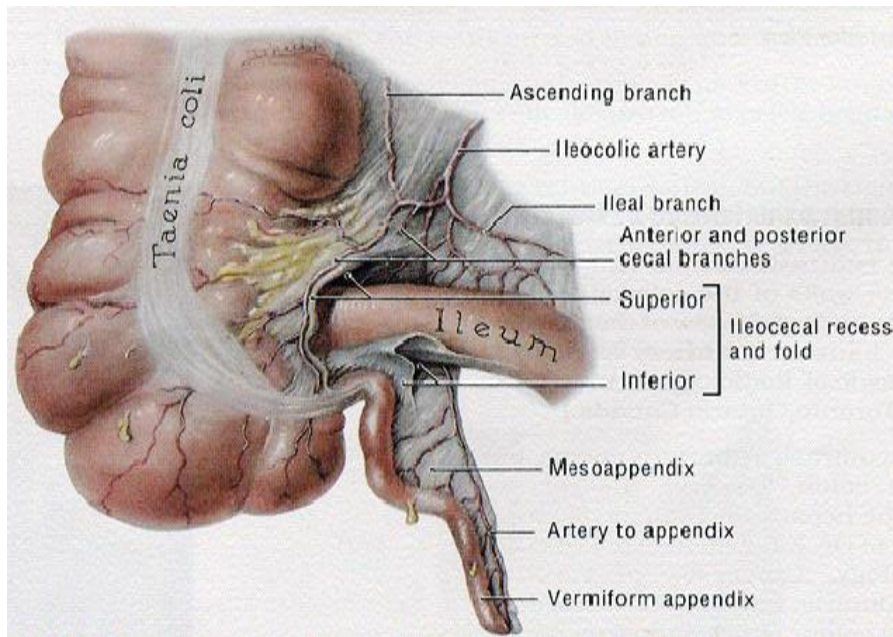


Figure 2: Anatomy of the pelvic appendix. The second most common site.

Pathophysiology

- *More than two-thirds* of the cases are due to obstruction of the lumen by
 1. Fecolith
 2. Lymphoid tissue in response to viral infection
 3. Kink or adhesion
 4. Parasite or tumor
- This will lead to a period of appendicular colic, with accumulation of secretion inside the lumen that will be infected causing inflammatory appendicitis, which easily perforates
- *Less than one-third* of the cases is due to non-obstructive acute appendicitis. Inflammation is usually catarrhal and stays like that with little liability to perforate

Fate of the Inflammatory Process

1. *Resolution* with or *without* future recurrent attacks
2. *Gangrene* and perforation (either at the tip or at the site of the fecolith impaction)
3. *Walling-off* by omentum, and small bowel loops forming a phlegmon (inflammatory mass), also known as *appendicular mass*
4. *Generalized peritonitis* in extremes of age and in patients with bad immunological defenses
5. *Localized perforation* within the appendicular mass leading to appendicular abscess, which can later perforate giving rise to generalized peritonitis
6. *Portal pyemia* with *septicemia* is rare

Clinical Presentation

Symptoms

- Pain that shifts is the presenting symptom.
- Visceral type of pain in the peri-umbilical region shifts to somatic type of pain localized to the right iliac fossa (RIF).
- Anorexia and nausea are constant, while vomiting is early and preceded by pain.
- Constipation is usual; however, diarrhea can be the start of the condition (usually viral), then acute appendicitis starts due to hypertrophy of the lymphoid follicles

Physical Signs

- Temperature and pulse are only slightly raised early in the disease, higher elevations indicate complications.
- Localized pain that increases with coughing.
- Localized tenderness and rebound usually over the McBurney's point (Figure 3) with guarding.
- Rigidity and restricted respiratory abdominal movements are present with local peritonitis.
- PR or PV examination can elicit tenderness in pelvic positions.

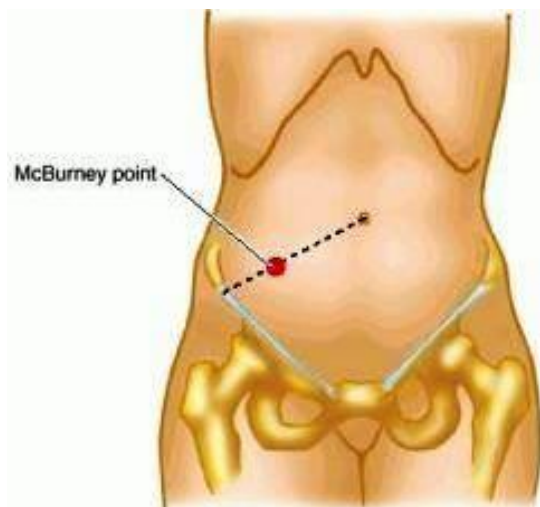


Figure 3: McBurney's point at the junction of the lateral third and medial two-thirds of the line between the umbilicus and the anterior superior iliac spine.

Special Signs

- *Rovsing sign*: contra-lateral pressure cause right-sided pain.
- *Psoas sign*: extension of the hip cause pain.
- *Obturator sign*: internal rotation of the hip cause pain.

Atypical Clinical Features

- Deep pelvic appendix causes tenesmus + dysuria and PR tenderness.
- Sub-hepatic cecum gives manifestations that simulate cholecystitis.
- Long retro-cecal appendix, gives pain simulating right colonic pain with minimal abdominal signs.
- A retro-ileal appendix can present with diarrhea.

Special Age Groups

Children

- Approximately, 80% of cases lead to early generalized perforation due to:
 1. Lack of well-developed omentum
 2. Lack of precise diagnosis due to difficulty in examining that age group
 3. Mis-interpretation of the case as gastroenteritis due to frequent vomiting

Elderly

- Acute appendicitis in the elderly also gives rise to early generalized perforations due to:
 1. Weak immune system
 2. Signs are not prominent although the case can be perforated

Child-Bearing Age / Pregnancy

- In such cases, early generalized perforation may occur due to
 1. The usual site is different (higher)
 2. Mis-diagnosis as pyelitis
 3. Weak omental action
 4. Plus the seriousness of the 50% abortion or the 30% premature labor

Differential Diagnosis

Thoracic Problems

- Right-sided pneumonia or pleurisy mimic the condition, yet the marked physical signs of the chest clinches the diagnosis

Upper Abdominal Problems

- *Perforated peptic ulcer* specially the leaking one; however, a long history of upper abdominal troubles and the free air under the diaphragm may help in diagnosis.
- *Acute cholecystitis*, yet the prominent vomiting, and the ultrasound (US) confirmation of the possibly palpable gall bladder (GB) usually differentiate between both conditions.
- *Acute pancreatitis* especially the edematous type, yet the type of pain and lack of localized tenderness and other special signs may clinch the diagnosis, which is confirmed by amylase.

Lower Abdominal Problems

- *Non-specific lymphadenitis*: it usually affects children, with higher fever, and may follow a sore throat, yet differentiation is difficult
- *Meckel's diverticulum*: it is almost impossible to diagnose except after exploration
- *Crohn's disease* is also impossible to diagnose before operation in the patient firstly presenting to the surgeon
- *Cancer cecum*: this is the usual differential diagnosis of an appendicular mass rather than appendicitis.

Pelvic Problems (Ultrasound is Usually Conclusive)

- *Rupture Graffian follicle* (mid-cyclic pain) is very common differential diagnosis
- *Disturbed ectopic pregnancy*: pallor and history of amenorrhea should elicit diagnosis

- *Complicated ovarian cyst* (infected or twisted), where the presence of a mass is the clue to diagnosis, this is *not* usually palpable due to the severe overlying rigidity
- *Pelvic inflammatory disease* (PID): it is characterized by bilateral tenderness, in addition to the vaginal discharge and higher degrees of fever.
- *Red degeneration of fibroid*: this is usually the differential diagnosis of appendicitis in pregnant women.

Urologic Problems

- *Ureteric colic*: this is typical visceral pain, with prominent vomiting, and lack of rigidity and rebound, with no high temperatures, and may be hematuria
- *Right pyelonephritis*: the temperature is very high with rigors, and pain is more localized to the loin except if it is ectopic kidney, and dysuria is very common

Neurologic Problems

- *Herpes Zoster* gives pain similar to appendicitis, usually the age is advanced, and the eruption that occurs after 36 to 48 hours clinches the diagnosis. No fever, no vomiting, and no rigidity or rebound tenderness

Investigations

Laboratory Investigations

- *Complete blood picture*: showing leukocytosis with shift to the left
- *Urine analysis*: for RBCs and pus cells (hematuria excludes non-ureteric problems)
- *Pregnancy test*: in females in childbearing age

Imaging Studies

Ultrasound (US) examination

- It is operator-dependent, yet it is very useful in children and in females in child-bearing age
- Signs of significance are:
 1. Non-compressible appendix
 2. Appendix > 6 mm in diameter
 3. Detection of a fecolith
 4. Presence of peri-appendicular fluid collection
 5. Probe tenderness
 6. Exclusion of tubo-ovarian pathology

CT with or without IV contrast

- It approaches a 97% accuracy rate, and is the main mode of investigating acute abdominal conditions
- Signs of significance are:
 1. Enlarged distended appendix > 6 mm
 2. Appendiceal wall thickening > 2 mm
 3. Peri-appendiceal fat stranding
 4. Focal cecal apical thickening
 5. Appendicolith (seen in 20-40% of patients with appendicitis)

6. Target structure (concentric thickening of the inflamed appendiceal wall)
7. Extra-luminal air
8. Collections of fluid in the RIF or pelvis

Magnetic resonance imaging (MRI)

- It is only used if ultrasound was not conclusive in a pregnant woman that cannot be exposed to irradiation of the CT scan and needs definitive diagnosis.

Types of Patients

- Patients with acute appendicitis are categorized as high-risk and non-high-risk patients.
- High-risk patients:
 1. Children due to high risk of perforation
 2. Elderly patients due to high risk of perforation and the existence of co- morbidity
 3. Women in the child-bearing period due to the presence of tubo-ovarian pathology that mimics appendicitis, and the high rate of complications in pregnancy
 4. Immuno-compromised patients due to high mortality and high morbidity

Perforation of Appendix in the Pediatric Age

- In children below two years, perforation approaches 100%
- From three to five years, perforation is around 70%
- From six to ten years, perforation is around 40%

Risk in Elderly Patients

- Elderly patients have higher morbidity and mortality rates
 1. Less typical presentation
 2. Cancer may be a possibility as an underlying cause.
 3. Perforation of 50% and mortality of 20% has been reported

Pregnancy with Acute Appendicitis

- 1: 2000 pregnancies.
- More common in the first two trimesters of pregnancy
- The appendix is pushed superiorly and laterally
- WBC > 15,000/mm³
- Premature labor: 10-15% with surgery
- Perforated appendix leads to fetal death in 20% of cases
- Rapid diagnosis and treatment is advised.

In Immuno-Compromised Patients

- Be aware of CMV or Kaposi sarcoma as the underlying cause
- WBC may not rise

Treatment

In Non-Complicated Acute Appendicitis

- Treatment is by appendectomy
- No delay is allowed specially in extremes of age, pregnancy and diabetics.

In Appendicular Mass

- It is recommended to start conservative treatment (*Oschsner–Sherren regimen*) although some surgeons use midline incisions to dismantle adhesions and remove the appendix
- *Oschsner-Sherren regimen*
 1. Bed rest
 2. Nil by mouth for few days then liquids for another few days
 3. IV fluids and antibiotics (double or triple regimen attack)
 4. Follow up as regards pain, mass size, temperature and pulse
- *Further Treatment: Interval appendectomy* is done after 3 months if the mass resolved, or otherwise one is dealing with appendicular abscess.

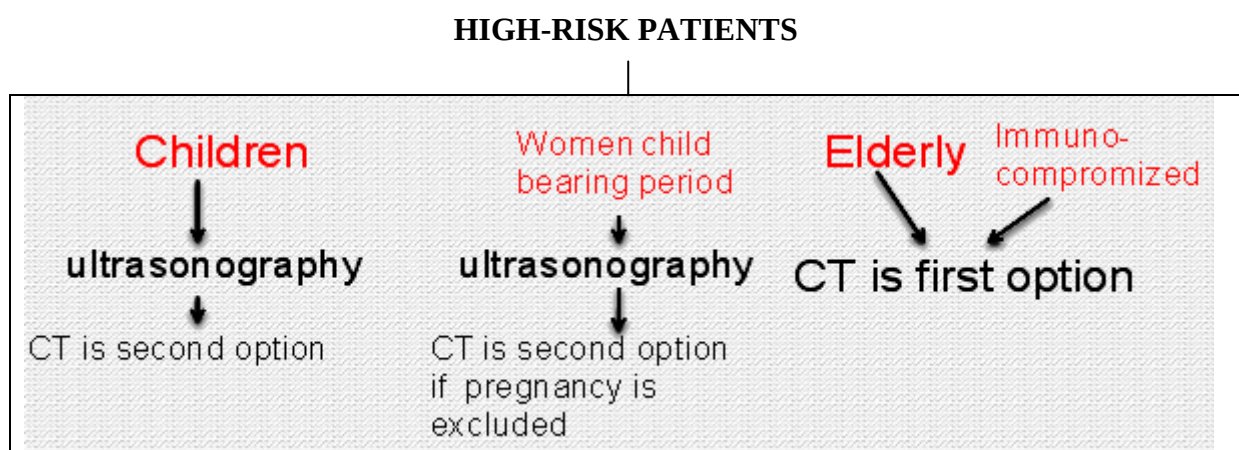
In Appendicular Abscess

- Ultrasound, or more accurately, CT scan confirms the presence of pus inside the mass, and can be used as guide to insert a pigtail catheter for drainage in case there is a single loculus.
- Surgical extra-peritoneal drainage is the classical treatment; all loculi are opened and an *overflow* or *suction* drainage is used
- No attempt is made to remove the appendix except if it is available without risk to the inflamed tissues. Interval appendectomy is otherwise done after 3 months to avoid recurrent inflammation like in appendicular mass.

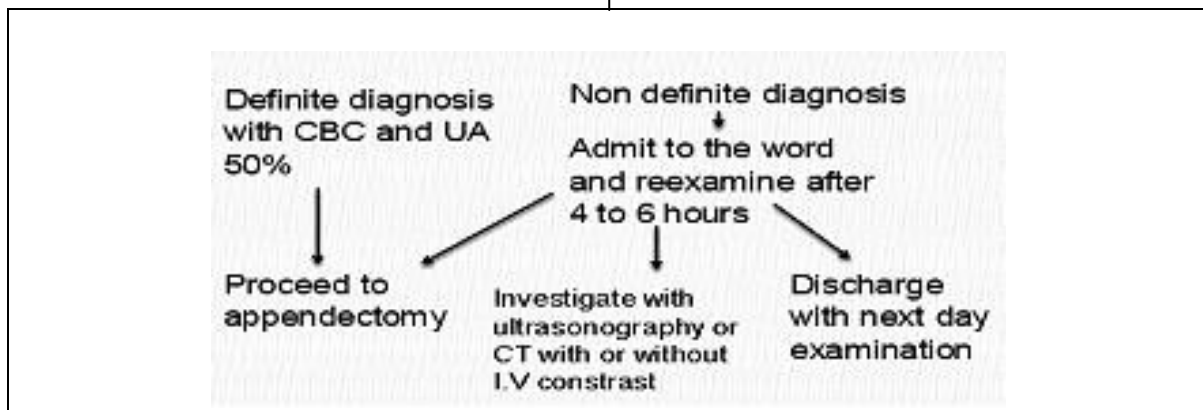
In Generalized Peritonitis

- Urgent surgery with appendectomy and peritoneal toilet and drainage is the sole line of treatment

Different Situations in Management



NON-HIGH-RISK PATIENTS



POTENTIAL COMPLICATED APPENDICITIS IN ANY PATIENT

- Delayed presentation, high fever, palpable mass, toxic look and signs (SIR)
- Prompt evaluation with US and CT is mandatory to evaluate the condition whether “pus” is present or it is only a “phlegmon”.
- Surgery through a midline exploration is a must if pus is present, while Oschsner technique is advised if no pus is present with interval appendectomy done after 3 months

Complications of Appendectomy

Mortality

- Death following appendectomy done for perforated appendix is usually attributable to sepsis. Delay by the patient in seeking medical attention or delay on the part of the physician in instituting appropriate treatment is an important factor in allowing perforation to occur and results in increased morbidity and mortality. Perforation can often be diagnosed pre-operatively. It should be suspected when the duration of symptoms exceeds 48 hours. A temperature higher than 38°C and a WBC count of >15,000 are both uncommon in non-perforated appendicitis. A history of diffuse abdominal pain following symptoms confined to the right lower quadrant accompanied by signs of diffuse peritonitis on physical examination indicates that free perforation has occurred. If perforation has been walled off into an appendiceal abscess, a tender mass can often be palpated in the right lower quadrant.
- Diagnosis should be suspected in a patient with a history suggestive of acute appendicitis of longer than 48 hours duration, who also has fever, leucocytosis with or without a tender mass in the right lower quadrant of the abdomen.
- General awareness of symptoms of appendicitis and a more aggressive surgical approach should reduce morbidity and mortality by decreasing the incidence of perforation.
- Age is also an important contributing factor to mortality, as death following appendectomy is more common in the very young and the elderly, with a mortality rate of 15% in patients above 70 years. This is most probably due to the presence of multiple associated medical problems, delay in seeking medical advice, and the surgeon's reluctance to operate until diagnosis is 100% certain.

The Most Important Complications are

1. Wound infection
2. Pelvic and other intra-abdominal abscesses
3. Appendiceal stump abscess and other problems of the stump
4. Fecal fistula
5. Intestinal obstruction, mainly adhesive intestinal obstruction, can be due to injury of the mesentery of the terminal ileum and gangrenous small bowel consequence.
6. Adhesions, sterility and increased rate of ectopic pregnancies or abortion if appendectomy is done in a pregnant woman.
7. Hemorrhage intra-peritoneal (slipped meso-appendiceal ligature) hypovolemic shock with urgent surgical intervention.
8. Bleeding per rectum (usually inverted non ligated or sloughed stump)
9. Right inguinal hernia (may be due to nerve injury (ilio-hypogastric nerve) or transversalis fascia or transverses aponeurosis injury during McBurney's grid iron incision)
10. Hepatic abscess and portal pyelo-thrombophlebitis
11. Intussusceptions of the appendiceal stump

Wound Infection

- It is the most common complication following appendectomy, occurring in up to 30% of patients. As expected the severity of the appendiceal inflammation correlates well with the incidence of wound infection, as it most common following appendectomy for perforated appendices (dirty wound) and least common after removal of a normal appendix (clean contaminated wound). It is usually a subcutaneous infection only; the muscles are really very strong barrier to infection on the condition of being viable.
- Frequently overlooked is the importance of the use of meticulous surgical technique in the prevention of postoperative wound infection. Care should be taken to prevent contact of the inflamed appendix with the wound, and the operative area should be carefully packed off to prevent inoculation of the wound when the appendix is removed. If perforation has occurred closure of the skin and subcutaneous tissue is an invitation to infection. These wound are best managed by packing open with Betadine –soaked gauze that is changed three times daily until 72 hours, when delayed closure is done if the wound appeared to be clean otherwise it is left to granulate and close by secondary intention.
- It is suspected in patients who has post-operative fever or complains of inordinate incisional discomfort .The signs of wound infection are well known and include erythema, tenderness, and purulent discharge. This is the clinical picture of the usual type of wound infection that appears as fever from the third postoperative day on and the usually offending organism is E. coli or mixture of E.coli and staphylo- or strepto-cocci. However, a severe form of wound infection starts as early as the first 24 hours postoperative and it affects patient with some sort of immune compromise, and its offending organism is streptococci with anaerobes and cause non-clostridial gas producing infection like necrotizing fascitits. Clostridia can also affect these wound early post-operatively with frank gas gangrene.

- Treatment consists of opening the wound and packing it with Betadine-soaked gauze to be changed three times daily. Adequate debridement of necrotic tissue is also necessary. Antibiotics are given for coexisting cellulites.

Pelvic Abscess

- It is not uncommon following appendectomy and its incidence also increases with the degree of appendiceal inflammation and is most common after perforation. Owing to its dependent position and proximity to the appendix, the pelvis is a frequent site of abscess formation. It usually becomes apparent 5-10 days after operation.
- Signs and symptoms include fever, abdominal discomfort, frequent loose bowel movement (irritative), lower abdominal deep tenderness and most importantly P.R tenderness and bogginess. Diagnosis is easily confirmed by US and/or CT scan through either of them PAD (per cutaneous abscess drainage) is done for treatment. Patients with multiple abscesses, or difficult locations of abscesses operative drainage is used.
- Abscesses may develop elsewhere in the abdomen according to the location of the appendix, most commonly the Morrison pouch and right sub-hepatic spaces.

Appendiceal Stump Problems

- Three basic methods have been advocated for handling the appendiceal stump: simple ligation, ligation and inversion, and inversion without ligation. The most commonly practiced technique is to ligate the appendix at its base and then invert the stump into the cecum using a purse string suture.
- The theoretical advantages of the method include
 1. Good control of hemorrhage from the stump
 2. Secure closure of the cecal wall
 3. Minimal contamination of the peritoneum by burial of the transected stump
 4. Serosal-to serosal approximation that minimizes the extent of raw surface exposed to create adhesions
- Simple ligation alone, on the other hand, can cause increase chance of peritoneal contamination, by exposure, sloughing of the stump or loosening of the ligature. Inversion is designed to prevent this possibility.
- Inversion is also criticized by being able to compromise the cecal wall vitality, and hence cause necrosis and /or perforation of the cecum with all imagined consequences. It can also cause formation of abscess in the wall of the cecum rupture of which can cause either generalized or localized peritonitis.
- Abscess formation due to appendicular stump rupture is uncommon 0.5% or even less. It typically occurs from the fifth day on. The patient who has been discharged from the hospital comes back with picture of generalized or localized peritonitis; US will demonstrate free fluid, which is usually a turbid offensive one, with fecal matter only present if sloughing of the cecal wall is present. Treatment is by re-exploration and refreshing of the edges of the cecum with closure in layers, cecostomy, or limited right hemicolectomy according to the anticipated healing power of the patient and tissues involved.

Fecal Fistula

- This complication is usually following the operation on appendicular abscess through a small McBurney's incision with too much manipulations of the inflamed tissues endangering the vitality of the cecal wall which if get adherent to the wound area will cause wound infection opening of which will disclose the presence of feces in the wound.
- A cecal fistula usually closes with conservative treatment provided that the patient is anabolic, the skin is healthy, the fistula is small, there is no associated intra-abdominal abscess and there is no distal obstruction or foreign bodies inside.
- If conservative treatment (NPO, IV alimentation and antibiotics, and somatostatin subcutaneously) surgery with limited right hemicolectomy or simple cecal wall closure after refreshment is indicated.

Intestinal Obstruction

- This is usually small bowel obstruction due to adhesions between a loop of the small bowel and the appendiceal stump if not inverted. Conservative treatment can usually help the patient to avoid an exploratory operation. However sometimes these adhesions cannot be treated except surgically. The typical mechanical small bowel obstruction with repeated copious vomiting as the patient tries to drink or eat with distension and multiple fluid levels in the plain standing X-ray clinches the diagnosis (with no fever, and no ileus).
- Another more serious type of intestinal obstruction is due aggressive manipulations in delivering the cecum out of the abdominal cavity with multiple injuries to the mesentery of the terminal ileum causing gangrenous and/or ischemic insults to the ileum giving clinically a picture of strangulated not pure mechanical obstruction, with ileus and signs of peritoneal irritation, together with a bad general condition of the patient. Surgery is a must in these cases to resect the non-viable bowel with end-to-end anastomosis.